

Notes to Lie algebra

October 1, 2025

When the northern sea was still covered with solid ice, I saw the plum blossoms in full bloom.
—————cited by cwz

This is my notes to Lie algebra, which is mainly referred to lecture notes by Wang Xiuling and [gtm9](#). In the first 6 sections, I added some things in GTM 9, but in the rest of the note, due to lack of time, I referred to Wang's lecture notes(which is similar to MIT 18.745) mainly. The statements marked by * is of highly importance.

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1 basic definitions and some examples

1.1 The concepts of Lie algebra

First, let's see some definitions concerning algebra.

Definition 1.1.1. An **F-algebra** $\mathcal{A}$ is a vector space over F , endowed with a bilinear operation: $\mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A}$, i.e.:

$$a(\lambda b + \mu c) = \lambda ab + \mu ac, (\lambda b + \mu c)a = \lambda ba + \mu ca \quad \forall a, b, c \in \mathcal{A}, \lambda, \mu \in F$$

Moreover, if the operation satisfies associativity, i.e.: $(ab)c = a(bc)$, the algebra will be called an **associative algebra**.

Definition 1.1.2. K is a **subalgebra** of L if:

- i. K is a subspace of L
- ii. $xy \in K$ for all x, y in K .¹

Now, let's begin definitions of Lie algebra:

Definition 1.1.3. Lie algebra: A Lie algebra over F is a **vector space** L over F with an operation **bracket** $L \times L \longrightarrow L, (x, y) \longrightarrow [xy]$, and the bracket should satisfy:

(L1) The bracket is bilinear

(L2) $[xx] = 0$ for all x in L

(L3) $[x[yz]] + [y[zx]] + [z[xy]] = 0$ (which is called Jacobi identity)

If we apply (L1) and (L2) to $[x+y, x+y]$, we can get **anticommutativity** (L2') $[xy] = -[yx]$.² Actually, (L1) ensures the vector space is an algebra, according to definition 1.1.1.

Definition 1.1.4. Two Lie algebras L, L' over F are **isomorphic** if there exists a **vector space isomorphism** $\phi : L \longrightarrow L'$, satisfying $\phi[xy] = [\phi(x)\phi(y)]$ for all x, y in L .³ Then ϕ is called an isomorphism of Lie algebra.

Definition 1.1.5. K is a **subalgebra** of L if:

- i. K is a subspace of L
- ii. $[xy] \in K$ for all x, y in K .⁴

1.2 Linear Lie algebra

In this subsection, we define a new operation $[x, y] = xy - yx$, called the bracket of x, y . We can easily find that: under this operation, $\text{End } V$ becomes a Lie algebra over F . In order to distinguish this new algebra structure from the old associative one, we write $\mathfrak{gl}(V)$ for $\text{End } V$ when it is viewed as Lie algebra and call it the **general linear algebra**. If we choose a set of bases, we can denote $\mathfrak{gl}(V)$ by $\mathfrak{gl}(n, F)$. Any subalgebra of $\mathfrak{gl}(V)$ is called **linear Lie algebra**. It is very important for we will see that every (finite dimensional) Lie algebra is isomorphic to some linear Lie algebra. (which is called Ado's theorem.)

Now, we concentrate on four families A_l, B_l, C_l, D_l , which are called the **classical algebras**:

$A_l: \mathfrak{sl}(l+1, F) = \{x \in \mathfrak{gl}(l+1, F) \mid \text{tr}(x) = 0\}$, which is a subalgebra of $\mathfrak{gl}(l+1, F)$, called **special linear**

¹which means K is closed under the operation

²Conversely, if $\text{char } F \neq 2$, (L2') will imply (L2)

³which simply means it communicates with the bracket operation

⁴which means K is closed under the operation

algebra . We can easily find that the dimension of $\mathfrak{sl}(l+1, F)$ is $(l+1)^2 - 1$.

C_l : Consider $s = \begin{pmatrix} 0 & I_l \\ -I_l & 0 \end{pmatrix}$ and $x = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$. Let x satisfy $sx = -x^t s$, i.e., that $n^t = n, p^t = p, m^t = -q$. (The last condition forces $\text{tr}(x)=0$.) Denote all x satisfying this by $\mathfrak{sp}(2l, F)$, called the **symplectic algebra**. The dimension is $2l^2 + l$.

B_l : Let $\dim V = 2l + 1$. Changing the s in C_l with $s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_l \\ 0 & I_l & 0 \end{pmatrix}$ and denote all x satisfying $sx = -x^t s$ by $\mathfrak{o}(2l+1, F)$, called the **orthogonal algebra**. The dimension is $2l^2 + l$.

D_l : When $\dim V$ is $2l$, we get another **orthogonal algebra**. In this case, s has a simpler form $\begin{pmatrix} 0 & I_l \\ I_l & 0 \end{pmatrix}$ and we can find that $\dim \mathfrak{o}(2l, F) = 2l^2 - l$.

Actually, we can find that: when $\text{ch} F = 0$, let $L = A_l, B_l, C_l, D_l$, we have $[LL] = L$.

Finally, we consider these special cases:

Let $\mathfrak{t}(n, F)$ be the set of all **upper triangular matrices**;

Let $\mathfrak{n}(n, F)$ be the set of all **strictly upper triangular matrices**;

Let $\mathfrak{d}(n, F)$ be the set of all **diagonal matrices**.

And we have: $\mathfrak{t}(n, F) = \mathfrak{n}(n, F) + \mathfrak{d}(n, F)$ (which is a vector space direct sum), with $[\mathfrak{d}(n, F), \mathfrak{n}(n, F)] = \mathfrak{n}(n, F)$, $[\mathfrak{d}(n, F), \mathfrak{d}(n, F)] = 0$ (Diagonal matrices are commutative), hence $[\mathfrak{t}(n, F), \mathfrak{t}(n, F)] = \mathfrak{n}(n, F)$. (If H, K are subalgebras of L , $[HK]$ denotes the subspace of L spanned by commutators $[xy]$, x in H , y in K .)

1.3 Lie algebras of derivations

Definition 1.3.1. Let $\mathcal{A}$ be an F -algebra. A **derivation** of $\mathcal{A}$ is a linear operation $\delta : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ and satisfies product rule $\delta(ab) = \delta(a)b + a\delta(b)$. **Der** $\mathcal{A}$ is the collection of all derivations of $\mathcal{A}$, which is a subspace of $\text{End } \mathcal{A}$. For the commutator $[\delta, \delta']$ of two derivations is again a derivation, **Der** $\mathcal{A}$ is a subalgebra of $\mathfrak{gl} \mathcal{A}$. For Lie algebra L is an F -algebra, the **Der** L is defined.

Definition 1.3.2. Let L be a Lie algebra. If x is in L , **ad** x is an endomorphism of L : **ad** $x(y) = [xy]$.¹ Derivations of this form are called **inner**, all others are called **outer**. The map **ad**: $L \rightarrow \text{Der } L, x \rightarrow \text{ad } x$ is called the **adjoint representation** of L , which is very important.²

Sometimes we need to view x simultaneously as an element of L and of a subalgebra K of L . To avoid ambiguity, the notation **ad** $_L x$ or **ad** $_K x$ will be used to indicate that x is acting on L (respectively, K). For example, if x is a diagonal matrix, for diagonal matrices are commutative, then **ad** $_{\mathfrak{d}(n, F)}(x) = 0$, whereas **ad** $_{\mathfrak{gl}(n, F)}(x)$ need not be zero.

1.4 Structure constants

Definition 1.4.1. Structure constants: Let L be a Lie algebra, $x_1, x_2, \dots, x_n$ is a basis of L . Then for all i, j , we have $[x_i x_j] = \sum_{k=1}^n a_{ij}^k x_k$, the a_{ij}^k are called **structure constants**. The multiplication table of L can be determined in this way, thus L is determined.

¹When $x \neq 0$, **ad** x is possible to be 0, e.g. $x \in C(L) = \{x \in L \mid [xy] = 0, \forall y \in L\}$.

²For **Der** $\mathcal{A}$ is a subalgebra of $\mathfrak{gl} \mathcal{A}$, perhaps we can use this representation to prove every (finite dimensional) Lie algebra is isomorphic to some linear Lie algebra.

Corollary 1.4.1. *Using $(L_2), (L_2')$, we can find that for $i \geq j$, a_{ij}^k can be deduced from the others. More precisely, we have: $a_{ij}^k = -a_{ji}^k$.*

Corollary 1.4.2. ** Using the thought of structure constants, we can find that 2-dimensional Lie algebra are only two cases: abelian Lie algebra and nonabelian Lie algebra, in which case let x, y be a basis of L , $[x, y] = x$.*

2 Ideals, homomorphisms and introduction on representations

2.1 Ideals

Definition 2.1.1. A subspace I of a Lie algebra L is called an **ideal** of L if $\forall x \in L, y \in I, [xy] \in I$. In fact, if $\forall x \in L, y \in I, [xy] \in I$, then I is closed under the bracket.

Example 2.1.1. The **center** $C(L) = \{x \in L \mid [xy] = 0, \forall y \in L\}$ ¹ is an ideal of L (Sometimes denoted $Z(L)$). $[LL]$, which is called **derived algebra** of L , is also an ideal of L , according to proposition 2.1.1. We usually denote the derived algebra of L by $L^{(1)}$, somewhere denoted L' . For example, $\mathfrak{sl}(n, F)$ is the derived algebra of $\mathfrak{gl}(n, F)$ and $\mathfrak{n}(n, F) = \mathfrak{t}(n, F)^{(1)}$. A Lie algebra L is abelian, i.e. $L=C(L)$, iff $[LL]=0$. (Take $\mathfrak{d}(n, F)$ as an example.)

Example 2.1.2. * Say D is a derivation, $a \in \mathfrak{g}$, then $[D, \text{ad}a] = \text{ad}D(a)$. Thus $\text{ad}(\mathfrak{g})$ is an ideal of $\text{Der}(\mathfrak{g})$.

Definition 2.1.2. A **Complete Lie algebra** is a Lie algebra $\mathfrak{g}$ with:

- (i). $Z(\mathfrak{g}) = 0$ and (ii). $\text{Der}(\mathfrak{g}) = \text{ad}\mathfrak{g}$.

Example 2.1.3. Let L be a Lie algebra, $\text{Der } L$ as in Def.1.3.1. Then we have: the set of all inner derivations $\text{adx}, x \in L$, is an ideal of $\text{Der } L$. More precisely, for any δ in $\text{Der } L$, we have: $[\delta, \text{adx}] = \text{ad}(\delta(x))$.

Definition 2.1.3. If L has no ideals except L and 0 , and $[LL] \neq 0$ ², then L is called a **simple** Lie algebra.

Example 2.1.4. If $\text{ch}F=2$, $\mathfrak{sl}(2, F)$ is not a simple Lie algebra.

If $\text{ch}F \neq 2$, $\mathfrak{sl}(2, F)$ is a simple Lie algebra.

Proposition 2.1.1. Let I, J be two ideals of L , then $I + J, I \cap J, [IJ]$ are ideals of L .

Definition 2.1.4. Quotient algebra: L is a Lie algebra, I is its ideal. Then L/I is a **quotient algebra**, where the multiplication is $[x+I, y+I] = [x, y] + I$.

Definition 2.1.5. The **normalizer** of a subalgebra (or just subspace) K of L is $N_L(K) = \{x \in L \mid [xK] \subseteq K\}$. By the Jacobi identity, $N_L(K)$ is a subalgebra of L , and it is the largest subalgebra of L which includes K as an ideal (in case K is a subalgebra to begin with). If $K = N_L(K)$, we call K **self-normalizing**.

Definition 2.1.6. The **centralizer** of a subset X of L is $C_L(X) = \{x \in L \mid [xX] = 0\}$. Again by the Jacobi identity, $C_L(X)$ is a subalgebra of L . Moreover, when X is an ideal of L , then $C_L(X)$ is also an ideal of L .

Example 2.1.5. $\mathfrak{t}(n, F)$ and $\mathfrak{d}(n, F)$ are self-normalizing subalgebras of $\mathfrak{gl}(n, F)$, whereas the normalizer of $\mathfrak{n}(n, F)$ is $\mathfrak{t}(n, F)$. $C_L(X) = C(L)$.

2.2 Homomorphisms and automorphisms

Definition 2.2.1. Homomorphism: A linear transformation $\phi : L \longrightarrow L'$ is a **homomorphism**, if $\phi([xy]) = [\phi(x)\phi(y)], \forall x, y \in L$. And we have definitions of monomorphisms, epimorphism, isomorphisms, canonical homomorphism the same as in abstract algebra.

¹Because of $(L2')$, $[xy]=0$ is equal to x is commutative with all y in L .

²which is imposed to exclude the one dimensional Lie algebra.

Proposition 2.2.1. (a) If $\phi : L \rightarrow L'$ is a homomorphism of Lie algebras, then $L/\ker \phi \cong \text{Im } \phi$.

(b) Denote $\mathfrak{n} \equiv \ker \phi$, which is an ideal of L . There is a bijective correspondence between the subalgebras (ideals) of L that contain $\mathfrak{n}$ and the subalgebras (ideals) of L' .

(c) Suppose that $\mathfrak{h}$ is an ideal of L , $\mathfrak{h} \supset \mathfrak{n}$, then $L/\mathfrak{h} \cong L'/\phi(\mathfrak{h})$.

(d) If I and J are ideals of L such that $I \subset J$, then J/I is an ideal of L/I and $(L/I)/(J/I)$ is naturally isomorphic to L/J .

(e) If I, J are ideals of L , there is a natural isomorphism between $(I + J)/J$ and $I/(I \cap J)$.

Example 2.2.1. Consider $\text{Tr}(x) : \mathfrak{gl}(n, F) \rightarrow F$, we will get $\mathfrak{gl}(n, F)/\mathfrak{sl}(n, F) \cong F$.

Example 2.2.2. Suppose that $\mathfrak{h}$ is an ideal of $\mathfrak{g}$. Denote derived algebra of $\mathfrak{g}$ as $\mathfrak{g}^{(1)}$. We can easily find that $\mathfrak{g}/\mathfrak{g}^{(1)}$ is abelian. Then using (d), we can get $\mathfrak{g}/\mathfrak{h}$ is abelian if and only if $\mathfrak{h}$ contains $\mathfrak{g}^{(1)}$. However $\mathfrak{g}/\mathfrak{g}^{(1)}$ is not always equal to $C(\mathfrak{g})$, take 2-dimensional nonabelian Lie algebra as a counterexample.

Definition 2.2.2. An automorphism is an isomorphism from L to itself. $\text{Aut } L$ forms a group.

Example 2.2.3. If δ is a nilpotent derivation of L , we have: $\exp(\delta) \in \text{Aut } L$. Using the Leibniz rule, we have: $\exp \delta(x) \cdot \exp \delta(y) = \exp \delta(xy)$ (Here we don't need the product of xy to be bracket, or, the concrete form of the product is not important.), which makes sure that $\exp \delta$ is a homomorphism. Furthermore, for δ is a nilpotent derivation, there are only finite terms, so $\exp \delta$ has an inverse.

Example 2.2.4. Let $\text{ch } F = 0$. Suppose x is an element in L and adx is nilpotent, say $(\text{adx})^n = 0$. Then we have $\exp(\text{adx}) = 1 + \text{adx} + \frac{(\text{adx})^2}{2!} + \dots + \frac{(\text{adx})^{n-1}}{(n-1)!}$. From last example, we know that $\exp(\text{adx}) \in \text{Aut } L$. Furthermore, an automorphism of the form $\exp(\text{adx})$, adx nilpotent, is called **inner automorphism**. The subgroup of $\text{Aut } L$ generated by these is denoted **Int L** . For $\phi \in \text{Aut } L$, we have $\phi(\text{adx})\phi^{-1} = \text{ad}\phi(x)$, whence we have $\phi(\exp(\text{adx}))\phi^{-1} = \exp(\text{ad}\phi(x))$. Therefore, The **Int L** is a normal subgroup of $\text{Aut } L$.

2.3 Representation

Definition 2.3.1. Let L be a Lie algebra over a field $\mathbb{F}$ and V a vector space over $\mathbb{F}$. A **representation** of L (on V) is a homomorphism $\phi : L \rightarrow \mathfrak{gl}(V)$. If the kernel is zero, the representation is said to be **faithful**.

Example 2.3.1. Trivial representation of L : $\phi(a) = 0$ for all $a \in L$.

Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$. The inclusion map $L \rightarrow \mathfrak{gl}(V)$ is trivially a Lie algebra homomorphism. The corresponding representation is known as the **natural representation** of L .

Example 2.3.2. * Adjoint representation of L on L is ad : $a \rightarrow \text{ad } a$. The kernel of ad is $C(L)$. Thus it defines an embedding of $L/C(L)$ in $\mathfrak{gl}(L)$, which proves Ado's Theorem in the case $C(L) = 0$.

Definition 2.3.2. Suppose that $\mathfrak{g}$ is a Lie algebra over a field $\mathbb{F}$. A **$\mathfrak{g}$ -module** is an $\mathbb{F}$ -vector space V together with a map

$$\mathfrak{g} \times V \rightarrow V \quad (a, v) \mapsto a \cdot v, \quad a \in \mathfrak{g}, v \in V$$

satisfying the conditions:

- (1) $(\lambda a + \mu b) \cdot v = \lambda(a \cdot v) + \mu(b \cdot v)$;
- (2) $a \cdot (\lambda v + \mu w) = \lambda(a \cdot v) + \mu(a \cdot w)$;
- (3) $[a, b] \cdot v = a \cdot (b \cdot v) - b \cdot (a \cdot v)$, for $a, b \in \mathfrak{g}$, $v, w \in V$, $\lambda, \mu \in \mathbb{F}$.

$\mathfrak{g}$ -modules and representations are two ways to describe the same structures. They are essentially the same thing.

Next, let's see some constructions of representations from given ones.

Definition 2.3.3. Representation from direct sum: Given representations ϕ_1, ϕ_2 of L on V_i . We have $\phi_1 \oplus \phi_2$ of L on $V_1 \oplus V_2 : (\phi_1 \oplus \phi_2)(a) = \phi_1(a) \oplus \phi_2(a), (\phi_1 \oplus \phi_2)(a)(v_1 + v_2) = \phi_1(a)(v_1) + \phi_2(a)(v_2)$.

Definition 2.3.4. Suppose that V is a $\mathfrak{g}$ -module. A **submodule of V** is a subspace U of V which is invariant under the action of $\mathfrak{g}$. Correspondingly, Given a representation of ϕ of $\mathfrak{g}$ on V , if $U \subset V$ is an invariant subspace under the action of $\mathfrak{g}$, we have the **subrepresentation** ϕ_U of $\mathfrak{g}$ on $U : a \longrightarrow \phi(a)|_U$. Furthermore, we have the **factor representation** $\phi_{V/U}$ of $\mathfrak{g}$ on $V/U : a \longrightarrow \phi(a)|_{V/U}$.

Definition 2.3.5. Let $\mathfrak{g}$ be a Lie algebra and let V and W be $\mathfrak{g}$ -modules. A **$\mathfrak{g}$ -module homomorphism** from V to W is a **linear map** $\theta : V \rightarrow W$ such that $\theta(x \cdot v) = x \cdot \theta(v)$ for all $v \in V, w \in W$, and $x \in \mathfrak{g}$. **Isomorphism.** Correspondingly, for $\phi_V : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ and $\phi_W : \mathfrak{g} \rightarrow \mathfrak{gl}(W)$, $\theta\phi_V = \phi_W\theta$.

3 Solvable and nilpotent Lie algebra

3.1 Solvable Lie algebra

Definition 3.1.1. Let L be a Lie algebra. **Derived series:** $L^{(0)} = L$, $L^{(1)} = [LL]$, $L^{(2)} = [L^{(1)}L^{(1)}]$, $\dots$. Call L **solvable** if $L^{(n)} = 0$ for some n . Using induction, we can get $(L^{(i)})^{(j)} = L^{(i+j)}$ quickly. We can find that if a Lie algebra is solvable, then its derived series is strictly monotone decreasing.

Example 3.1.1. *Abelian implies solvable. Simple Lie algebras are nonsolvable. $\mathfrak{t}(n, F)$ is solvable, so as $\mathfrak{n}(n, F)$.*

Example 3.1.2. *All 1-dimensional and 2-dimensional Lie algebras are all solvable.*

Proposition 3.1.1. * Let L be a Lie algebra.

(a) *If L is solvable, then so are all subalgebras and homomorphic images of L . Especially, if I is an ideal of L , then L/I is solvable for $L/I = \pi(L)$, where π is the canonical homomorphism.*

(b) *If I is a solvable ideal of L such that L/I is solvable, then L itself is solvable.*

(c) *If I, J are solvable ideals of L , then so is $I + J$.*

sketch of proof. (a). Let K be a subalgebra of L , then for all i , $K^{(i)} \subset L^{(i)}$. $\phi(L^{(i)}) = \phi([L^{(i-1)}, L^{(i-1)}]) = [\phi(L^{(i-1)}), \phi(L^{(i-1)})] = \phi(L^{(i-1)})^{(1)}$. Thus $\phi(L)^{(i)} = \phi(L^{(i)})$. (b). For L/I is solvable, $\pi(L)^{(n)} = 0$, i.e. $(L)^{(n)} \subset \ker \pi = I$. $(L)^{(m+n)} = (L^{(n)})^{(m)} \subset I^{(m)} = 0$. (c). $(I + J)/J \cong I/(I \cap J)$. $\square$

Corollary 3.1.1. *L is a Lie algebra, S is its maximal solvable ideal, if I is any other solvable ideal of L , the (c) forces $S+I=S$, i.e. $I \subset S$. Thus the maximal solvable ideal is unique, called the **radical** of L and denoted $\text{Rad}L$.*

Definition 3.1.2. If $\text{Rad}L = 0$, L is called **semisimple**. For example, simple Lie algebras are semisimple.

Corollary 3.1.2. *For any Lie algebra L , $L/\text{Rad}L$ is semisimple.*

This suggests that we might have a chance to understand all finite-dimensional Lie algebras. The radical $\text{Rad}L$ of any Lie algebra L is solvable, and $L/\text{Rad}L$ is semisimple, so to figure out L it is necessary to figure out solvable Lie algebra(i) and semisimple Lie algebra(ii).

Working over $\mathbb{C}$, an answer to (i) was found by Lie, who proved (in essence) that every solvable Lie algebra appears as a subalgebra of a Lie algebras of upper triangular matrices.

For (ii) we shall show that every semisimple Lie algebra is a direct sum of simple Lie algebras.

We then need to find all simple Lie algebras over $\mathbb{C}$. This is a major theorem; the proof is based on work by Killing, Engel, and Cartan.

3.2 Nilpotent Lie algebra

Definition 3.2.1. Let L be a Lie algebra. **lower central series/descending central series :** $L^0 = L$, $L^1 = [L, L^0]$, $L^2 = [L, L^1]$, $\dots$, $L^n = [L, L^{n-1}]$, $\dots$. (Somewhere the series begin with $L^1 = L$.) Clearly, $L^{(i)} \subset L^i$. Thus nilpotent algebra is always solvable, but solvable algebra is not always nilpotent. A Lie algebra L is called **nilpotent**, if $L^n = 0$ for some $n > 0$. We can find that if a Lie algebra is nilpotent, then its descending central series is strictly monotone decreasing.

Example 3.2.1. *$\mathfrak{t}(n, \mathbb{F})$ is a solvable, but not nilpotent Lie algebra. $\mathfrak{n}(n, \mathbb{F})$ is a nilpotent Lie algebra. 2-dimensional nonabelian Lie algebra is solvable but not nilpotent.*

Example 3.2.2. *Define Heis_{2n+1} to be the Lie algebra with basis p_i, q_i, c where $[p_i, q_i] = c = -[q_i, p_i]$, $1 \leq i \leq n$, and all other bracketed pairs are 0. Using the Engle's theorem, we can easily find that Heis_{2n+1} is nilpotent.*

Proposition 3.2.1. * Let L be a Lie algebra.

- (a) If L is nilpotent, then so are all subalgebras and homomorphic images of L .
- (b) If $L/Z(L)$ is nilpotent, then so is L .
- (c) If L is nilpotent and nonzero, then $Z(L) \neq 0$.
- (d) If I, J are nilpotent ideals of L , then so is $I + J$, and samely, L has a unique maximal nilpotent ideal.
- (e) I and L/I are nilpotent can't ensure L is nilpotent. A counterexample is $\mathfrak{t}(n, F)$. However, if $I \subset Z(L)$, L is nilpotent.

sketch of proof. The proofs of (a) is trivial. The only thing needs attention is if ϕ is a homomorphism, then we have $\phi(L^i) = \phi(L)^i$ similarly. For (b), we just need to notice that if $L^k \subseteq Z(L)$, then we have $L^{k+1} = [L, L^k] \subseteq [L, Z(L)] = 0$. For (c), say $L^k \neq 0$ and $L^{k+1} = 0$, then $L^k \subseteq Z(L)$. For (d), Th1: Using induction, we have $(I + J)^n \subset I^n + J^n + I \cap J$, then for some n , $(I + J)^n \subset I \cap J$. Then:

$$(I + J)^{n+1} = [I + J, (I + J)^n] \subset [I + J, I \cap J] = [I, I \cap J] + [J, I \cap J] \subset I^1 \cap J + I \cap J^1$$

$$(I + J)^{n+2} = [I + J, (I + J)^{n+1}] \subset [I + J, I^1 \cap J + I \cap J^1] \subset I^2 \cap J + I^1 \cap J^1 + I \cap J^2$$

By induction, we have $(I + J)^{n+m} \subset I^m \cap J + I^1 \cap J^1 + I \cap J^m$, then for some m , we have $(I + J)^{n+m} \subset I^1 \cap J^1$. Then using induction, $(I + J)^{n+k} \subset I^n \cap J^n = 0$.

For (e), we need to notice that $\mathfrak{n}(n, F)$ is a nilpotent ideal of $\mathfrak{t}(n, F)$ and the derived algebra of $\mathfrak{t}(n, F)$. $\square$

Definition 3.2.2. We call x **ad-nilpotent** if adx is a nilpotent endomorphism.

Lemma 3.2.1. Let $x \in \mathfrak{gl}(V)$ be an nilpotent endomorphism. Then adx is also nilpotent.

sketch of proof. Through computation we know that: $\text{adx} = \lambda_x - \rho_x$, where $\lambda_x(y) = xy$, $\rho_x(y) = yx$. When x is nilpotent, they are nilpotent and commute. $\square$

Theorem 3.2.1. (Which is also called Engel's theorem.) Let L be a subalgebra of $\mathfrak{gl}(V)$, V finitely dimensional. If L consists of nilpotent endomorphisms and $V \neq 0$, then there exists nonzero $v \in V$, $L.v = 0$.

There are three steps to prove this. The keypoints will be marked in bold.

proof. Using induction on $\dim L$.¹ When $\dim L = 0, 1$, the case are simple. (When $\dim L = 1$, $L = Fx$, where x is nilpotent. It always has an eigenvector belonging to 0.)

Let K be a proper subalgebra of L . K acts (via ad) on L (who is a vector space) as a Lie algebra of nilpotent linear transformations, so K can act on L/K samely. For $\dim K < \dim L$, we can use the induction hypothesis, which guarantees the existence of a vector $x + K \neq K$ killed by the action of K , i.e. $\text{ad}_K(x) = [K, x] \in K$, i.e. $x \in N_L(K)$. Thus we can get **K is a proper subalgebra of $N_L(K)$** .

Let K be the maximal subalgebra in L . Then $N_L(K) = L$, thus K is an ideal of L . If $\dim L/K$ is bigger than 1, then take a $\dim 1$ subalgebra of L/K , using the homomorphism theorem, we know that there exists a proper subalgebra in L which contains K properly. Thus we get a contradiction. So **K has codimension one**. This allows us write $L = K + Fz$.

Let $W = \ker K$. For K is an ideal of L , **W is a L -subspace** (which is stable under L): Let $v \in W, x \in K, y \in L$, then $0 = [x, y]v = xyv - yxv = xyv$. z is a nilpotent linear transformation, then it has an eigenvector in W , i.e. $\exists v \in W, z.v = 0$. This v is desired. $\square$

Corollary 3.2.1. Let L be nilpotent, K is an ideal of L . If $K \neq 0$, then $K \cap C(L) \neq 0$.

Definition 3.2.3. A **flag** in V is a chain of subspaces: $0 = V_0 \subset V_1 \subset \dots \subset V_i \subset \dots \subset V_n$, $\dim V_i = i$. If $x \in \text{End } V$, we say x **stabilizes the flag** provided $x.V_i \subset V_i$.

Corollary 3.2.2. Under the hypothesis of this theorem, there exists an flag (V_i) in V stable under L , i.e. there exist a basis of V relative to which the matrices in L are all in $\mathfrak{n}(n, F)$.

sketch of proof. Using induction. Set $V_1 = Fv$, where $L.v = 0$. Consider the induced action of L on $V/V_1 \equiv W$. Then W has a flag stabilized by L . $\square$

¹In the inductive assumption, we only request on $\dim L$, whereas $\dim V$ and how the product acts can be optional, as long as the element of L can act on $v \in V$.

3.3 Engel's theorem

Theorem 3.3.1. (Engel's theorem): L is nilpotent iff all elements of L are ad-nilpotent.

sketch of proof. Using induction on $\dim L$. The algebra $\text{ad}L \subset \mathfrak{gl}(L)$. Then $C(L) \neq 0$. Then consider $L/C(L)$, which is nilpotent, using the induction hypothesis. Using Proposition 3.2.1(b), we finish the proof.

□

3.4 Some problems

In this subsection, the use of derived algebra and inner derivations is highly stressed.

1、 Show that all derivations of 2-dimensional nonabelian Lie algebra are inner. Thus this Lie algebra is a complete Lie algebra.

sketch of proof. We just need to consider the image of the derivation under the chosen basis. □

2、 Let $\mathfrak{g}$ be a Lie algebra with 2-dimensional nonabelian Lie algebra $\mathfrak{g}_2$ as its ideal, then exists an ideal M of $\mathfrak{g}$ s.t. $\mathfrak{g} = \mathfrak{g}_2 \oplus M$. Moreover, $M = C_{\mathfrak{g}}(\mathfrak{g}_2)$.¹

proof. We have known that M is an ideal of $\mathfrak{g}$. If $x \in \mathfrak{g}_2 \cap M$, extend it to a basis of $\mathfrak{g}_2$ x, y . Then we have $[x, y] = 0$, which leads to a contradiction. Thus the sum of $\mathfrak{g}_2$ and M is a direct sum.

Let y, z be a basis of $\mathfrak{g}_2$, x is an element in $\mathfrak{g}$.

Method 1: For $\mathfrak{g}_2$ is an ideal of $\mathfrak{g}$, $[x, y], [x, z]$ is in $\mathfrak{g}_2$, say $[x, y] = a_1y + b_1z$, $[x, z] = a_2y + b_2z$. Then using the method of undetermined coefficients, we can find an element x_0 in $\mathfrak{g}_2$, s.t. $x - x_0 \in M$, then we have $\mathfrak{g} = \mathfrak{g}_2 \oplus M$.

Method 2: We can use the conclusion in 1. For an element in $\mathfrak{g}$, $\text{adx}|_{\mathfrak{g}_2}$ is a derivation of $\mathfrak{g}_2$, thus is an inner derivation of $\mathfrak{g}_2$, i.e., there is an element x_0 in $\mathfrak{g}_2$, s.t. $\text{adx}|_{\mathfrak{g}_2} = \text{adx}_0|_{\mathfrak{g}_2}$. Consider $x - x_0$, we will find that it falls in $M = C_{\mathfrak{g}}(\mathfrak{g}_2)$, which is also an ideal of $\mathfrak{g}$. Thus we have $\mathfrak{g} = \mathfrak{g}_2 \oplus M$. □

3、 Determine the classification of three-dimensional Lie algebra $\mathfrak{g}$ with $\dim \mathfrak{g}' = 1$.

proof. Let's consider two cases: $\mathfrak{g}' \subseteq C(\mathfrak{g})$ and $\mathfrak{g}' \not\subseteq C(\mathfrak{g})$.²

I、 If $\mathfrak{g}' \subseteq C(\mathfrak{g})$ then as $\mathfrak{g}' = \mathbb{F}z$, for some $z \in \mathfrak{g}$, and we can extend to a basis x, y, z of $\mathfrak{g}$, with the multiplication in $\mathfrak{g}$ is defined by:

$$[x, y] = z, [x, z] = 0, [y, z] = 0$$

The Jacobi identity can be easily verified and thus $\mathfrak{g}$ is a well defined Lie algebra. Moreover, it is exactly $Heis_3$, and we observe that the Lie algebra of strictly upper triangular 3×3 matrices over $\mathbb{F}$ has this form if one takes the basis $x = e_{12}, y = e_{23}, z = e_{13}$.

It has properties: (i)、 Non-abelian; (ii)、 Solvable as $\mathfrak{g}^{(2)} = [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}] = [\mathbb{F}z, \mathbb{F}z] = 0$ and thus $\text{Rad} \mathfrak{g} = \mathfrak{g}$; (iii)、 Nilpotent since $\mathfrak{g}^2 = [\mathfrak{g}', \mathfrak{g}] = [C(\mathfrak{g}), \mathfrak{g}] = 0$; (iv)、 $C(\mathfrak{g}) = \mathfrak{g}'$.

II、 If $\mathfrak{g}' \not\subseteq C(\mathfrak{g})$ then $\mathfrak{g}' = \mathbb{F}x$ and there exists a $y \in \mathfrak{g}$ such that $[x, y] \neq 0$. Moreover as $[x, y] \in \mathfrak{g}'$ we have $[x, y] = \alpha x$ for some $\alpha \in \mathbb{F}^*$. Thus by choosing a proper y , we can get $[x, y] = x$.

Let $\mathfrak{g}_2 \equiv \mathbb{F}x + \mathbb{F}y$. $[\mathfrak{g}, \mathfrak{g}_2] \subset \mathfrak{g}' \subset \mathfrak{g}_2$, thus $\mathfrak{g}_2$ is an ideal of $\mathfrak{g}$. Then using the conclusion in 2, we have $\mathfrak{g} = \mathfrak{g}_2 \oplus C_{\mathfrak{g}}(\mathfrak{g}_2)$. Then choose any $z \in C_{\mathfrak{g}}(\mathfrak{g}_2)$, gives rise to a basis x, y, z of $\mathfrak{g}$ such that:

$$[x, y] = x, [x, z] = 0, [y, z] = 0. \quad (3.1)$$

It has properties: (i)、 Non-abelian; (ii)、 Solvable as $\mathfrak{g}^{(2)} = [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}] = [\mathbb{F}x, \mathbb{F}x] = 0$ and thus $\text{rad}(\mathfrak{g}) = \mathfrak{g}$; (iii)、 Not nilpotent since by induction one can show $\mathfrak{g}^n = \mathbb{F}x \neq 0, \forall n \in \mathbb{N}$; (iv)、 $C(\mathfrak{g}) = \mathbb{F}z$ is one-dimensional; (v)、 $\mathfrak{g}' = \mathbb{F}x$. □

¹In fact we just need to verify that $\mathfrak{g} - \mathfrak{g}_2$ falls in M .

²In fact, for $\dim \mathfrak{g}' = 1$, the relation between $\mathfrak{g}'$ and $C(\mathfrak{g})$. So this way is convenient.

Remark: These two kinds are good examples. In the second case, the dimension of the ideal $\mathfrak{g}_2$ is bigger than $\mathfrak{g}'$, which indicates $\mathfrak{g}'$ is not the biggest ideal. Using the method of undetermined coefficients and the method of variable substitution, we can also get (3.1).

4、From [W.A. de Graaf, Classification of solvable Lie algebras, Exp. Math. 14\(1\) \(2005\), 15-25.](#)

The strategy that we use for constructing solvable Lie algebras of dimension n is to extend a solvable Lie algebra of dimension $n - 1$ by a derivation. More precisely, let K be a solvable Lie algebra of dimension $n - 1$, and let $D : K \rightarrow K$ be a derivation. Then we construct a solvable Lie algebra of dimension n by setting $\mathfrak{g} = \mathbb{F}x_D \oplus K$. The Lie bracket on $\mathfrak{g}$ is defined by:

$$[\alpha D + y_1, \beta D + y_2] = \alpha D(y_2) - \beta D(y_1) + [y_1, y_2]$$

where $\alpha, \beta \in \mathbb{F}$ and $y_1, y_2 \in K$. We recall that the derivations of a Lie algebra $\mathfrak{g}$ that are of the form ad_x for $x \in \mathfrak{g}$ are called *inner*. Derivations not of that form are called *outer*. Prove that:

Lemma Let $\mathfrak{g}$ be a solvable Lie algebra. Then there is a subalgebra $K \subset \mathfrak{g}$ of codimension 1, and a derivation D of K such that $\mathfrak{g} = \mathbb{F}x_D \oplus K$. Moreover, if $\mathfrak{g}$ is not Abelian, then D and K can be chosen such that D is an outer derivation of K .

proof. We know that the derived algebra of $\mathfrak{g}$ is properly contained in $\mathfrak{g}$. Thus let L be a $(n-1)$ -dimensional subspace of $\mathfrak{g}$ which contains $\mathfrak{g}'$, we will find L is a subalgebra of $\mathfrak{g}$, for $[L, L] \subset \mathfrak{g}' \subset L$, and moreover an ideal of $\mathfrak{g}$. Let $x \in \mathfrak{g}$ span a complementary subspace of L . $D = \text{ad}_x$ and $x = x_D$ is what we want.

The proof of the second statement needs induction on $\dim \mathfrak{g}$. If $\dim \mathfrak{g} = 2$, the situation is clear. Suppose the statement is true for $\dim \mathfrak{g} = k$. Let's see when $\dim \mathfrak{g} = k+1$. We have known that $\mathfrak{g} = \mathbb{F}x_1 \oplus K_1$, where $\dim K_1 = k$ and K_1 is solvable. Suppose x_1 is an inner derivation of K_1 , then there exists $y \in K_1$, s.t. $\text{ad}_{x_1}|_{K_1} = \text{ad}_y|_{K_1}$. Let $z = x_1 - y$, $\mathfrak{g} = \mathbb{F}z \oplus K_1$ and $[z, K_1] = 0$. If K_1 is abelian, then $\mathfrak{g}$ is abelian, thus K_1 is nonabelian. Thus $K_1 = \mathbb{F}x_2 \oplus K_2$, where ad_{x_2} is an outer derivation of K_2 according to induction hypothesis. Let $K = \mathbb{F}z \oplus K_2$, for $\text{ad}_{x_2}(z) = 0$, if there exists $x'_2 \in K$, s.t. $\text{ad}_K x_2 = \text{ad}_K x'_2$, we can assume $x'_2 \in K_2$ with $\text{ad}_K x'_2$ invariant for $z \in C(\mathfrak{g})$. Then for an element in K , say $az + k$, we have $\text{ad}_{x_2}(k) = \text{ad}_{x_2}(az + k) = \text{ad}_{x'_2}(az + k) = \text{ad}_{x'_2}(k)$ ¹. Then $\text{ad}_{K_2} x_2 = \text{ad}_{K_2} x'_2$, ad_{x_2} is an inner derivation of K_2 , which is a contradiction. Thus ad_{x_2} is an outer derivation of K . $\square$

Remark: The way to find a $(n-1)$ -dimensional subalgebra is worth memorizing. In fact, the subalgebra containing $\text{ad } \mathfrak{g}$ is an ideal of $\mathfrak{g}$.

5、Give the classification of all 3-dimensional nilpotent Lie algebras.

proof. Consider the dimension of $[L, L]$. We know that **the center of the Lie algebra is not 0**.

If $[L, L] = 0$, then L is abelian, which is without doubt a nilpotent Lie algebra.

If $\dim[L, L] = 3$, it's apparent that it couldn't be a nilpotent Lie algebra.

If $\dim[L, L] = 2$, we know that 2-dimensional Lie algebra has two cases: abelian and nonabelian. If $[L, L]$ is nonabelian, we can find an element x in L' , ad_x is not nilpotent, which lead to a contradiction. Then let's consider L' is abelian. We know $L' \cap C(L) \neq 0$. If $\dim(L' \cap C(L)) = 2$, i.e., $L' \subset C(L)$, let x, y be a basis of L' , extend then to a basis of L x, y, z . Then we will find that $[x, y] = [y, z] = [x, z] = 0$, i.e., $L' = 0$, which leads to a contradiction. If $\dim(L' \cap C(L)) = 1$, let $L' \cap C(L) \equiv \mathbb{F}x$. Extend x to a basis of L' x, y , then a basis of L x, y, z . Then we have $[x, y] = [x, z] = 0$. So the basis of L' is $[y, z]$, in which case $\dim L' = 1$, which leads to a contradiction. So $\dim[L, L]$ can't be 2.

If $\dim[L, L] = 1$, then $L' \subset C(L)$, let $L' = \mathbb{F}x$. Then extend it to a basis of L x, y, z . In this case, it is exactly Heis_3 , which is a nilpotent Lie algebra.

So there are only 2 kinds 3-dimensional nilpotent Lie algebra. $\square$

6、Let $\mathfrak{g}$ be a complex Lie algebra. Show that $\mathfrak{g}$ is nilpotent iff every 2-dimensional subalgebra of $\mathfrak{g}$ is abelian.

¹Here we use $z \in C(\mathfrak{g})$, thus $[x'_2, z] = 0$.

proof. If $\mathfrak{g}$ is nilpotent and L is a 2-dimensional subalgebra of $\mathfrak{g}$, we know that L has only 2 kinds, abelian and nonabelian. If L is nonabelian, say x, y is a basis of L and $[x, y] = x$, then $\text{ad } y$ is not nilpotent, which leads to a contradiction. Thus every 2-dimensional subalgebra of $\mathfrak{g}$ is abelian.

If every 2-dimensional subalgebra of $\mathfrak{g}$ is abelian, let's prove adx is nilpotent for all x in $\mathfrak{g}$. Find a basis $x_1, x_2, \dots, x_n$ of $\mathfrak{g}$, and the action of adx can be represented like:

$$\text{adx}(x_1, x_2, \dots, x_n) = (x_1, x_2, \dots, x_n)A$$

where A is a matrix over the field $\mathbb{C}$. According to the theory of Jordan canonical forms, we have $A = PA_0P^{-1}$, where A_0 is the Jordan canonical form of A . Thus we get:

$$\text{adx}(x_1, x_2, \dots, x_n)P = (x_1, x_2, \dots, x_n)PA_0$$

For P is invertible, $(x_1, x_2, \dots, x_n)P$ is then a basis of $\mathfrak{g}$. Let's denote as $e_1, e_2, \dots, e_n$. Without loss of generality, we can assume A_0 is $aI + J$, $a \in \mathbb{F}$, where I is the identity and J is the basic cyclic matrix. Then we get $[x, e_1] = ae_1$ and x, e_1 forms an 2-dimensional subalgebra of $\mathfrak{g}$, thus $a = 0$. Then adx is nilpotent for all $x \in \mathfrak{g}$. Then using Engel's theorem, we have $\mathfrak{g}$ is nilpotent. $\square$

4 Lie theorem

4.1 Lie theorem

First, to make things simpler, let's introduce the concept of weight and weight space.

Definition 4.1.1. Let $\mathfrak{h}$ be a Lie algebra, $\phi : \mathfrak{h} \rightarrow \mathfrak{gl}(V)$ a representation of $\mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$, the dual space of linear maps from $\mathfrak{h}$ to $\mathbb{F}$. We define **the weight space of $\mathfrak{h}$ attached to λ** as

$$V_\lambda^\mathfrak{h} = \{v \in V \mid \phi(h)v = \lambda(h)v, \forall h \in \mathfrak{h}\}.$$

If $V_\lambda^\mathfrak{h} \neq 0$, we will say that λ is a **weight for ϕ** .

The next lemma is similar with Engel's theorem in nature, but requires algebraic closure, to assure $\mathbb{F}$ will contain all required eigenvalues. And we will find $\text{ch}\mathbb{F} = 0$ is necessary.

Lemma 4.1.1. (Lie's Lemma). Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{h} \subset \mathfrak{g}$ an ideal both over an algebraically closed field $\mathbb{F}$ of characteristic 0. Let ϕ be a representation of $\mathfrak{g}$ on a finite dimensional $\mathbb{F}$ -vector space V . Then each weight space $V_\lambda^\mathfrak{h}$ for the restricted representation $\phi|_\mathfrak{h}$ is invariant under $\mathfrak{g}$.

Remark: The keypoint is to show $\lambda([h, a]) = 0, \forall h \in \mathfrak{h}, \forall a \in \mathfrak{g}$. We will prove this by finding a suitable basis such that the corresponding matrix is an upper triangular matrix, thus we can finish our proof by analyzing the trace of that matrix.

proof. We need to proof: If $v \in V_\lambda^\mathfrak{h}$, then $\phi(a)v \in V_\lambda^\mathfrak{h}, \forall a \in \mathfrak{g}$, i.e.:

$$\phi(h)\phi(a)v = \lambda(h)\phi(a)v, \forall h \in \mathfrak{h}, \forall a \in \mathfrak{g} \quad (4.1)$$

It's obvious that:

$$\phi(h)\phi(a)v = [\phi(h), \phi(a)]v + \phi(a)\phi(h)v = \phi([h, a])v + \phi(a)\lambda(h)v. \quad (4.2)$$

Since $\mathfrak{h}$ is an ideal, $[h, a] \in \mathfrak{h}$, then (4.2) becomes:

$$\phi(h)\phi(a)v = \lambda([h, a])v + \phi(a)\lambda(h)v. \quad (4.3)$$

Comparing (4.1) with (4.3), we find that it's sufficient to show that $\lambda([h, a]) = 0, \forall h \in \mathfrak{h}, \forall a \in \mathfrak{g}$.

Then, we fix $a \in \mathfrak{g}$ and let v be a nonzero element in $V_\lambda^\mathfrak{h} \neq 0$. We define:

$$W_m = \text{span}\{v, \phi(a)v, \phi(a)^2v, \dots, \phi(a)^mv\}, \forall m \geq 0, W_{-1} = 0.$$

Since V is finite dimensional, then W_m is finite dimensional for all m . We can prove that if $\phi(a)^k v$ can be linearly expressed, $\phi(a)^m v$ can also be linearly expressed for all m bigger than k . Thus, there exists $N \in \mathbb{N}$ s.t. N is the maximal integer for which all the generators of W_N are linearly independent. Moreover we have $W_N = W_{N+1} = \dots$ and $\phi(a)W_N \subset W_N$.

Now we claim that $\forall m \geq 0, W_m$ is invariant under $\phi(\mathfrak{h})$ and furthermore

$$\forall h \in \mathfrak{h}, \quad \phi(h)\phi(a)^m v - \lambda(h)\phi(a)^m v \in W_{m-1}. \quad (4.4)$$

Actually, the statement that W_m is invariant under $\phi(\mathfrak{h})$ is contained in (4.4), just using a transposition. So we can only prove (4.4). We use induction on m . Suppose that we have proved the assumption for $m - 1$: we want to prove it for m .

$$\begin{aligned} & \phi(h)\phi(a)^m v - \lambda(h)\phi(a)^m v \\ &= [\phi(h), \phi(a)]\phi(a)^{m-1}v + \phi(a)\phi(h)\phi(a)^{m-1}v - \lambda(h)\phi(a)^m v \\ &= \phi([h, a])\phi(a)^{m-1}v + \phi(a)\phi(h)\phi(a)^{m-1}v - \phi(a)\lambda(h)\phi(a)^{m-1}v. \end{aligned}$$

By induction hypothesis, we have that $w = \phi(h)\phi(a)^{m-1}v - \lambda(h)\phi(a)^{m-1}v \in W_{m-2}$ and thus $\phi(a)w \in W_{m-1}$. We know that $\phi([h, a]) \in \phi(\mathfrak{h})$ and by inductive hypothesis, we have $[\phi(h), \phi(a)]\phi(a)^{m-1}v \in W_{m-1}$, thus this concludes the proof of the inductive step.

Then by knowledge in linear algebra, choosing the basis $\{v, \phi(a)v, \dots, \phi(a)^N v\}$, we have:

$$\phi(h)(v, \phi(a)v, \dots, \phi(a)^N v) = (v, \phi(a)v, \dots, \phi(a)^N v) \begin{pmatrix} \lambda(h) & * & \dots & * \\ 0 & \lambda(h) & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda(h) \end{pmatrix}$$

Substitute $\phi(h)$ with $\phi([h, a])$, then we have:

$$\text{tr}_{W_N}(\phi([h, a])) = 0 = \text{tr}_{W_N}(\phi[h, a]) = (N+1)\lambda([h, a])$$

which implies that $\lambda([h, a]) = 0$, since $\text{char}\mathbb{F} = 0$. This concludes the proof. $\square$

Then we can get into Lie Theorem:

Theorem 4.1.1. Let $\mathfrak{g}$ be a finite dimensional solvable Lie algebra and ϕ a representation of $\mathfrak{g}$ on a finite dimensional vector space $V \neq 0$, over an algebraically closed field $\mathbb{F}$ of characteristic 0. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for ϕ , that is $V_\lambda^\mathfrak{g} \neq 0$. (i.e., V contains a common eigenvector for all the endomorphisms in L , which is the expression in *gtm9*.)

proof. Using induction on the dimension of $\mathfrak{g}$. The case $\dim \mathfrak{g} = 0$ is trivial. Suppose now that we have proved Lie's theorem $\dim \mathfrak{g} = m - 1$, we want to show that the theorem holds also for $\mathfrak{g}$, $\dim \mathfrak{g} = m \geq 1$.

We can find an $(n-1)$ -dimensional ideal of $\mathfrak{g}$, say $\mathfrak{h}$, which is still a solvable Lie algebra. Thus $V_\lambda^\mathfrak{h} \neq 0$. Find a nonzero element in $\mathfrak{g}/\mathfrak{h}$ and find its preimage a in $\mathfrak{g}$. Then the sum of $\mathfrak{h}$ and $\mathbb{F}a$ is a direct sum. Considering the dimension, we have $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{F}a$. According to the Lie lemma, we know $V_\lambda^\mathfrak{h}$ is invariant under $\phi(\mathfrak{g})$, especially under $\phi(a)$, i.e. $\phi(a)V_\lambda^\mathfrak{h} \subset V_\lambda^\mathfrak{h}$. Thus there exists $0 \neq v \in V_\lambda^\mathfrak{h}$ such that $\phi(a)v = lv$, for some $l \in \mathbb{F}$. We define a linear functional $\lambda' \in \mathfrak{g}^*$ on $\mathfrak{g}$ by

$$\lambda'(h + \mu a) = \lambda(h) + \mu l, \forall h \in \mathfrak{h}, l \in \mathbb{F}.$$

Thus, by construction, we see that v belongs to $V_{\lambda'}^\mathfrak{g}$. In particular $V_{\lambda'}^\mathfrak{g} \neq \{0\}$. $\square$

Then we can figure out the structure of solvable Lie algebras., which is also called Lie theorem.

Theorem 4.1.2. For all representations ϕ of a solvable Lie algebra $\mathfrak{g}$ on a finite dimensional vector space V over an algebraically closed field $\mathbb{F}$, $\text{char}\mathbb{F} = 0$, there exists a basis of V in which the matrices of $\phi(\mathfrak{g})$ are upper triangular. (i.e., $\mathfrak{g}$ stabilizes a flag in V .)

proof. Using induction on $\dim V$. when $\dim V = 1$, the case is obvious. Suppose now that we have proved this for $\dim V = m - 1$, we want to show that the theorem holds also for $\mathfrak{g}$, $\dim V = m > 1$.

For $\phi(\mathfrak{g})$ is solvable, there exists a in V , s.t. a is a common eigenvector for $\phi(\mathfrak{g})$. Consider the quotient sapce $V/\mathbb{F}a$, whose dimension is $n-1$, thus by induction can give a basis such that $\phi(\mathfrak{g})$ are upper triangular matrices. Taking any preimages of that basis in V , and extending it to a basis of V by including a , we obtain a basis of V on which $\phi(\mathfrak{g})$ are upper triangular matrices. $\square$

Corollary 4.1.1. Let L be a solvable Lie algebra. Then there exists a chain of ideals of L , $0 = L_1 = \dots = L_n = L$, s.t. $\dim L_i = i$.

sketch of proof. Consider the adjoint representation of L . Using theorem 4.1.2. $\square$

Corollary 4.1.2. L is solvable $\Leftrightarrow L'$ is nilpotent.

proof. The $\Leftarrow$ is obvious, for the $\Rightarrow$, for L is solvable, then there exists $x_1, \dots, x_n$, s.t. $\phi(L) \subset \mathfrak{t}(n, F)$, where ϕ is the adjoint representation. Then $\phi(L') = [\phi(L), \phi(L)] \subset [\mathfrak{t}(n, F), \mathfrak{t}(n, F)] = \mathfrak{n}(n, F)$. Thus $\phi(L')$ is nilpotent, according to proposition 2.2.1(a), so as $L'/C(L')$. According to proposition 3.2.1(c), L' is nilpotent. $\square$

4.2 Some problems

1、Set $\mathfrak{h} = \mathfrak{t}(n, F)$. Show that e_1 is an eigenvector for $\mathfrak{h}$. Find the corresponding weight and determine its weight space.

proof. Let x be a matrix in $\mathfrak{h}$, then $x.e_1$ means the first column of x , which is an element in $\mathbb{F}e_1$. Thus e_1 is an eigenvector of $\mathfrak{h}$. $\lambda(x) = a_{11}$ is the weight and the weight space is $\mathbb{F}e_1$. $\square$

2、Let p be prime and let $\mathbb{F}$ be a field of characteristic p , consider the following $p \times p$ matrices:

$$x = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & p-2 & 0 \\ 0 & 0 & \cdots & 0 & p-1 \end{pmatrix}$$

Check that $[x, y] = x$. Deduce that x and y span a 2-dimensional solvable subalgebra $\mathfrak{g}$ of $\mathfrak{gl}(p, \mathbb{F})$. Show that x and y have no common eigenvector, and so the conclusions of Lie's Theorem fail.

proof. Do some computation:

$$\begin{aligned} [x, y] &= xy - yx \\ &= \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & p-2 & 0 \\ 0 & 0 & \cdots & 0 & p-1 \end{pmatrix} - \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & p-2 & 0 \\ 0 & 0 & \cdots & 0 & p-1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & p-1 \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & p-2 \\ p-1 & 0 & \cdots & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix} = x \end{aligned}$$

If $v = (x_1, x_2, \dots, x_p)$ is the eigenvector of x , then we have $xv = av$, a is an element in $\mathbb{F}$. Then we have $(x_2, x_3, \dots, x_n, x_1) = (ax_1, \dots, ax_n)$. Thus we have $x_k = a^{k-1}x_1, k = 1, \dots, p$, i.e., $v = x_1(1, a, \dots, a^{p-1}), x_1 \in \mathbb{F}$. So the eigen-subspace of x is 1-dimensional. Then let's prove $u \equiv (1, a, \dots, a^{p-1})$ is not an eigenvector of y . If u is an eigenvector of y , then there exists an element b in $\mathbb{F}$, s.t.: $yu = (0, a, \dots, (p-1)a^{p-1}) = bu = (b, ba, \dots, ba^{p-1})$. Then b doesn't exist. Thus conclude the proof. $\square$

2、If there is a basis s.t. under which all elements in L are in $\mathfrak{t}(n, F)$, then L is solvable.

proof. If all elements in L are in $\mathfrak{t}(n, F)$, i.e. $L \subseteq \mathfrak{t}(n, F)$. Then $adL \subseteq [\mathfrak{t}(n, F), \mathfrak{t}(n, F)] = \mathfrak{n}(n, F)$, which is nilpotent. According to corollary 4.1.2., we finish the proof. $\square$

5 Generalized Eigenspaces & Generalized Weight Spaces

5.1 Jordan-Chevalley decomposition

Definition 5.1.1. Generalized Eigenspaces: Let A be a linear operator on a vector space V over field $\mathbb{F}$ and let $\lambda \in \mathbb{F}$, then the subspace

$$V_\lambda = \{v \mid (A - \lambda I)^N v = 0 \text{ for some positive integer } N\}$$

is called a generalized eigenspace of A with eigenvalue λ . (which is the same as root subspace in linear algebra.) And the vectors in generalized eigenspace are called generalized eigenvectors.

Example 5.1.1. A is a nilpotent operator if and only if $V = V_0$.

Lemma 5.1.1. Decomposing lemma¹:

- (a). A, B are two linear operations on V and $AB = BA$, then $\ker B$ and BV are A invariant subspaces.
- (b). f and g are coprime polynomials, then $\ker f(A)g(A) = \ker f(A) \oplus \ker g(A)$.

Proposition 5.1.1. Let A be a linear operator on a finite dimensional vector space V over an algebraically closed field $\mathbb{F}$, and let $\lambda_1, \dots, \lambda_s$ be all eigenvalues of A , $n_1, n_2, \dots, n_s$ be their multiplicities. Then one has the generalized eigenspace decomposition: (which is the same as root subspace decomposition in linear algebra.)

$$V = \bigoplus_{i=1}^s V_{\lambda_i} \text{ where } \dim V_{\lambda_i} = n_i.$$

sketch of proof. One can prove this by considering the Jordan canonical forms or using the decomposing lemma (b). □

Definition 5.1.2. An element $x \in \text{End}(V)$ is called **semisimple** if the roots of its minimal polynomial over $\mathbb{F}$ (algebraically closed) are distinct. According to the knowledge in linear algebra we know that:

- (a). x is semisimple iff x is diagonalizable.
- (b). two commuting semisimple endomorphisms can be simultaneously diagonalized.
- (c). If x semisimple and W is x -subspace, then $x|_W$ is also semisimple.

Example 5.1.2. We have known that if x is nilpotent, then is $\text{ad } x$. Moreover, we have if x is semisimple, then is $\text{ad } x$. Choose a basis $(v_1, \dots, v_n)$ of V , s.t. under which $x = \text{diag}(a_1, \dots, a_n)$. A basis $\{e_{ij}\}$ of $\mathfrak{gl}(V)$ relative to $(v_1, \dots, v_n)$ is $e_{ij}(v_k) = \delta_{jk}v_i$. Thus $\text{ad } x(e_{ij}) = (a_i - a_j)e_{ij}$. Thus $\text{ad } x$ is diagonal under $\{e_{ij}\}$. (One can consider this example under matrix forms, which may be more visually.)

Theorem 5.1.1. (Chinese Remainder Theorem) Let $A_1, \dots, A_n$ be ideals in a ring R such that $R^2 + A_i = R$ for all i and $A_i + A_j = R$ for all $i \neq j$ (which means they are "coprime"). If $b_1, \dots, b_n \in R$, then there exists $b \in R$ such that

$$b \equiv b_i \pmod{A_i} \quad (i = 1, 2, \dots, n).$$

Furthermore, b is uniquely determined up to congruence modulo the ideal

$$A_1 \cap A_2 \cap \dots \cap A_n.$$

REMARK. If R has an identity, then $R^2 = R$, whence $R^2 + A = R$ for every ideal A of R . The proof can be found in gtm 73, page 131.²

¹which is actually a lemma in linear algebra, but is very useful

²Actually, the content in wiki pedia is also highly recommended for the comprehensiveness.

Proposition 5.1.2. * Let V be a finite dimensional vector space over $\mathbb{F}$, $x \in \text{End } V$.

(a). There exist unique $x_s, x_n \in \text{End } V$ satisfying the conditions: $x = x_s + x_n$, x_s is semisimple, x_n is nilpotent, and x_s and x_n commute.

(b). There exist polynomials $p(t), q(t)$ in one indeterminate, without constant term, such that $x_s = p(x)$, $x_n = q(x)$. In particular, x_s and x_n commute with any endomorphism commuting with x .

(c). If $A \subseteq B \subseteq V$ are subspaces, and x maps B into A , then x_s and x_n also map B into A .

This decomposition is called **Jordan-Chevalley decomposition**, also called (classical) Jordan decomposition. We should notice that here is nothing to do with Lie algebra.

proof. Consider the generalized eigenspace decomposition $V = \bigoplus_{i=1}^s V_{\lambda_i}$ $\dim V_{\lambda_i} = n_i$ and a_i is the corresponding eigenvalues. Then $(t - a_i)^{n_i}$ is the characteristic polynomial of $x|_{V_i}$. Then using the Chinese Remainder Theorem (on the ring $\mathbb{F}[x]$) to locate a polynomial $p(t)$ satisfying: $p(t) \equiv a_i \pmod{(t - a_i)^{n_i}}$, $p(t) \equiv 0 \pmod{t}$. Set $q(t) = t - p(t)$, then $p(t)$ and $q(t)$ has zero constant term, since $p(t) \equiv 0 \pmod{t}$.

Let $x_s = p(x)$, $x_n = q(x)$, for they are polynomials of x , then x_s and x_n commute and commute with any endomorphism commuting with x . And the statement (c) is obvious in this case.

For $p(t) \equiv a_i \pmod{(t - a_i)^{n_i}}$, i.e. $x_s - a_i I = f(x)(x - a_i)^{n_i}$, thus $V_i \subseteq \ker(x_s - a_i I)$, hence x_s acts diagonally on V_i with single eigenvalue a_i . Then it clear that x_s is semisimple, x_n is nilpotent.

Then it only remains to prove the uniqueness. Let $x = s + n$ is another decomposition satisfying the requires, where s is semisimple and n is nilpotent. Then $x_s - s = x_n - n$ and these four elements are commute with each other. Then $x_s - s$ is also semisimple and $x_n - n$ is nilpotent. Then it's both semisimple and nilpotent, thus $x_s - s = x_n - n = 0$, which concludes the proof. $\square$

Example 5.1.3. Let $x \in \text{End}(V)$ and $x = x_s + x_n$ is the Jordan decomposition of x , then $\text{ad } x = \text{ad } x_s + \text{ad } x_n$ is the Jordan decomposition of $\text{ad } x$.

sketch of proof. We have seen that $\text{ad } x_s$ is semisimple and $\text{ad } x_n$ is nilpotent. And we conclude the proof by the existence and uniqueness. ¹ $\square$

5.2 More about Generalized Eigenspaces

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and ϕ its representation on a finite-dimensional vector space V , over an algebraically closed field $\mathbb{F}$ of characteristic 0. We have the following generalized eigenspace decompositions for a **fixed** $a \in \mathfrak{g}$ (which is essentially the proposition 5.1.1):

$$\begin{aligned} V &= \bigoplus_{\lambda \in \mathbb{F}} V_{\lambda}^a, \quad V_{\lambda}^a = \{v \in V \mid (\phi(a) - \lambda I)^N v = 0 \text{ for some } N \in \mathbb{N}\} \\ \mathfrak{g} &= \bigoplus_{\alpha \in \mathbb{F}} \mathfrak{g}_{\alpha}^a, \quad \mathfrak{g}_{\alpha}^a = \{g \in \mathfrak{g} \mid (\text{ad } a - \alpha I)^N g = 0 \text{ for some } N \in \mathbb{N}\} \end{aligned} \tag{5.1}$$

Lemma 5.2.1. Suppose U is a unital associative algebra over $\mathbb{F}$, and let $a, b \in U$ and $\lambda, \alpha \in \mathbb{F}$. Then we have (which is very similar to binomial expansion)

$$(a - \alpha - \lambda)^N b = \sum_{j=0}^N \binom{N}{j} ((\text{ad } a - \alpha I)^j b) (a - \lambda)^{N-j}.$$

sketch of proof. Write $\text{ad } a = L_a - R_a$, and we have

$$L_{a-\alpha-\lambda} = (\text{ad } a - \alpha I) + R_{a-\lambda}$$

¹In fact, we can find the commutativity from another way by the corollary of Jordan decomposition $[\text{ad } x_s, \text{ad } x_n] = \text{ad}[x_s, x_n] = 0$.

For any given $a, b \in U$, the operators L_a and R_b commute by associativity of U , thus $\mathbf{ad}a$ commutes with both L_a and R_a . Then since $\alpha I, \lambda I \in \mathbb{F}I \subset C(U)$, the terms $(\mathbf{ad}a - \alpha I)$ and $R_{a-\lambda}$ commute. Thus we can applying the binomial expansion. $\square$

Then we have:

Theorem 5.2.1. $\phi(\mathfrak{g}_\alpha^a)V_\lambda^a \subset V_{\lambda+\alpha}^a$, i.e. for all $g \in \mathfrak{g}_\alpha^a$, we have $\phi(g)V_\lambda^a \subset V_{\lambda+\alpha}^a$. Moreover, consider the adjoint representation, we have $\phi(\mathfrak{g}_\alpha^a)\mathfrak{g}_\beta^a \subset \mathfrak{g}_{\beta+\alpha}^a$, where $\phi(\mathfrak{g}_\alpha^a) = \mathbf{ad}\mathfrak{g}_\alpha^a$, thus $[\mathfrak{g}_\alpha^a, \mathfrak{g}_\beta^a] \subseteq \mathfrak{g}_{\alpha+\beta}^a$.

proof. For $g \in \mathfrak{g}_\alpha^a, v \in V_\lambda$, we need to find an N s.t. $(\phi(a) - \alpha I - \lambda I)^N \phi(g)v = 0$. Consider:

$$(\phi(a) - \alpha I - \lambda I)^N \phi(g)v = \sum_{j=0}^N \binom{N}{j} ((\mathbf{ad}\phi(a) - \alpha I)^j \phi(g)) ((\phi(a) - \lambda I)^{N-j}v)$$

Thus conclude the proof. $\square$

Then we look at Lemma B on gtm 9.

Lemma 5.2.2. Let $\mathcal{A}$ be a finite dimensional F -algebra. Then $\text{Der } \mathcal{A}$ contains the semisimple and nilpotent part (in $\text{End } \mathcal{A}$) of all its elements.

proof. In fact we only need to show that the semisimple part of an element is in $\text{Der } \mathcal{A}$, then by using subtraction we can get the nilpotent part of the element. Let $x \in \text{Der } \mathcal{A}$, say x_s is its semisimple part. We need to prove x_s is still a derivation. Consider the generalized eigenspace decomposition of $\mathcal{A}$ under x : $\mathcal{A} = \bigoplus_{\lambda \in F} \mathcal{A}_\lambda^x = \bigoplus_{i=1}^n \mathcal{A}_{\lambda_i}^x$. Then $x_s = \sum_{i=1}^n x_i$, where $x_i = \delta_{ij} \lambda_i I$ when acts on $\mathcal{A}_{\lambda_j}^x$. It's obvious that x_i are linear, then x_s is also linear. For $a, b \in \mathcal{A}$, say $a = \sum_i a_i, b = \sum_i b_i$, then we have:

$$x_s(ab) = x_s\left(\sum_{i,j} a_i b_j\right) = \sum_{i,j} x_s(a_i b_j) = \sum_{i,j} x_s(a_i) b_j + \sum_{i,j} a_i x_s(b_j) = x_s(a)b + ax_s(b)$$

Thus we only need to prove for $a_i \in \mathcal{A}_{\lambda_i}^x, b_j \in \mathcal{A}_{\lambda_j}^x$, $x_s(a_i b_j) = x_s(a_i) b_j + a_i x_s(b_j)$. Using Lemma 5.2.1, we can get $\mathcal{A}_{\lambda_i}^x \mathcal{A}_{\lambda_j}^x \subseteq \mathcal{A}_{\lambda_i + \lambda_j}^x$, thus $x_s(a_i b_j) = (\lambda_i + \lambda_j) a_i b_j$. And we have $x_s(a_i) b_j + a_i x_s(b_j) = \lambda_i a_i b_j + \lambda_j a_i b_j = (\lambda_i + \lambda_j) a_i b_j$, thus we conclude the proof. $\square$

5.3 Generalized weight space

Definition 5.3.1. Let $\mathfrak{g}$ be a Lie algebra with a representation ϕ on a vector space V , and let $\lambda \in \mathfrak{g}^*$ be a linear function on $\mathfrak{g}$. The generalized weight space of $\mathfrak{g}$ in V attached to λ is

$$V_\lambda^\mathfrak{g} = \{v \in V \mid (\phi(g) - \lambda(g)I)^N v = 0 \text{ for some } N \text{ depending on } g, \forall g \in \mathfrak{g}\}.$$

And we can easily check that $\bigcap_{a \in \mathfrak{g}} V_{\lambda(a)}^a = V_\lambda^\mathfrak{g}(*),$ which means for a weight λ , the intersection of all generalized eigenspaces concerning λ equals to the generalized weight space of $\mathfrak{g}$ in V attached to λ .

Theorem 5.3.1. ¹ Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and ϕ its representation on a finite-dimensional vector space V , over an algebraically closed field $\mathbb{F}$ of characteristic 0. Let $\mathfrak{h}$ be a **nilpotent** subalgebra of $\mathfrak{g}$. Then the following equalities hold.²

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda^\mathfrak{h} \quad (5.2)$$

¹The keypoint is to find the weight λ .

²This decomposition should be unique.

$$\phi(\mathfrak{g}_\alpha^\mathfrak{h})V_\lambda^\mathfrak{h} \subseteq V_{\lambda+\alpha}^\mathfrak{h} \quad (5.3)$$

Moreover, in the case of the adjoint representation, we may express these as:

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha^\mathfrak{h} \quad \text{and} \quad [\mathfrak{g}_\alpha^\mathfrak{h}, \mathfrak{g}_\beta^\mathfrak{h}] \subseteq \mathfrak{g}_{\alpha+\beta}^\mathfrak{h} \quad (5.4)$$

proof. Case 1. For each $a \in \mathfrak{h}$, $\phi(a)$ has only one eigenvalue.

In this case, V is a generalized eigenspace $V_{\lambda(a)}^a$ of every $a \in \mathfrak{h}$, so we just need to check the linearity of λ . Since $\mathfrak{h}$ is nilpotent, it is solvable. Thus applying Lie's theorem¹, we get a weight λ' with some nonzero weight space $V_{\lambda'}^\mathfrak{h}$. Then $\lambda'(a)$ must be the eigenvalue of $\phi(a)$ with which $\phi(a)$ acts on $V_{\lambda'}^\mathfrak{h}$, so $\lambda' = \lambda$. Therefore λ is linear, so V is the generalized weight space $V_\lambda^\mathfrak{h}$.

Case 2. For some $a_0 \in \mathfrak{h}$, $\phi(a_0)$ has at least two distinct eigenvalues.

Since $\mathfrak{h}$ is nilpotent, $\mathbf{ad}a$ is a nilpotent operator on $\mathfrak{h}$ for all $a \in \mathfrak{h}$. Thus $\mathfrak{h} \subseteq \mathfrak{g}_0^a$. Then by Theorem 5.2.1, $\phi(\mathfrak{h})V_\lambda^a \subseteq V_\lambda^a$ for any $a \in \mathfrak{h}$, $\lambda \in F$, which means every general eigenspaces of a is invariant under $\phi(\mathfrak{h})$. Since $\mathbb{F}$ is algebraically closed, V can be written as a direct sum of the generalized eigenspaces of a_0 . Since each $V_\lambda^{a_0}$ is invariant under the action of $\mathfrak{h}$, each $V_\lambda^{a_0}$ is also a representation of $\mathfrak{h}$. Since $\dim V_\lambda^{a_0} < \dim V$, we may apply induction on $\dim V$. This establishes the equality (5.2), i.e. $V = \bigoplus_{i=1}^n V_{\lambda_i}^{a_0} = \bigoplus_{i=1}^n \bigoplus_{j=1}^n V_{\lambda_{ij}}^\mathfrak{h}$, where

$\lambda_i \in F$, $\lambda_{ij} \in \mathfrak{h}^*$ and $\lambda_{ij}(a_0) = \lambda_i$.

To finish, we'll prove the inclusion (5.3). Suppose $\alpha, \lambda \in \mathfrak{h}^*$, and suppose $g \in \mathfrak{g}_\alpha^\mathfrak{h}$. Then $g \in \mathfrak{g}_{\alpha(a)}^a$ for all $a \in \mathfrak{h}$. By Theorem 5.2.1, $\phi(g)V_{\lambda(a)}^a \subseteq V_{\lambda(a)+\alpha(a)}^a$ for all $a \in \mathfrak{h}$. Then

$$v \in \bigcap_{a \in \mathfrak{h}} V_{\lambda(a)}^a \Rightarrow \phi(g)v \in \bigcap_{a \in \mathfrak{h}} V_{\lambda(a)+\alpha(a)}^a$$

Since $\bigcap_{a \in \mathfrak{h}} V_{\lambda(a)}^a = V_\lambda^\mathfrak{h}$ by the definition of a generalized weight space, this establishes (5.3). □

Example 5.3.1. Take $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{F})$ and $\mathfrak{h} = \{\text{diagonal matrices}\}$. Find the generalized weight space decomposition in both the tautological and the adjoint representations, and check the inclusions (5.3) and (5.4) in Theorem 5.3.1.

solution. The generalized eigenvectors of $h \in \mathfrak{h}$ are actual eigenvectors, and every standard basis element of $\mathbb{F}^n$ is an eigenvector. Given any linear combination $ae_i + be_j$ of more than one basis element, there is some diagonal matrix that takes e_i to e_i and e_j to zero, so these linear combinations are not generalized eigenvectors of everything in $\mathfrak{h}$. Thus the only candidates for generalized weight spaces are the n axes, each of which is the span of a single standard basis element of $\mathbb{F}^n$.

For the axis V_i spanned by e_i , the linear functional on $\mathfrak{h}$ that takes h to the component h_i is a weight making V_i a weight space. $\mathbb{F}^n$ decomposes as a direct sum of n copies of $\mathbb{F}$.

To do the same with the adjoint representation, suppose the diagonal entries of $h \in \mathfrak{h}$ are h_i . Then for $a \in \mathfrak{g}$, we have: $((\mathbf{ad}h)a)_{i,j} = (ha - ah)_{i,j} = (h_i - h_j)a_{i,j}$. This shows that $\mathbf{ad}h$ is diagonalizable, which again implies that its generalized eigenvectors are actual eigenvectors, and so its generalized weight spaces are actually weight spaces. By appropriate choice of h , we can always make the basis elements $e_{i,j}$ of $\mathfrak{g}$ satisfying $i < j$ lie in distinct eigenspaces for some h , and thus they lie in distinct candidate weight spaces. This weight space can be achieved with the linear functional $\lambda_{i,j}$ taking h to $h_i - h_j$. Moreover, the zero linear functional has $\mathfrak{h}$ as its weight space. Thus the generalized weight space decomposition of $\mathfrak{g}$ in the adjoint representation is $\mathfrak{h}$ plus some 1-dimensional weight spaces:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{i \neq j} \text{span}\{e_{i,j}\}$$

The inclusions are trivial. □

¹The use of Lie theorem is beautiful.

5.4 Some problems

1、 By the example of the adjoint representation of a nonabelian solvable Lie algebra, show that the generalized weight space decomposition fails if the Lie algebra is solvable but not nilpotent.

proof. For 2-dimensional nonabelian Lie algebra $\mathfrak{g} = \langle x, y \rangle$ and $[x, y] = x$, it's solvable but not nilpotent. Consider the adjoint representation of $\mathfrak{g}$ on $\mathfrak{g}$. If there exists a weight λ , say $\lambda(x) = c, \lambda(y) = d$. Consider the dimension of $V_\lambda^{\mathfrak{g}}$. If $\dim V_\lambda^{\mathfrak{g}} = 2$, i.e. $\mathfrak{g} = V_\lambda^{\mathfrak{g}}$. For $\mathfrak{g}$ is solvable, according to Lie theorem, the weight space is non-empty. Then for $\text{adx}(x) = \text{ady}(y) = 0$, we have $d=c=0$. However, $(\text{ady})^n(x) \neq 0$, for all n , thus we get a contradiction.

If $\dim V_\lambda^{\mathfrak{g}} = 1$, say $V_\lambda^{\mathfrak{g}} = \mathbb{F}(ax + by)$. According to Lie theorem, $\mathbb{F}(ax + by)$ is the weight space of adg . Thus we have $(\text{adx} - cI)(ax + by) = (\text{ady} - dI)(ax + by) = 0$. $(\text{adx} - cI)(ax + by) = (b - ac)x - bcy = 0$, $(\text{ady} - dI)(ax + by) = (-a - ad)x - bdy = 0$. If $b \neq 0$, for $(b - ac)x - bcy = 0$ we have $c=0$, thus get a contradiction. Hence $b=0$, then $a \neq 0$ and $V_\lambda^{\mathfrak{g}} = \mathbb{F}x$. If the decomposition is true, then $\mathfrak{g} = \mathbb{F}x \oplus V_{\lambda_1}^{\mathfrak{g}}$ where $\dim V_{\lambda_1}^{\mathfrak{g}} = 1$, i.e. $V_{\lambda_1}^{\mathfrak{g}} = \mathbb{F}(ax + by)$, where $b \neq 0$. Thus we get an contradiction. So we conclude the proof. $\square$

anotherproof. ¹ For 2-dimensional nonabelian Lie algebra $\mathfrak{g} = \langle x, y \rangle$ and $[x, y] = x$, it's solvable but not nilpotent. Let's consider the matrix of adx and ady under the basis x, y :

$$\text{adx} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{ady} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Thus the eigenvalue of adx can only be 0 and the eigenvalue ady can only be 0, 1. Thus weight λ can only be $\lambda_1(x) = \lambda_1(y) = 0$ or $\lambda_2(x) = 0, \lambda_2(y) = 1$. Thus if the decomposition is right, we have $\mathfrak{g} = \mathfrak{g}_{\lambda_1}^{\mathfrak{g}} \oplus \mathfrak{g}_{\lambda_2}^{\mathfrak{g}}$ (if λ is not a weight, we assume $\mathfrak{g}_\lambda = 0$). $\mathfrak{g}_{\lambda_1}^{\mathfrak{g}} = \bigcap_{g \in \mathfrak{g}} V_{\lambda_1(g)}^{\mathfrak{g}} \subseteq V_0^{\mathfrak{g}} = \mathbb{F}y$. However, $\text{ad}(x + y)^N(y) \neq 0, \forall N$.

Thus $\mathfrak{g}_{\lambda_1}^{\mathfrak{g}} = 0$, $\mathfrak{g} = \mathfrak{g}_{\lambda_2}^{\mathfrak{g}} \subseteq V_1^{\mathfrak{g}}$, whereas $y \notin V_1^{\mathfrak{g}}$, thus we get a contradiction. $\square$

¹This proof is from one of my classmates, shown in our class. I think this method is quite clear.

6 Cartan Subalgebra

6.1 Cartan Subalgebra

Lemma 6.1.1. Let $\mathfrak{g}$ be a nilpotent Lie algebra and $\mathfrak{h} \subset \mathfrak{g}$ a subalgebra such that $\mathfrak{h} \neq \mathfrak{g}$. Then, $\mathfrak{h} \subsetneq N_{\mathfrak{g}}(\mathfrak{h})$.¹

sketch of proof. Consider the action of $\text{ad}\mathfrak{h}$ on $\mathfrak{g}/\mathfrak{h}$. Then using the Engel's theorem, we can find an element in $N_{\mathfrak{g}}(\mathfrak{h}) - \mathfrak{h}$. Or one can prove this by:

Consider the lower central series: $\mathfrak{g} = \mathfrak{g}^1 \supset \mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}] \supset \mathfrak{g}^3 \supset \cdots \supset \mathfrak{g}^n = 0$. Take j to be the maximal possible positive integer such that: $\mathfrak{g}^j + \mathfrak{h} \neq \mathfrak{h}$, $\mathfrak{g}^{j+1} + \mathfrak{h} = \mathfrak{h}$.² Then we have $[\mathfrak{g}^j + \mathfrak{h}, \mathfrak{h}] \subseteq \mathfrak{g}^{j+1} + \mathfrak{h} = \mathfrak{h}$. Thus $\mathfrak{h} \subsetneq \mathfrak{g}^j + \mathfrak{h} \subset N_{\mathfrak{g}}(\mathfrak{h})$. $\square$

Definition 6.1.1. A Cartan subalgebra(CSA) of a Lie algebra $\mathfrak{g}$ is a subalgebra $\mathfrak{h}$, satisfying the following two conditions:

- (i) $\mathfrak{h}$ is a nilpotent Lie algebra;
- (ii) $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$, which means $\mathfrak{h}$ is self-normalizing.

Corollary 6.1.1. Using the Lemma 6.1.1., one can easily find that A CSA of a Lie algebra $\mathfrak{g}$ is a maximal nilpotent subalgebra, which means there is no other proper nilpotent subalgebra containing it. Moreover, if $\mathfrak{g}$ is not nilpotent, its CSA is itself.

Example 6.1.1. Let $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{F})$ with $\text{char}(\mathbb{F}) \neq 2$. Let $\mathfrak{h} = \mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n$. Then this is a maximal nilpotent subalgebra but not a Cartan subalgebra.

proof. As $[\mathfrak{t}(n, \mathbb{F}), \mathfrak{n}(n, \mathbb{F})] \subseteq \mathfrak{n}(n, \mathbb{F})$, we have $\mathfrak{t}(n, \mathbb{F}) \subset N_{\mathfrak{g}}(\mathfrak{n}(n, \mathbb{F})) \subset N_{\mathfrak{g}}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n)$. Thus $\mathfrak{h}$ is not self-normalizing, so $\mathfrak{h}$ is not a CSA.

Obviously, $\mathfrak{h}^1 = [\mathfrak{h}, \mathfrak{h}] \subseteq [\mathfrak{t}(n, \mathbb{F}), \mathfrak{t}(n, \mathbb{F})] = \mathfrak{n}(n, \mathbb{F})$, $\mathfrak{h}^2 = [\mathfrak{h}, \mathfrak{h}^1] \subseteq [\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n, \mathfrak{n}(n, \mathbb{F})]$. $\forall a \in [\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n, \mathfrak{n}(n, \mathbb{F})]$, $\exists x, y \in \mathfrak{n}(n, \mathbb{F})$, s.t. $a = [x + kI_n, y] = [x, y] \in \mathfrak{n}^1(n, \mathbb{F})$. Thus $\mathfrak{h}^2 \subseteq \mathfrak{n}^1(n, \mathbb{F})$. Samely, $\mathfrak{h}^k \subseteq \mathfrak{n}^{k-1}(n, \mathbb{F})$. Thus $\mathfrak{h}$ is nilpotent.

If $\mathfrak{h} \subsetneq L$ and L is a nilpotent subalgebra, then $\exists x \in L - \mathfrak{h}$ and we can assume the strictly upper triangular part of x is 0 for L is a linear space and $\mathfrak{n}(n, \mathbb{F}) \subseteq L$. If the diagonal elements of x are not equal to each other, then x has different eigenvalues and we can assume x has eigenvalue 0 for the same reason. For $z \in \mathfrak{gl}(n, \mathbb{F})$, $\text{adx}(y) = xy - yx = z^{-1}(zxz^{-1}zyz^{-1} - zy z^{-1}zxz^{-1})z$. Using induction, we can find that $\text{adx}^N(y) = 0 \iff \text{ad}_{zxz^{-1}}^N(zyz^{-1}) = 0$, thus whether x is ad-nilpotent is invariant under similar elementary transformations. Then we can just consider the Jordan canonical form of x , say $x = \text{diag}(J_0, J_a, \cdots)$, where

$J_0 = J$ and $J_a = aI + J$. Consider block matrix $A = \begin{pmatrix} 0 & B & 0 & \cdots \\ 0 & 0 & & \\ \vdots & & & \end{pmatrix}$ where $B: b_{ij} = \delta_{ij}$ and the block s.t. A

is able to multiply with x . Then $\text{adx}(A) = xA - Ax = \begin{pmatrix} 0 & J_0B - BJ_a & 0 & \cdots \\ 0 & 0 & & \\ \vdots & & & \end{pmatrix} = \begin{pmatrix} 0 & -aB & 0 & \cdots \\ 0 & 0 & & \\ \vdots & & & \end{pmatrix}$.

Thus x can't be ad-nilpotent. According to Engel's theorem, L is not nilpotent.

Thus for element $b \in L - \mathfrak{h}$, we can assume $b = \sum_{i,j} c_{ij}e_{ij}$, where $i > j$. Say $c_{i'j'} \neq 0$ for $i' > j'$, then $[b, e_{j'i'}] = \sum_i c_{ij'}e_{ii'} - \sum_j c_{ij'}e_{j'j} \in L$. Note that the (i', i') th and the (j', j') th entries are $c_{i'j'}$ and $-c_{i'j'}$ respectively, both of which are nonzero. Then by some subtraction, we can get an element in L with at least 2 different eigenvalues, thus we get a contradiction. $\square$

anotherproof. We only prove for $\mathfrak{h}$ is a maximal nilpotent subalgebra. We have $\mathfrak{t}(n, \mathbb{F}) \subset N_{\mathfrak{g}}(\mathfrak{n}(n, \mathbb{F})) \subset N_{\mathfrak{g}}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n)$. Suppose there exists some nilpotent subalgebra $\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n \subsetneq \mathfrak{h}'$.

First note that in fact we have $\mathfrak{t}(n, \mathbb{F}) = N_{\mathfrak{gl}(n, \mathbb{F})}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n)$, since if $b = \sum_{i,j} c_{ij}e_{ij} \in N_{\mathfrak{g}}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n)$ with $c_{i'j'} \neq 0$ for $i' > j'$, then $e_{j'i'} \in \mathfrak{n}(n, \mathbb{F}) \subset \mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n$ and $[b, e_{j'i'}] = \sum_i c_{ij'}e_{ii'} - \sum_j c_{ij'}e_{j'j}$.

¹Which is mentioned in the proof of Engel's theorem.

²which means $\mathfrak{g}^j \subsetneq \mathfrak{h}$, $\mathfrak{g}^{j+1} \subseteq \mathfrak{h}$.

Note that the $(i', i')^{th}$ and the $(j', j')^{th}$ entries are $c_{i'j'}$ and $-c_{i'j'}$ respectively; both of which are nonzero. Thus, $[b, e_{j'i'}] \notin \mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n$, unless $\text{char}(\mathbb{F}) = 2$.

Now, by the Lemma above, we must have that $\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n \subsetneq N_{\mathfrak{h}'}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n) \cap N_{\mathfrak{gl}(n, \mathbb{F})}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n) \subseteq \mathfrak{h}' \cap N_{\mathfrak{gl}(n, \mathbb{F})}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n)$.¹ Thus, $\mathfrak{h}'$ contains some element of $\mathfrak{t}(n, \mathbb{F}) \setminus \mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n$; but all elements of $\mathfrak{t}(n, \mathbb{F}) \setminus \mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n$ have at least two distinct eigenvalues, $\dots$ $\square$

Proposition 6.1.1. Let $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{F})$ be a subalgebra containing a diagonal matrix $a = \text{diag}(a_1, \dots, a_n)$ with distinct a_i , and let $\mathfrak{h}$ be the subspace of all diagonal matrices in $\mathfrak{g}$. Then $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$.
sketch of proof. The nilpotent is easy to verify. For $x \in N_{\mathfrak{g}}(\mathfrak{h})$, $[x, a] = xa - ax \in \mathfrak{h}$. Say $x = (x_{ij})$, $a = \text{diag}(a_1, \dots, a_n)$, then $xa - ax = ((a_j - a_i)x_{ij})$. $\square$

Proposition 6.1.2. Let $\mathfrak{g}$ be a finite dimensional Lie algebra over an algebraically closed field $\mathbb{F}$ of characteristic zero and let $\mathfrak{h} \subset \mathfrak{g}$ be a Cartan subalgebra. Consider the generalized weight space decomposition (called **root space decomposition**) with respect to $\mathfrak{h} : \mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha}$. Then $\mathfrak{g}_0 = \mathfrak{h}$.

proof. Since by Engel's Theorem $\text{ad}h|_{\mathfrak{h}}$ is nilpotent for all $h \in \mathfrak{h}$, it follows that $\mathfrak{h} \subset \mathfrak{g}_0$.

But by definition of $\mathfrak{g}_0$, for all elements $h \in \mathfrak{h}$, $\text{ad}h|_{\mathfrak{g}_0}$ is a nilpotent operator. Hence $\text{ad}h|_{\mathfrak{g}_0/\mathfrak{h}}$ is a nilpotent operator for all $h \in \mathfrak{h}$. Therefore, by Engel's Theorem, there exists a non-zero $\bar{b} \in \mathfrak{g}_0/\mathfrak{h}$ which is annihilated by all $\text{ad}h|_{\mathfrak{g}_0/\mathfrak{h}}$. Taking a pre-image $b \in \mathfrak{g}_0$ of $\bar{b}$, this means that $[b, \mathfrak{h}] \subset \mathfrak{h}$ (i.e. $\mathfrak{h} \neq N_{\mathfrak{g}}(\mathfrak{h})$), which contradicts the fact that $\mathfrak{h}$ is a Cartan subalgebra. $\square$

6.2 Regular Element

Let $\mathfrak{g}$ be a finite dimensional Lie algebra of dimension d and consider the characteristic polynomial of an endomorphism $\mathbf{ad}a$ for some $a \in \mathfrak{g}$: $\det_{\mathfrak{g}}(\mathbf{ad}a - \lambda I) = (-\lambda)^d + c_{d-1}(a)(-\lambda)^{d-1} + \dots + \det(\mathbf{ad}a)$.

This is a polynomial of degree d . Because $(\mathbf{ad}a)(a) = [a, a] = 0$, we know $\det(\mathbf{ad}a) = 0$ and the constant term in the polynomial vanishes.

Let $a \in \mathfrak{g}$ and $\mathfrak{g}_0^a = \{b \in \mathfrak{g} \mid (\mathbf{ad}a)^N b = 0 \text{ for some } N\}$. The number $n_a = \dim \mathfrak{g}_0^a > 0$ coincides with the multiplicity of the zero root of the characteristic polynomial of the operator $\mathbf{ad}a$ in $\mathfrak{g}$. (Consider the algebraic multiplicity and the dimension of root spaces in linear algebra.²)

Definition 6.2.1. The number $\text{rk} \mathfrak{g} = \min\{n_a \mid a \in \mathfrak{g}\}$ is called **the rank of the Lie algebras $\mathfrak{g}$** . An element $a \in \mathfrak{g}$ is said to be **regular** if $n_a = \text{rk} \mathfrak{g}$ and **singular** if $n_a > \text{rk} \mathfrak{g}$.

We can restate the definition using the language of characteristic polynomials. The smallest positive integer r such that $c_r(a)$ is not the zero polynomial on $\mathfrak{g}$ is called **the rank of $\mathfrak{g}$** , write $r = \text{rk}(\mathfrak{g})$. Note that $1 \leq r \leq n$. An element $a \in \mathfrak{g}$ is called **regular** if $c_r(a) \neq 0$. The nonzero polynomial $c_r(a)$ of degree $d - r$ is called **the discriminant of $\mathfrak{g}$** . And there is a simple proposition:

Proposition 6.2.1. Let $\mathfrak{g}$ be a Lie algebra of dimension d , rank r over $\mathbb{F}$,

(1) $r = d$ if and only if $\mathfrak{g}$ is nilpotent, namely, $\mathfrak{g}$ is nilpotent iff the characteristic polynomial of $\mathfrak{g}$ has only one term.

(2) If $\mathfrak{g}$ is nilpotent, the set of regular elements is $\mathfrak{g}$.

(3) If $\mathfrak{g}$ is not nilpotent, the set of regular elements is infinite if $\mathbb{F}$ is.

sketch of proof. (1), (2). Just analyzing every element in $\mathfrak{g}$; (3). easy. $\square$

Example 6.2.1. By “polynomial in $\mathfrak{g}$ ” we mean that if we fix a basis $x_1, x_2, \dots, x_d$ of $\mathfrak{g}$ and write a general element $a = \sum_i a_i x_i$, then $c_k(a) \in k[a_1, \dots, a_d]$. Show that $c_j(a)$ is a homogeneous polynomial on $\mathfrak{g}$ of degree $d - j$.

¹We know that $N_{\mathfrak{h}'}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n) \subsetneq N_{\mathfrak{gl}(n, \mathbb{F})}(\mathfrak{n}(n, \mathbb{F}) + \mathbb{F}I_n)$

²One should always mind that the root space in linear algebra and Lie algebra are different things.

proof. Observe that $\mathbf{ada}$ is a d by d matrix whose entries b_{ij} are linear combinations of the a_i . Now consider $\det_{\mathfrak{g}}(\mathbf{ada} - \lambda I)$. This is a polynomial in the entries of the matrix $\mathbf{ada} - \lambda I$. A general element in the expansion of this determinant is $b_{**}b_{**} \cdots (b_{ii} - \lambda) \cdots (b_{jj} - \lambda)$. In order to get a term with λ^k we have to choose k λ 's and $d - k$ of the b_{**} in the expansion of this product. Therefore the coefficient of the polynomial λ^k is homogeneous of degree $d - k$ in the b_{**} , so it is also homogeneous of degree $d - k$ in the a_i since the b_{ij} are linear in the a_i . $\square$

The next lemma is obvious:

Proposition 6.2.2. Take $\mathfrak{g}$ is a subalgebra of $\mathfrak{gl}_n$ and a is an element in $\mathfrak{g}$. We have:

- (1) The Jordan decomposition of $\mathbf{ada}$ is given by $\mathbf{ada} = (\mathbf{ada}_s) + (\mathbf{ada}_n)$ in $\mathfrak{gl}_n$
- (2) If $\lambda_1, \dots, \lambda_n$ are eigenvalues of a_s , then $\lambda_i - \lambda_j$ are eigenvalues of $\mathbf{ada}_s$
- (3) $\mathbf{ada}_s$ has the same eigenvalues as $\mathbf{ada}$.

sketch of proof. (1) has been proved in subsection Jordan-Chevalley decomposition; (2) Just select n eigenvectors as a basis of $\mathbb{F}$, then consider the basis e_{ij} . And analyzing the dimension we know they are all eigenvalues of a_s ; (3) Consider the Jordan canonical form. $\square$

Example 6.2.2. For $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{F})$, the rank is n and we will find its discriminant. Also, the set of regular elements consists of all matrices with distinct eigenvalues.

Then we can verify that $\text{rk } \mathfrak{g} = n$ and the regular elements are all matrices with distinct eigenvalues using the lemma. We have known that $\dim \mathfrak{g}_0^a = n$ if a has distinct eigenvalues. If $a \in \mathfrak{g}$ has eigenvalue $\lambda_1, \dots, \lambda_k$, consider the generalized eigenspace decomposition of a , then it's obvious that $e_{ii} \in \mathfrak{g}_0^a, \forall i$, thus the dimension will not less than n .

If $\lambda_1, \dots, \lambda_n$ are eigenvalues of a , the discriminant is given by $c_n(a) = \prod_{i \neq j} (\lambda_i - \lambda_j)$. One can get this by considering the connection between the characteristic polynomial and the algebraic multiplicity of each eigenvalue.

6.3 Cartan theorem

We will use the fact that Zariski topology is highly non-Hausdorff to prove the Cartan theorem. For more, one can refer to [AN INTRODUCTION TO THE ZARISKI TOPOLOGY](#) and Lecture 7 in [18.745](#).

Definition 6.3.1. Zariski topology: Let $\mathbb{F}$ be a field and $X = \mathbb{F}^n$. A set F in X is closed if and only if F is the set of common zeroes of a collection (possibly infinite) of polynomials $P_\lambda(x)$ on $\mathbb{F}^n$. We can verify that any two non-empty Zariski open sets have a non-empty intersection.

Lemma 6.3.1. Let $\mathfrak{g}$ be a Lie algebra, $\dim \mathfrak{g} = n$ and $a \in \mathfrak{g}$. For $\forall b \in \mathfrak{g}$, there exists infinite many $\mu \in \mathbb{C}$ such that $\text{ad}(a + \mu b)$ are nilpotent operators on $\mathfrak{g}$, then $\mathfrak{g}$ is a nilpotent Lie algebra.

sketch of proof. Consider the characteristic polynomial of $\text{ad}(a + \mu b)$, which is a polynomial in $\mathbb{F}[\mu, \lambda]$. One can view it as $\mathbb{F}[\mu][\lambda]$ where $\mathbb{F}[\mu]$ is the ring and in this case the highest degree coefficient of λ is 1:

$$\det(\text{ad}(a + \mu b) - \lambda I) = \lambda^n + a_1(\mu)\lambda^{n-1} + \cdots + a_n(\mu)$$

For $\text{ad}(a + \mu b)$ is nilpotent, $a_i(\mu) = 0$, according to proposition 6.2.1. for all $\mu \in \mathbb{C}$. Thus $\det(\text{ad}(a + \mu b) - \lambda I) = \lambda^n$ for all μ . Using Engel's theorem, we conclude the proof. $\square$

We will first give two proofs of this theorem and for the second one, we will supplement the reason U and R are Zariski open subsets of $\mathfrak{h}$ after the proof. The thought of two proofs are the same, but the techniques are different.

Theorem 6.3.1. (E.Cartan) Let $\mathfrak{g}$ be a finite dimensional Lie algebra over an algebraically closed field $\mathbb{F}$ of characteristic zero. Let $a \in \mathfrak{g}$ be a regular element, and let $\mathfrak{g} = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_\lambda^a$ be the generalized eigenspace decomposition of $\mathfrak{g}$ with respect to ada . Then $\mathfrak{h} = \mathfrak{g}_0^a$ is a Cartan subalgebra.¹

proof. Say a is a regular element of $\mathfrak{g}$ and consider the generalized eigenspace decomposition attached to ada : $\mathfrak{g} = \mathfrak{g}_0^a \oplus \bigoplus_{\alpha \in \mathbb{F}} \mathfrak{g}_\alpha^a$. Say $V \equiv \bigoplus_{\alpha \in \mathbb{F}} \mathfrak{g}_\alpha^a$ and $\mathfrak{h} \equiv \mathfrak{g}_0^a$. We just need to prove that $\mathfrak{h}$ is a CSA of $\mathfrak{g}$.

To prove $\mathfrak{h}$ is nilpotent, we want to prove that for all $x \in \mathfrak{h}$, there are infinite many $\mu \in \mathbb{F}$, s.t. $\text{ad}(a + \mu b)$ is nilpotent. We just need to prove that there are infinite many $\mu \in \mathbb{F}$, s.t. all eigenvalue of $\text{ad}(a + \mu b)|_{\mathfrak{h}}$ are 0, i.e. $n_{(a+\mu b)|_{\mathfrak{h}}} = n$. Recall the fact that $[\mathfrak{g}_\lambda^a, \mathfrak{g}_\mu^a] \subset \mathfrak{g}_{\lambda+\mu}^a$ and in particular, if $\lambda = 0$ then $[\mathfrak{h}, \mathfrak{g}_\mu^a] \subset \mathfrak{g}_\mu^a$, which says V is an invariant subspace of x , for all x in $\mathfrak{h}$. For a is a regular element of $\mathfrak{g}$, $n_{a+\mu b} \geq n_a = n$. Say $D(a + \mu b)$ is the determinant of $(a + \mu b)|_V$, which is a polynomial of μ . Thus it has only finite roots, say $\mu_1, \dots, \mu_k$. Choose $\mu \neq \mu_i$, then $D(a + \mu b) \neq 0$, i.e. $(a + \mu b)|_V$ has no eigenvalue 0. Thus all eigenvalue 0 falls in $\mathfrak{h}$, i.e. $n_{(a+\mu b)|_{\mathfrak{h}}} \geq n$. Moreover, we have $n_{(a+\mu b)|_{\mathfrak{h}}} \leq \dim \mathfrak{h} = n$. Thus $\mathfrak{h}$ is nilpotent.

For $x \in N_{\mathfrak{g}}(\mathfrak{h})$, $\text{ad}_{\mathfrak{h}}(x) \subseteq \mathfrak{h}$, in particular, $\text{ada}(x) \in \mathfrak{h}$. Thus $x \in \mathfrak{g}_0^a = \mathfrak{h}$. So $\mathfrak{h}$ is self-normalizing. $\square$

anotherproof. The proof for this theorem uses the fact that Zariski Topology is highly non-Hausdorff, namely any two non-empty Zariski open sets have a non-empty intersection. Recall the fact that $[\mathfrak{g}_\lambda^a, \mathfrak{g}_\mu^a] \subset \mathfrak{g}_{\lambda+\mu}^a$, in particular, if $\lambda = 0$ then $[\mathfrak{h}, \mathfrak{g}_\mu^a] \subset \mathfrak{g}_\mu^a$. Let $V = \bigoplus_{\lambda \neq 0} \mathfrak{g}_\lambda^a$. Then $\mathfrak{g} = \mathfrak{h} \oplus V$ and $[\mathfrak{h}, V] \subset V$.

Consider the following two subsets of $\mathfrak{h}$:

$$\begin{aligned} U &= \{h \in \mathfrak{h} \mid \text{adh}|_{\mathfrak{h}} \text{ is not a nilpotent operator}\} \\ R &= \{h \in \mathfrak{h} \mid \text{adh}|_V \text{ is a non-singular operator}\} \end{aligned}$$

Both U and R are Zariski open subsets of $\mathfrak{h}$. Next we note that $a \in R$ since all zero eigenvalues of ada lie in $\mathfrak{h}$, hence R is non-empty.

Now, we shall prove by contradiction that $\mathfrak{h}$ is a nilpotent subalgebra. Suppose the contrary is true. Then, by Engel's Theorem there exists $h \in \mathfrak{h}$ such that $\text{adh}|_{\mathfrak{h}}$ is not nilpotent. Thus $h \in U$ and $U \neq \emptyset$. Therefore $U \cap R \neq \emptyset$. We now take $b \in U \cap R$. Then $\text{adh}|_{\mathfrak{h}}$ is not nilpotent and $\text{adb}|_V$ is invertible. Hence $\mathfrak{g}_0^b \subsetneq \mathfrak{h}$, which contradicts the fact that a is a regular element. Thus, this contradicts the assumption made; and we find that $\mathfrak{h}$ is a nilpotent Lie algebra.

Finally, we need to proof that $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. Now, if $b \in N_{\mathfrak{g}}(\mathfrak{h})$, so that $[b, \mathfrak{h}] \subset \mathfrak{h}$, then we have that, in particular, $[b, a] \in \mathfrak{h}$. But since $a \in \mathfrak{h}$ and $\mathfrak{h}$ is a nilpotent Lie algebra, then $\text{ada}|_{\mathfrak{h}}$ is a nilpotent operator. In particular, $0 = (\text{ada})^N((\text{ada})b) = (\text{ada})^{N+1}(b)$. Hence $b \in \mathfrak{g}_0^a = \mathfrak{h}$, which completes the proof of the theorem. $\square$

SUPPLEMENT:² We will only prove U is a Zariski open set, the proof for R is almost the same. We just need to prove that $\mathfrak{h} - U$ is a closed set, i.e. there exists some polynomials s.t. $U^c = V(f_\alpha)$. Say $x_1, \dots, x_n$ is a basis of $\mathfrak{h}$ and the matrix of $\text{ad}x_i$ under this basis is A_i . We know that all elements in $\mathfrak{h}$ can be written in the form $a_1x_1 + \dots + a_nx_n$.

$$\begin{aligned} U^c &= \{a_1x_1 + \dots + a_nx_n \in \mathfrak{h} \mid \text{ad}(a_1x_1 + \dots + a_nx_n) \text{ is nilpotent}\} \\ &= \{a_1x_1 + \dots + a_nx_n \in \mathfrak{h} \mid \det(a_1A_1 + \dots + a_nA_n - \lambda I) = (-\lambda)^n\} \\ &= \{a_1x_1 + \dots + a_nx_n \in \mathfrak{h} \mid f_1(a_1, \dots, a_n)\lambda^{n-1} + \dots + f_n(a_1, \dots, a_n) = 0, \forall \lambda\} \\ &= \{a_1x_1 + \dots + a_nx_n \in \mathfrak{h} \mid f_i(a_1, \dots, a_n) = 0\} = V(f_1, \dots, f_n) \end{aligned}$$

¹For the existence of regular element, the CSA of a Lie algebra is always exists.

²With the help of my senior classmates.

6.4 Conjugacy of Cartan Subalgebras

We state the main result on conjugacy of Cartan subalgebras— Chevalley's Theorem:

Theorem 6.4.1. (Chevalley) *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra of an algebraically closed field $\mathbb{F}$ of characteristic 0. Denote by G the subgroup of the group of automorphisms of $\mathfrak{g}$ which is generated by automorphisms of the form $e^{\text{ad}a}$ for $a \in \mathfrak{g}$ such that $\text{ad}a$ is nilpotent (which is actually $\text{Int } \mathfrak{g}$). Then any two Cartan subalgebras $\mathfrak{h}_1$ and $\mathfrak{h}_2$ are conjugate by G , i.e. there exists $\sigma \in G$ such that $\sigma(\mathfrak{h}_1) = \mathfrak{h}_2$.*

Corollary 6.4.1. *Let $\mathbb{F}$ be an algebraically closed field of characteristic 0 and let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{F}$. Then any Cartan subalgebra $\mathfrak{h}$ in $\mathfrak{g}$ is of the form $\mathfrak{g}_0^a$ for some regular element $a \in \mathfrak{g}$, and in particular $\dim \mathfrak{h} = \text{rk } \mathfrak{g}$. Also, all such subalgebras $\mathfrak{g}_0^a$ are isomorphic.*

6.5 The classification of all 3-dimensional Lie algebra

Assume $\mathfrak{g}$ is over an algebraically closed field $\mathbb{F}$ of characteristic zero. We classify 3-dimensional Lie algebra $\mathfrak{g}$ according to $\text{rk } \mathfrak{g}$.

If $\text{rk } \mathfrak{g} = 3$, then $\mathfrak{g}$ is nilpotent, whose classification we have given in subsection 3.4 (5).

If $\text{rk } \mathfrak{g} = 2$, In this case $\dim \mathfrak{h} = 2$. Since $\mathfrak{h}$ must be a nilpotent Lie algebra, we can conclude that $\mathfrak{h}$ is abelian. Hence the root space decomposition is $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{F}b$, where $[\mathfrak{h}, b] \subset \mathbb{F}b$. Since $\mathfrak{h} = N_{\mathfrak{g}}(\mathfrak{h})$ then $[\mathfrak{h}, b] \neq 0$. Hence there exist $a \in \mathfrak{h}$ such that $[a, b] = b$. Also since $\mathfrak{h}$ is 2-dimensional, then there exists $c \in \mathfrak{h}$ such that $[c, b] = 0$ (by some subtraction). We also know that $[a, c] = 0$.

Thus, we can conclude that the only 3-dimensional Lie algebra of rank 2 is a direct sum of a 2-dimensional non-abelian algebra and one dimensional central subalgebra: $\mathfrak{g} = (\mathbb{F}a + \mathbb{F}b) \oplus \mathbb{F}c$.

If $\text{rk } \mathfrak{g} = 1$, We assert that:

$\mathfrak{g}$ is isomorphic to one of the following Lie algebras with basis h, a, b :

- (i) $[h, a] = a, [h, b] = a + b, [a, b] = 0$;
- (ii) $[h, a] = a, [h, b] = \lambda b$, where nonzero $\lambda \in \mathbb{F}, [a, b] = 0$;
- (iii) $[h, a] = a, [h, b] = -b, [a, b] = h$.

Let $\mathfrak{h} = \mathbb{F}h$, where $h \in \mathfrak{g}$, be a Cartan subalgebra; we have: $\mathfrak{g} = \mathfrak{h} \oplus V$,¹ where $[h, V] \subset V$, $\dim V = 2$, and $\text{ad}h$ is non-singular on V . We may assume one of the eigenvalues of $\text{ad}h|_V$ is 1, if we scale h accordingly. Then we just need to consider the Jordan canonical form of $\text{ad}h|_V$.

If there is only one Jordan block in $\text{ad}h|_V$. Thus, $\text{ad}h|_V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ in some basis $\{a, b\}$. Thus, $[h, a] = a$ and $[h, b] = a + b$. Also, by using Jacobian Identity, we have $[h, [a, b]] = [[h, a], b] + [a, [h, b]] = [a, b] + [a, a + b] = 2[a, b]$. But we know that the eigenvalue 2 cannot be an eigenvalue of $\text{ad}h$, thus $[a, b] = 0$. Thus, this Lie algebra corresponds to (i).

Now, assume there are two Jordan block in $\text{ad}h|_V$. Thus, we have that $\text{ad}h|_V = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}$ in some basis $\{a, b\}$. We then have that $[h, a] = a$ and $[h, b] = \lambda b$. By following the same procedure as above, we note that $[h, [a, b]] = [[h, a], b] + [a, [h, b]] = [a, b] + [a, \lambda b] = (1 + \lambda)[a, b]$. Thus, we must have that either $[a, b] = 0$, which corresponds to (ii), or $(1 + \lambda)$ is an eigenvalue of $\text{ad}h$, case (iii).

Note that for the case of $[a, b] = 0$, then λ is arbitrary and uniquely defined by $\mathfrak{g}$ up to inverting it by swapping a and b and scaling h accordingly.

In the case where $(1 + \lambda)$ is an eigenvalue of $\text{ad}h$, we must have that $1 + \lambda = 0$ and that $[a, b]$ be a multiple of h . Thus $\lambda = -1$ and, scaling a accordingly, we may assume that $[a, b] = h$. Thus, we get option (iii) with the latter case.

¹One should notice that the subspace of this decomposition could not be subalgebra of $\mathfrak{g}$.

Proposition 6.5.1. *Show that all Lie algebras in this example are non-isomorphic. Those from (i) and (ii) are solvable, and the one from (iii) is isomorphic to $\mathfrak{sl}(2, \mathbb{F})$, which is not solvable.*

proof. *We first show that the Lie algebra from (iii) is isomorphic to $\mathfrak{sl}(2, \mathbb{F})$. The standard basis for $\mathfrak{sl}(2, \mathbb{F})$ consist of $\left\{ h' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, a' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, b' = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}$. Thus, we have $[h', a'] = 2a'$, $[h', b'] = -2b'$ and $[a', b'] = h'$. Now if we scale and set $h = \frac{h'}{2}$, $a = \frac{a'}{\sqrt{2}}$ and $b = \frac{b'}{\sqrt{2}}$, then we get the Lie algebra of case (iii). Thus, Lie algebra (iii) is isomorphic to $\mathfrak{sl}(2, \mathbb{F})$, which is not solvable.*

In contrast, it is clear that the algebras (i) and (ii) are solvable by construction. (One can check this by verifying the matrices are upper triangular.)

By conjugacy of Cartan subalgebras, the isomorphism class of $\mathfrak{g}$ needs to be independent of the choice of the Cartan subalgebra. It follows that the algebras of the three types are non-isomorphic to each other. It also follows that the algebras of type (ii), corresponding to parameters λ and λ' are isomorphic if and only if $\lambda\lambda' = 1$. $\square$

Proposition 6.5.2. *Show that the algebras of type (ii) in this example, corresponding to parameters λ and λ' are isomorphic if and only if $\lambda' = \lambda$ or $\lambda' = \frac{1}{\lambda}$.*

7 Trace Form & Cartan's Criterion

7.1 Trace form, Killing form

Definition 7.1.1. Let $\mathfrak{g}$ be a Lie algebra and ϕ a representation of $\mathfrak{g}$ on a finite dimensional vector space V . The **trace form** is a bilinear form on $\mathfrak{g}$, given by the following formula:

$$(a, b)_V = \text{tr}(\phi(a)\phi(b)), \quad \forall a, b \in \mathfrak{g}.$$

It's trivial that:

Proposition 7.1.1. (i) The trace form is symmetric, i.e. $(a, b)_V = (b, a)_V, \forall a, b \in \mathfrak{g}$;

(ii) The trace form is invariant(associate), i.e. $([a, b], c)_V = (a, [b, c])_V \forall a, b, c \in \mathfrak{g}$, which is equivalent to $0 = (\text{adb}(a), c)_V + (a, \text{adb}(c))_V \forall a, b, c \in \mathfrak{g}$.

Definition 7.1.2. If $\dim \mathfrak{g} < \infty$, then the trace form of the adjoint representation is called the **Killing form**:

$$\kappa(a, b) = \text{tr}(\text{ad}a \text{ad}b).$$

The Killing form is compatible with restriction to ideals. Suppose that $\mathfrak{g}$ is a Lie algebra and $\mathfrak{h}$ is an ideal of $\mathfrak{g}$. We write κ for the Killing form on $\mathfrak{g}$ and $\kappa_{\mathfrak{h}}$ for the Killing form on $\mathfrak{h}$, considered as a Lie algebra in its own right.

Lemma 7.1.1. If $a, b \in \mathfrak{h}$, then $\kappa_{\mathfrak{h}}(a, b) = \kappa(a, b), \forall a, b \in \mathfrak{h}$.

proof. Take a basis for $\mathfrak{h}$ and extend it to a basis of $\mathfrak{g}$. If $a \in \mathfrak{h}$, then $\text{ad}a$ maps $\mathfrak{g}$ into $\mathfrak{h}$, so the matrix of $\text{ad}a$ in this basis is of the form $\begin{pmatrix} A_a & B_a \\ 0 & 0 \end{pmatrix}$, where A_a is the matrix of $\text{ad}a$ restricted to $\mathfrak{h}$.

If $b \in \mathfrak{h}$, then $\text{ad}a\text{ad}b$ has matrix $\begin{pmatrix} A_a A_b & A_a B_b \\ 0 & 0 \end{pmatrix}$, where $A_a A_b$ is the matrix of $\text{ad}a\text{ad}b$ restricted to $\mathfrak{h}$.

Only the block $A_a A_b$ contributes to the trace of this matrix, so $\kappa(a, b) = \text{tr}(A_a A_b) = \kappa_{\mathfrak{h}}(a, b)$. $\square$

Definition 7.1.3. A symmetric bilinear form $\beta(x, y)$ is called **nondegenerate** if its **radical** S is 0, where $S = \{x \in \mathfrak{g} \mid \beta(x, y) = 0 \text{ for all } y \in \mathfrak{g}\}$, we can verify that S is an ideal of $\mathfrak{g}$ if β is killing form or trace form. Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{h}$ an ideal of $\mathfrak{g}$. We will have $\mathfrak{h}^\perp = \{a \in \mathfrak{g} \mid \kappa(a, \mathfrak{h}) = 0\}$ is an ideal of $\mathfrak{g}$.

From linear algebra, a practical way to test nondegeneracy is as follows: Fix a basis $x_1, x_2, \dots, x_n$ of $\mathfrak{g}$. Then κ is nondegenerate if and only if the $n \times n$ matrix whose i, j entry is $\kappa(x_i, x_j)$ has nonzero determinant (or the matrix has a row or entry whose elements are all 0.). The method that analyzing on a basis will also be useful in Lie algebra.

Example 7.1.1. Let $\mathfrak{g}$ be the 2-dimensional non-abelian Lie algebra with basis a, b such that $[a, b] = a$. In this basis we have

$$\text{ad}a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{ad}b = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $\kappa(a, a) = \kappa(a, b) = \kappa(b, a) = 0$ and $\kappa(b, b) = 1$. The matrix of κ in the basis a, b is therefore $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$.

Example 7.1.2. Show that the trace form of $\mathfrak{gl}(n, \mathbb{F})$ and $\mathfrak{sl}(n, \mathbb{F})$ associated to the standard representation is non-degenerate and the Killing form on $\mathfrak{sl}(n, \mathbb{F})$ is also non-degenerate, provided $\text{char} \mathbb{F} \nmid 2n$. Find the kernel of the Killing form on $\mathfrak{gl}(n, \mathbb{F})$ (which is again a big problem...).

The solution for the standard representations are quite trivial, just consider the action on a basis. Purely a linear algebra question.

solution. To show the trace form on $\mathfrak{gl}(n, \mathbb{F})$ for the standard representation is non-degenerate, if $\sum_{i,j} a_{ij} e_{ij}$ lies in the kernel where for some $r, s, a_{rs} \neq 0$, then $\text{tr}((\sum_{i,j} a_{ij} e_{ij})e_{sr}) = \text{tr}(a_{rs} e_{rs} e_{sr}) = a_{rs} \neq 0$ since $\text{tr}(e_{ij} e_{sr}) = 0$ unless $j = s, i = r$, which is a contradiction.

To show the trace form on $\mathfrak{sl}(n, \mathbb{F})$ for the standard representation is non-degenerate, if $x = \sum_{i,j} a_{ij} e_{ij}$ lies in the kernel and some $a_{rs} \neq 0, r \neq s$, then by the same argument we have a contradiction. So without loss of generality we can say $a_{rs} = 0$ for $r \neq s$, $x = \sum a_{ii} e_{ii}$. If $\exists a_{jj} \neq a_{kk}$ for some j, k , then $\text{tr}(\sum a_{ii} e_{ii}, e_{jj} - e_{kk}) = a_{jj} - a_{kk} \neq 0$, which is a contradiction. So $a_{jj} = a_{kk}, \forall j, k$, and $\text{tr}(x) = na_{11} = 0$. So $a_{11} = 0$ (since $\text{char } \mathbb{F} \nmid n$), and $x = 0$.

For the killing form on $\mathfrak{gl}(n, \mathbb{F})$, consider the basis of $\mathfrak{gl}(n, \mathbb{F})$, $\{e_{ij}\}$. Then:

$$\text{ad}_{e_{ij}} \text{ad}_{e_{kl}}(e_{gh}) = [\delta_{ij} e_{lg} e_{kh} - \delta_{hk} e_{gl}] = \delta_{jk} \delta_{lg} e_{ih} - \delta_{hi} \delta_{lg} e_{kj} - \delta_{jg} \delta_{hk} e_{il} + \delta_{li} \delta_{hk} e_{gj}$$

The coefficient of e_{gh} in this expansion is $a_{gh} = \delta_{gi} \delta_{jk} \delta_{lg} - \delta_{jh} \delta_{kg} \delta_{hi} \delta_{lg} - \delta_{gi} \delta_{lh} \delta_{jg} \delta_{hk} + \delta_{hj} \delta_{li} \delta_{hk}$. So:

$$\begin{aligned} \kappa_{\mathfrak{gl}_n}(e_{ij}, e_{kl}) &= \sum_{g,h} a_{gh} = \sum_{g,h} (\delta_{gi} \delta_{jk} \delta_{lg} - \delta_{gk} \delta_{jh} \delta_{hi} \delta_{lg} - \delta_{gi} \delta_{lh} \delta_{jg} \delta_{hk} + \delta_{hj} \delta_{li} \delta_{hk}) \\ &= n \delta_{il} \delta_{jk} - \delta_{kl} \delta_{ij} - \delta_{ij} \delta_{kl} + n \delta_{jk} \delta_{il}^1 = 2n \delta_{il} \delta_{jk} - 2 \delta_{ij} \delta_{kl} = 2n \text{tr}(e_{ij} e_{kl}) - 2 \text{tr}(e_{ij}) \text{tr}(e_{kl}) \end{aligned}$$

It follows that $\kappa_{\mathfrak{gl}_n}(x, y) = 2n \text{tr}(xy) - 2 \text{tr}(x) \text{tr}(y)$ by bilinearity (One can verify this by two steps: first $\kappa_{\mathfrak{gl}_n}(x, e_{ij}) = 2n \text{tr}(x e_{ij}) - 2 \text{tr}(x) \text{tr}(e_{ij})$ and then $\kappa_{\mathfrak{gl}_n}(x, y) = 2n \text{tr}(xy) - 2 \text{tr}(x) \text{tr}(y)$).

To calculate the radical of $\kappa_{\mathfrak{gl}_n}$, note if $x = \sum_{i,j} x_{ij} e_{ij}$, $\kappa_{\mathfrak{gl}_n}(x, e_{kl}) = 2n x_{lk} - 2(\sum x_{ii}) \delta_{kl} \equiv 0$. Thus we have $x_{lk} = 0$ when $k \neq l$, and $n x_{kk} = \sum_i x_{ii} \forall k$, so $x = \lambda I$ for some λ (since $\text{char } F \nmid 2n$). So the radical of $\kappa_{\mathfrak{gl}_n}$ is $\mathbb{F}I$.

Since $\mathfrak{sl}(n, \mathbb{F})$ is an ideal of $\mathfrak{gl}(n, \mathbb{F})$, $\kappa_{\mathfrak{sl}_n}$ is the **restriction**² of $\kappa_{\mathfrak{gl}_n}$ to $\mathfrak{sl}(n, \mathbb{F})$; hence $\kappa_{\mathfrak{sl}_n}(x, y) = 2n \text{tr}(xy)$. Since it is a scalar multiple of the trace form of the standard representation, which is non-degenerate, it follows that the radical of $\kappa_{\mathfrak{sl}_n}$ is trivial. $\square$

7.2 Cartan's Criterion

Lemma 7.2.1. (Cartan). Let $\mathfrak{g}$ be a finite dimensional Lie algebra over $\mathbb{F} = \bar{\mathbb{F}}$, a field of characteristic 0 (so that $\mathbb{Q} \subset \mathbb{F}$). Let ϕ be a representation of $\mathfrak{g}$ in a finite dimensional vector space V . Let $\mathfrak{h}$ be a Cartan subalgebra of $\mathfrak{g}$, and consider the generalized weight space decomposition of V and the generalized root space decomposition of $\mathfrak{g}$ with respect to $\mathfrak{h}$:

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_{\lambda}, \quad \mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha}, \quad \phi(\mathfrak{g}_{\alpha}) V_{\lambda} \subseteq V_{\lambda+\alpha}, \quad [\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

Pick $e \in \mathfrak{g}_{\alpha}$, $f \in \mathfrak{g}_{-\alpha}$, so that $h = [e, f] \in \mathfrak{g}_0 = \mathfrak{h}$. Suppose that $V_{\lambda} \neq 0$. Then $\lambda(h) = r\alpha(h)$, where $r \in \mathbb{Q}$ depends only on λ and α but not on h .

remark: The proof is quite simple, but it seems that the lemma isn't really meaningful (I cannot find out anyway). Some lemmas or propositions are mentioned to conquer some technical issues rather than raise something meaningful.

proof. Let $U = \bigoplus_{n \in \mathbb{Z}} V_{\lambda+n\alpha} \subset V$. Then $\dim U < \infty$, and U is $\phi(e)$, $\phi(f)$ and $\phi(h)$ invariant. But $[\phi(e), \phi(f)] = \phi(h)$, hence $\text{tr}_U(\phi(h)) = 0$. Thus we have the following:

$$0 = \text{tr}_U(\phi(h)) = \sum_n \text{tr}_{V_{\lambda+n\alpha}}(\phi(h)) = \sum_n (\lambda + n\alpha)(h) \dim V_{\lambda+n\alpha}$$

¹One can check this by assuming blablabla

²The use of restriction is beautiful.

In the last line we have used the fact that the matrix of $\phi(h)|_{V_{\lambda+n\alpha}}$ takes the following form:

$$A = \begin{pmatrix} (\lambda + n\alpha)(h) & & * \\ & (\lambda + n\alpha)(h) & * \\ & & \ddots \\ & & & (\lambda + n\alpha)(h) \end{pmatrix}$$

$$\Rightarrow \lambda(h) \left(\sum_n \dim V_{\lambda+n\alpha} \right) = -\alpha(h) \sum_n n \dim V_{\lambda+n\alpha}$$

$$\Rightarrow \lambda(h) = r\alpha(h), r = -\frac{\sum_n n \dim V_{\lambda+n\alpha}}{\sum_n \dim V_{\lambda+n\alpha}}$$

Note in the above that $V_\lambda \neq 0$, so $\sum_n \dim V_{\lambda+n\alpha} \neq 0$. □

Lemma 7.2.2. If $\mathfrak{g} \subset \mathfrak{gl}(V)$, such that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, then $(a, a)_V \neq 0$ for some $a \in \mathfrak{g}$.

proof. Proof by contradiction. Choose a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$, and let:

$$V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda, \quad \mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha,$$

Since $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$, we obtain that $\mathfrak{h} = \sum_{\alpha \in \mathfrak{h}^*} [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$.

Hence by Cartan's Lemma $\lambda(h) = r_{\lambda, \alpha_0} \alpha_0(h)$ ¹ for $r_{\lambda, \alpha_0} \in \mathbb{Q}, \neq 0$ if $V_\lambda \neq 0$. The assumption that $(a, a)_V = 0$ means that, for all $h \in [\mathfrak{g}_{\alpha_0}, \mathfrak{g}_{-\alpha_0}]$:

$$0 = (h, h)_V = \text{tr}_V(\phi(h)^2) = \sum_{\lambda \in \mathfrak{h}^*} \text{tr}_{V_\lambda}(\phi(h)^2) = \sum_{\lambda \in \mathfrak{h}^*} \lambda(h)^2 \dim V_\lambda = \alpha_0(h)^2 \sum_{\lambda} r_{\lambda, \alpha_0}^2 \dim V_\lambda$$

It follows from this calculation that $\alpha_0(h) = 0$, and hence $\lambda(h) = 0$ for all $h \in [\mathfrak{g}_{\alpha_0}, \mathfrak{g}_{-\alpha_0}]$ (One should notice that $\lambda(h)$ is probably not in $\mathbb{R}$, thus $\lambda(h)^2$ could not be non-negative.). Since $\mathfrak{h} = \sum_{\alpha \in \mathfrak{h}^*} [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, it follows that $\lambda(h) = 0$ for all $h \in \mathfrak{h}$. Since $\lambda \in \mathfrak{h}^*$ was arbitrary, this means $V = V_0$.

Since $\phi(\mathfrak{g}_\alpha)V = \phi(\mathfrak{g}_\alpha)V_0 \subset V_\alpha = 0$ for $\alpha \neq 0$, it follows that $\mathfrak{g}_\alpha = 0$ for $\alpha \neq 0$. Hence $\mathfrak{g} = \mathfrak{g}_0$, and $\mathfrak{g}$ is nilpotent. This contradicts the fact that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. □

Theorem 7.2.1. (Cartan's criterion). Let $\mathfrak{g}$ be a subalgebra of $\mathfrak{gl}(V)$, where V is a finite dimensional vector space over $\mathbb{F} = \bar{\mathbb{F}}$, a field of characteristic 0. Then the following are equivalent:

- (1) $(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}])_V = 0$, i.e. $(a, b)_V = 0$ for $a \in \mathfrak{g}, b \in [\mathfrak{g}, \mathfrak{g}]$.
- (2) $(a, a)_V = 0$ for all $a \in [\mathfrak{g}, \mathfrak{g}]$.
- (3) $\mathfrak{g}$ is solvable.

proof. (1) $\Rightarrow$ (2): Obvious.

(3) $\Rightarrow$ (1): By Lie's theorem, in some basis of V , all matrices of $\mathfrak{g}$ are upper triangular, and thus $[\mathfrak{g}, \mathfrak{g}]$ is strictly upper triangular. Thus $\phi(ab)$ is strictly upper triangular and $(a, b)_V = 0$ if $a \in \mathfrak{g}, b \in [\mathfrak{g}, \mathfrak{g}]$.

(2) $\Rightarrow$ (3): Suppose not. Then the derived series of $\mathfrak{g}$ stabilizes, so suppose $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] = \mathfrak{g}^{(k)}$ for some k with $\mathfrak{g}^{(k)} \neq 0$. Then $(a, a)_V = 0$ for $a \in \mathfrak{g}^{(k)}$ and $[\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] = \mathfrak{g}^{(k)}$. We reach the desired contradiction using Lemma 7.2.2. □

There is a direct corollary of this criterion:

Corollary 7.2.1. A finite dimensional Lie algebra $\mathfrak{g}$ over $\mathbb{F} = \bar{\mathbb{F}}$ ($\text{char } \mathbb{F} = 0$) is solvable iff $\kappa(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]) = 0$.

The next example shows that there are other non-degenerate symmetric invariant bilinear forms except for trace form.

¹Here, the α_0 is fixed.

Example 7.2.1. Consider the following 4-dim solvable Lie algebra $D = \text{Heis}_3 + \mathbb{F}d$, where $\text{Heis}_3 = \mathbb{F}p + \mathbb{F}q + \mathbb{F}c$, with the relations $[d, p] = p$, $[d, q] = -q$, $[d, c] = 0$. Define on D the bilinear form $B(p, q) = B(q, p) = B(c, d) = B(d, c) = 1$, rest = 0. Show that $B(,)$ is a non-degenerate symmetric invariant bilinear form, but $B(D, [D, D]) \neq 0$, so Cartan's criterion fails for this bilinear form.

proof. It is symmetric by construction. It is nondegenerate since if $a_1p + a_2q + a_3c + a_4d$ lies in the kernel, taking the bilinear form with q, p, d, c respectively gives $a_1 = a_2 = a_3 = a_4 = 0$. Cartan's criterion fails since $B(p, q) \neq 0$ and $p \in D, q \in [D, D]$. To check that it is invariant:

$$\begin{aligned} & B([a_1p + a_2q + a_3c + a_4d, b_1p + b_2q + b_3c + b_4d], c_1p + c_2q + c_3c + c_4d) \\ &= B((a_1b_2 - a_2b_1)c + (a_4b_1 - a_1b_4)p + (a_2b_4 - a_4b_2)q, c_1p + c_2q + c_3c + c_4d) \\ &= -(a_1b_2 - a_2b_1)c_4 + (a_4b_1 - a_1b_4)c_1 + (a_2b_4 - a_4b_2)c_1 \\ & B(a_1p + a_2q + a_3c + a_4d, [b_1p + b_2q + b_3c + b_4d, c_1p + c_2q + c_3c + c_4d]) \\ &= B(a_1p + a_2q + a_3c + a_4d, (b_1c_2 - b_2c_1)c + (b_4c_1 - b_1c_4)p + (b_2c_4 - b_4c_2)q) \\ &= a_1(b_2c_4 - b_4c_2) + a_2(b_4c_1 - b_1c_4) + a_4(b_1c_2 - b_2c_1) \end{aligned}$$

By comparison, it is clear $B([r, s], t) = B(r, [s, t])$ for $r = a_1p + a_2q + a_3c + a_4d$, $s = b_1p + b_2q + b_3c + b_4d$, $t = c_1p + c_2q + c_3c + c_4d$, so B is invariant, as required. $\square$

7.3 Some relax on base field

Very often one can remove the condition $\mathbb{F} = \bar{\mathbb{F}}$ by the following trick. Let $\mathbb{F} \subset \bar{\mathbb{F}}$ be the algebraic closure. Given a Lie algebra $\mathfrak{g}$ over $\mathbb{F}$, let $\bar{\mathfrak{g}} = \bar{\mathbb{F}} \otimes_{\mathbb{F}} \mathfrak{g}$ be a Lie algebra over $\bar{\mathbb{F}}$.¹

Proposition 7.3.1. 1. $\mathfrak{g}$ is solvable (resp. nilpotent) iff $\bar{\mathfrak{g}}$ is.

2. Derive Cartan's criterion and Corollary for $\text{char}\mathbb{F} = 0$ but not $\mathbb{F} = \bar{\mathbb{F}}$.

3. Show that $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent iff $\mathfrak{g}$ is solvable when $\text{char}\mathbb{F} = 0$.

4. $\mathfrak{g}_0^a$ is a Cartan sub-algebra for every regular element of $a \in \mathfrak{g}$ for any field $\mathbb{F}$.

proof. By construction of $\bar{\mathfrak{g}}$, if we pick a basis $a_1, \dots, a_n$ of $\mathfrak{g}$, so that $\mathfrak{g} = \mathbb{F}a_1 + \dots + \mathbb{F}a_n$, then $\bar{\mathfrak{g}} = \bar{\mathbb{F}}a_1 + \dots + \bar{\mathbb{F}}a_n$ with the same bracket relations holding.

1. Note $[\bar{\mathfrak{g}}, \bar{\mathfrak{g}}] = [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]$. To see this, $[\mathfrak{g}, \mathfrak{g}]$ is the $\mathbb{F}$ -span of $[a_i, a_j]$ with certain linear relations holding between them, so both $[\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]$ and $[\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]$ are the $\bar{\mathbb{F}}$ -span of $[a_i, a_j]$ with certain linear relations holding between them; and the Lie algebra structure is the same. Iterating this, we have $\overline{\mathfrak{g}^{(i)}} = \bar{\mathfrak{g}}^{(i)}$. So $\mathfrak{g}$ is solvable $\Leftrightarrow \exists i, \mathfrak{g}^{(i)} = 0 \Leftrightarrow \exists i, \bar{\mathfrak{g}}^{(i)} = 0 \Leftrightarrow \bar{\mathfrak{g}}$ is solvable. A similar argument shows that $\overline{[\mathfrak{g}, \mathfrak{g}]^{(i)}} = [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]^{(i)}$, and more generally, $\bar{\mathfrak{g}}_i = \bar{\mathfrak{g}}_i$. So $\mathfrak{g}$ is nilpotent $\Leftrightarrow \exists i, \mathfrak{g}^i = 0 \Leftrightarrow \exists i, \bar{\mathfrak{g}}^i = 0 \Leftrightarrow \bar{\mathfrak{g}}$ is nilpotent.

2. In Cartan's Criterion, note that the second condition $(a, a)_V = 0 \forall a \in [\mathfrak{g}, \mathfrak{g}]$ is equivalent to the condition $(a, b)_V = 0 \forall a, b \in [\mathfrak{g}, \mathfrak{g}]$, since $(\cdot, \cdot)_V$ is symmetric (to see this, expand $(a + b, a + b)_V = 0$ and note it is characteristic 0). Thus by Cartan's criterion, we have $(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}])_V = 0 \Leftrightarrow (\bar{\mathfrak{g}}, [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}])_V = 0 \Leftrightarrow (a, b)_V = 0 \forall a, b \in [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}] \Leftrightarrow (a, b)_V = 0 \forall a, b \in [\mathfrak{g}, \mathfrak{g}]$. So the first two conditions of Cartan's criterion are equivalent for $\text{char}\mathbb{F} = 0$. By Cartan's criterion, $(a, b)_V = 0 \forall a, b \in [\mathfrak{g}, \mathfrak{g}] \Leftrightarrow (a, b)_V = 0 \forall a, b \in [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}] \Leftrightarrow \bar{\mathfrak{g}}$ is solvable $\Leftrightarrow \mathfrak{g}$ is solvable, so the last two conditions of Cartan's criterion are equivalent for $\text{char}\mathbb{F} = 0$. For the Corollary, $\mathfrak{g}$ is solvable $\Leftrightarrow \bar{\mathfrak{g}}$ is solvable $\Leftrightarrow \kappa(\bar{\mathfrak{g}}, [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]) = 0 \Leftrightarrow \kappa(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]) = 0$, so the Corollary is true for all $\text{char}\mathbb{F} = 0$.

3. This was proven in a previous exercise when $\mathbb{F} = \bar{\mathbb{F}}$. So $\mathfrak{g}$ is solvable $\Rightarrow \bar{\mathfrak{g}}$ is solvable $\Rightarrow [\bar{\mathfrak{g}}, \bar{\mathfrak{g}}]$ is nilpotent $\Rightarrow [\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

¹For the lake of time, I will(?) make some changes in time. This part is just a copy of lecture notes of my teacher so far.

4. a is a regular element of $\mathfrak{g} \Rightarrow a$ is a regular element of $\bar{\mathfrak{g}}$ (since the discriminant of a is same in both $\mathfrak{g}$ and $\bar{\mathfrak{g}}$). Hence $\bar{\mathfrak{g}}_0^a$ is a Cartan subalgebra of $\bar{\mathfrak{g}}$, and hence $\mathfrak{g}_0^a$ is a Cartan subalgebra of $\mathfrak{g}$. To see the last step, a subalgebra is Cartan iff it is nilpotent and self-normalizing, and $\bar{\mathfrak{g}}_0^a = \bar{\mathfrak{g}}_0^a$, so since $\bar{\mathfrak{g}}_0^a$ is nilpotent, $\mathfrak{g}_0^a$ is nilpotent, and $N_{\bar{\mathfrak{g}}}(\bar{\mathfrak{g}}_0^a) = \bar{\mathfrak{g}}_0^a \Rightarrow N_{\mathfrak{g}}(\mathfrak{g}_0^a) = \mathfrak{g}_0^a$. $\square$

7.4 Some problems

1. Let $\mathbb{F}$ be a field of characteristic 0. Suppose $(\cdot, \cdot)$ be an invariant bilinear form on $\mathfrak{g}$. Show that if $v \in \mathfrak{g}$ is such that $\text{ad} v$ is nilpotent, then $(e^{\text{ad} v} a, e^{\text{ad} v} b) = (a, b)$. In other words, the bilinear form is invariant with respect to the group G generated by $e^{\text{ad} v}$, $\text{ad} v$ nilpotent.

proof. For $\text{ad} v$ is nilpotent, thus there are only finite terms in $e^{\text{ad} v}$, say $e^{\text{ad} v} = f(\text{ad} v)$. Then we have:

$$(e^{\text{ad} v} a, e^{\text{ad} v} b) = (f(\text{ad} v) a, f(\text{ad} v) b) = (a, b) + \sum_{i \geq 1} \sum_{j+k=i} \frac{1}{j!k!} ((\text{ad} v)^j a, (\text{ad} v)^k b)$$

Thus we only need to prove for $i \geq 1$, $\sum_{j+k=i} \frac{1}{j!k!} ((\text{ad} v)^j a, (\text{ad} v)^k b) = 0$. Notice that

$$((\text{ad} v)^j a, (\text{ad} v)^k b) = ((\text{ad} v)(\text{ad} v)^{j-1} a, (\text{ad} v)^k b) = -((\text{ad} v)^{j-1} a, (\text{ad} v)^{k+1} b)$$

Thus we have this formula and for $\sum_{k=0}^i (-1)^k \binom{i}{k} = 0$, we conclude the proof.

$$\sum_{j+k=i} \frac{1}{j!k!} ((\text{ad} v)^j a, (\text{ad} v)^k b) = \sum_{j+k=i} \frac{1}{j!k!} (-1)^k ((\text{ad} v)^i a, b) = \frac{1}{i!} ((\text{ad} v)^i a, b) \sum (-1)^k \binom{i}{k} = 0$$

$\square$

2. Compute the Killing form of $\mathfrak{sl}(2, \mathbb{C})$, using the standard basis consist of

$$\left\{ h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, a = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, b = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \right\}.$$

Check that the Killing form is nondegenerate.

proof. We just need to compute $\kappa(x, y)$ for $x, y = a, b, h$ and analyze the matrix under the basis a, b, h .

First we list the multiplication table of a, b, h : $[h, a] = 2a$, $[h, b] = -2b$ and $[a, b] = h$. Thus we can compute the representation matrices:

For h , we have $\text{ad}_h(h) = [h, h] = 0$, $\text{ad}_h(a) = [h, a] = 2a$, $\text{ad}_h(b) = [h, b] = -2b$, thus we have:

$$\text{ad}_h \begin{pmatrix} h \\ a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} h \\ a \\ b \end{pmatrix}$$

Samely, we have:

$$\text{ad}_a \begin{pmatrix} h \\ a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h \\ a \\ b \end{pmatrix} \quad \text{ad}_b \begin{pmatrix} h \\ a \\ b \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 2 & -1 & 0 \end{pmatrix} \begin{pmatrix} h \\ a \\ b \end{pmatrix}$$

Thus we can compute the killing form of h, a, b and thus have the Killing form is nondegenerate.

$$\kappa = \begin{pmatrix} \text{Tr}(\text{ad}_h^2) & \text{Tr}(\text{ad}_h \text{ad}_a) & \text{Tr}(\text{ad}_h \text{ad}_b) \\ \text{Tr}(\text{ad}_a \text{ad}_h) & \text{Tr}(\text{ad}_a^2) & \text{Tr}(\text{ad}_a \text{ad}_b) \\ \text{Tr}(\text{ad}_b \text{ad}_h) & \text{Tr}(\text{ad}_b \text{ad}_a) & \text{Tr}(\text{ad}_b^2) \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}$$

$\square$

8 The Radical and Semisimple Lie Algebras

8.1 Killing form and decomposition

Recall the radical of a finite-dimensional Lie algebra $\mathfrak{g}$, denoted $\text{Rad}\mathfrak{g}$, is the maximal solvable ideal of $\mathfrak{g}$. If $\mathfrak{g}$ is a finite dimensional solvable Lie algebra, then $\text{Rad}\mathfrak{g} = \mathfrak{g}$. The opposite case is when $\text{Rad}\mathfrak{g} = 0$, in which case $\mathfrak{g}$ is called semisimple.

We also remember that for any finite dimensional Lie algebra $\mathfrak{g}$, $\mathfrak{g}/\text{Rad}\mathfrak{g}$ is a semisimple Lie algebra. So an arbitrary finite dimensional Lie algebra “reduces” to a solvable Lie algebra $\text{Rad}\mathfrak{g}$ and a semisimple Lie algebra $\mathfrak{g}/\text{Rad}\mathfrak{g}$. In the case $\text{char}\mathbb{F} = 0$ a stronger result holds (We give it without a proof):

Theorem 8.1.1. (*Levi decomposition*) *If $\mathfrak{g}$ is a finite dimensional Lie algebra over a field $\mathbb{F}$ of characteristic 0, then there exists a semisimple subalgebra $\mathfrak{s} \subset \mathfrak{g}$, complementary to the radical $\text{Rad}\mathfrak{g}$, such that $\mathfrak{g} = \mathfrak{s} \oplus \text{Rad}\mathfrak{g}$ as vector spaces, where $\mathfrak{s}$ is called Levi subalgebra.*

Proposition 8.1.1. (1) *A finite dimensional Lie algebra $\mathfrak{g}$ is semisimple.*

(2) *Any solvable ideal of $\mathfrak{g}$ is 0.*

(3) *Any abelian ideal of $\mathfrak{g}$ is 0.*

(4) *Any nilpotent ideal of $\mathfrak{g}$ is 0.*

proof. (1) $\Leftrightarrow$ (2), (2) $\Rightarrow$ (3) is obvious.

(3) $\Rightarrow$ (2): Suppose that $\mathfrak{g}$ contains a non-zero solvable ideal $\mathfrak{r}$. For some k , we have $\mathfrak{r} \supset \mathfrak{r}^{(1)} \supset \mathfrak{r}^{(2)} \supset \dots \supset \mathfrak{r}^{(k)} = 0$, hence $\mathfrak{r}^{(k-1)}$ is a nonzero abelian ideal, since all the $\mathfrak{r}^{(i)}$ are ideals of $\mathfrak{g}$.

(4) $\Rightarrow$ (2) is obvious and similar to (3) $\Rightarrow$ (2) we can prove (3) $\Rightarrow$ (4). Thus complete the proof. $\square$

Definition 8.1.1. A decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}$ (direct sum of vector spaces), where $\mathfrak{h}$ is a subalgebra and $\mathfrak{r}$ is an ideal is called a semi-direct sum of $\mathfrak{h}$ and $\mathfrak{r}$ and is denoted by $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{r}$. The special case when $\mathfrak{h}$ is an ideal as well corresponds to the direct sum: $\mathfrak{g} = \mathfrak{h} \times \mathfrak{r} = \mathfrak{h} \oplus \mathfrak{r}$.

To prove theorem 8.1.2., we should first see two lemmas.

Lemma 8.1.1. *Let V be a finite-dimensional vector space with a symmetric bilinear form $(,)$. Let U be a subspace such that the restriction $(,)|_{U \times U}$ is non-degenerate. Denote $U^\perp = \{v \in V | (v, U) = 0\}$. Then $V = U \oplus U^\perp$. Moreover, if $(,)|_{U \times U}$ is degenerate, $V = U + U^\perp$, where the intersection of U and $U^\perp$ is nonempty. (which has been mentioned in linear algebra, i would like to ignore the proof.)*

Lemma 8.1.2. (which is the direct result of the former lemma.) *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and $(,)$ be a symmetric invariant bilinear form on $\mathfrak{g}$. Then:*

(1) *If $\mathfrak{a} \subset \mathfrak{g}$ is an ideal, then $\mathfrak{a}^\perp$ is also an ideal.*

(2) *If $(,)|_{\mathfrak{a} \times \mathfrak{a}}$ is non-degenerate, then $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$, a direct sum of Lie algebras.*

Then we have the most important theorem in this subsection:

Theorem 8.1.2. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $\mathbb{F} = \bar{\mathbb{F}}$, a field of characteristic 0. Then the Killing form on $\mathfrak{g}$ is non-degenerate if and only if $\mathfrak{g}$ is semisimple. Moreover, if $\mathfrak{g}$ is semisimple and $\mathfrak{a} \subset \mathfrak{g}$ is an ideal, then the restriction of the Killing form to $\mathfrak{a}$, $\kappa|_{\mathfrak{a} \times \mathfrak{a}}$, is also non-degenerate and coincides with the Killing form of $\mathfrak{a}$.*

proof. Suppose κ is non-degenerate on $\mathfrak{g}$, but $\mathfrak{g}$ is not semisimple. Hence there exists an abelian ideal $\mathfrak{a} \subset \mathfrak{g}$. But then $\kappa(\mathfrak{a}, \mathfrak{g}) = 0$. Indeed, if $x \in \mathfrak{g}$ and $y \in \mathfrak{a}$, then $\text{ad}x \text{ad}y(z) = [x, [y, z]] \in \mathfrak{a}$ for all $z \in \mathfrak{g}$ (and 0 for all $z \in \mathfrak{a}$). It follows that in the basis $e_1, \dots, e_k$ of $\mathfrak{a}$, $e_1, \dots, e_k, e_{k+1}, \dots, e_n$ basis of $\mathfrak{g}$, the matrix of $\text{ad}x \text{ad}y$ is of the form $\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$. But trace of this matrix is 0, so $\kappa(x, y) = 0$.

Conversely, let $\mathfrak{g}$ be semisimple. Let $\mathfrak{a}$ be an ideal of $\mathfrak{g}$. If $\kappa_{\mathfrak{a} \times \mathfrak{a}}$ is degenerate, so that $\mathfrak{a} \cap \mathfrak{a}^\perp \neq 0$, hence $\mathfrak{b} = \mathfrak{a} \cap \mathfrak{a}^\perp$ is an ideal of $\mathfrak{g}$ such that $\kappa(\mathfrak{b}, \mathfrak{b}) = 0$. By considering the adjoint representation of $\mathfrak{b}$ in $\mathfrak{g}$ and

applying the Cartan's criterion we conclude that $\mathfrak{b}$ is solvable. Since $\mathfrak{g}$ is semisimple, we deduce that $\mathfrak{b} = 0$. Thus if $\mathfrak{g}$ is semisimple, the Killing form is non-degenerate, by taking $\mathfrak{a} = \mathfrak{g}$.

As for the second part, we already proved that $\kappa|_{\mathfrak{a} \times \mathfrak{a}}$ is non-degenerate. Hence $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$. By Lemma 8.1.2, it is a direct sum of ideals, so $[\mathfrak{a}, \mathfrak{a}^\perp] = 0$. Hence κ for $\mathfrak{a}$ equals κ of $\mathfrak{g}$ restricted on $\mathfrak{a}$. (Or one can get this conclusion from Lemma 7.1.1. directly.) $\square$

Recall that a Lie algebra $\mathfrak{g}$ is called **simple** if its only ideals are 0 and $\mathfrak{g}$, and $\mathfrak{g}$ is not abelian. Then we have this corollary:

Corollary 8.1.1. *Any semisimple, finite-dimensional Lie algebra over a field $\mathbb{F}$ of characteristic 0 is a direct sum of simple Lie algebras. In fact, the simple Lie algebras are actually ideals of $\mathfrak{g}$.*

proof. We know that an ideal $\mathfrak{a}$ of a semisimple Lie algebra is still semisimple, by considering the killing form of $\mathfrak{a}$, which is exactly the killing form of $\mathfrak{g}$ restricted to $\mathfrak{a}$. It is of course non-degenerate, and from the former theorem, we proved this claim. If $\mathfrak{g}$ is semisimple, but not simple, and if $\mathfrak{a}$ is an ideal, then by the theorem, then $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$, where $\mathfrak{a}$ and $\mathfrak{a}^\perp$ are also semisimple. For $\dim \mathfrak{g}$ is finite, after finitely many steps it can be decomposed into simple algebras.

To prove the second result, we just need to verify the ideal of $\mathfrak{a}$, say $\mathfrak{h}$, is an ideal of $\mathfrak{g}$. For $x \in \mathfrak{h}, y \in \mathfrak{g}$, for $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$. Then we can assume $y = y_1 + y_2, y_1 \in \mathfrak{a}, y_2 \in \mathfrak{a}^\perp$. $[x, y] = [x, y_1 + y_2] = [x, y_1] + [x, y_2]$, where $[x, y_1] \in \mathfrak{h}, [x, y_2] \in \mathfrak{a} \cap \mathfrak{a}^\perp = 0$. Thus conclude the proof. $\square$

We can also relax the restriction on the field as follow:

Corollary 8.1.2. *Let $\mathfrak{g}$ denote a Lie algebra over $\mathbb{F}$ with $\text{char} \mathbb{F} = 0$. Show that $\mathfrak{g}$ is semisimple if and only if the Killing form on $\mathfrak{g}$ is nondegenerate.*

proof. When we proved the theorem 8.1.2, we used the algebraic closure of the base field $\mathbb{F}$ only to apply Cartan's criterion. We know from exercise 8.14, however, that Cartan's criterion is valid assuming merely that $\mathbb{F}$ has characteristic 0. So the same proof applies in this case. $\square$

8.2 Some problems

1、 Let $\mathfrak{h}$ and $\mathfrak{r}$ be Lie algebras and let $\gamma : \mathfrak{h} \rightarrow \text{Der}(\mathfrak{r})$ be a Lie algebra homomorphism. Let $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}$ be the direct sum of vector spaces and extend the bracket on $\mathfrak{h}$ and on $\mathfrak{r}$ to the whole of $\mathfrak{g}$ by letting

$$[h, r] = -[r, h] = \gamma(h)(r)$$

for $h \in \mathfrak{h}$ and $r \in \mathfrak{r}$. Show that this provides $\mathfrak{g}$ with a Lie algebra structure, $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{r}$, and that any semidirect sum of $\mathfrak{h}$ and $\mathfrak{r}$ is obtained in this way. Finally, show that $\mathfrak{g} = \mathfrak{h} \times \mathfrak{r}$ if and only if $\gamma = 0$.

proof. We need to verify that the bracket defined in this way is bilinear, anticommutative and satisfying Jacobi identity. For $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}$, then for all $x \in \mathfrak{g}, \exists h \in \mathfrak{h}, r \in \mathfrak{r}$, s.t. $x = h + r$. We can easily check that bilinear and anticommutative holds. As for the Jacobi identity, we have $[h_1, [r_1, r_2]] + [r_1, [r_2, h_1]] + [r_2, [h_1, r_1]] = \gamma(h_1)([r_1, r_2]) - [r_1, \gamma(h_1)(r_2)] + [r_2, \gamma(h_1)(r_1)] = [\gamma(h_1)(r_1), r_2] + [r_1, \gamma(h_1)(r_2)] - [r_1, \gamma(h_1)(r_2)] + [r_2, \gamma(h_1)(r_1)] = 0$, where we used the fact that $\gamma(h_1)$ is a derivation. $[h_1, [h_2, r_1]] + [h_2, [r_1, h_1]] + [r_1, [h_1, h_2]] = [h_1, \gamma(h_2)(r_1)] - [h_2, \gamma(h_1)(r_1)] - \gamma([h_1, h_2])(r_1) = \gamma(h_1)\gamma(h_2)(r_1) - \gamma(h_2)\gamma(h_1)(r_1) - (\gamma(h_1)\gamma(h_2)(r_1) - \gamma(h_2)\gamma(h_1)(r_1)) = 0$, where we used the fact that $\gamma : \mathfrak{h} \rightarrow \text{Der}(\mathfrak{r})$ is a Lie algebra homomorphism. For general case, we can prove first for $x_1, h, r, x_1, h_1, h_2, x_1, r_1, r_2$ then for $x_1, x_2, h/r$, then for x_1, x_2, x_3 , in the process using the bilinear and anticommutative. Thus this gives a Lie bracket.

$\mathfrak{r}$ is an ideal of $\mathfrak{g}$, for $x \in \mathfrak{g}$, say $x = h + r, h \in \mathfrak{h}, r \in \mathfrak{r}$. Then $[x, r_1] = [h, r_1] + [r, r_1] = \gamma(h)(r_1) + [r, r_1] \in \mathfrak{r}$. Thus we have $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{r}$.

If $\mathfrak{h}$ is also an ideal of $\mathfrak{g}$, then for all $h \in \mathfrak{h}, r \in \mathfrak{r}, [h, r] \in \mathfrak{h} \cap \mathfrak{r}$, i.e. $[h, r] = 0, \forall h \in \mathfrak{h}, r \in \mathfrak{r}$. Thus $\gamma = 0$. $\square$

2、 Recall that a semisimple Lie algebra $\mathfrak{g} = \bigoplus_{j=1}^N \mathfrak{g}_j$ where $\mathfrak{g}_j$ are simple Lie algebras. Prove that this decomposition is unique up to permutation of the summands and any ideal $\mathfrak{a}$ of $\mathfrak{g}$ is a subsum of this sum.

proof. We first prove the second part, take any ideal $\mathfrak{a}$, then $\mathfrak{a} \cap \mathfrak{g}_j$ is an ideal of the subalgebra $\mathfrak{g}_j$. $\mathfrak{g}_j$ is simple so this is either 0 or $\mathfrak{g}_j$. So

$$\mathfrak{a} = \bigoplus_{j=1}^N (\mathfrak{a} \cap \mathfrak{g}_j) = \mathfrak{g}_{i_1} \oplus \mathfrak{g}_{i_2} \oplus \cdots \oplus \mathfrak{g}_{i_s}, \quad \mathfrak{g}_{i_j} \in \{\mathfrak{g}_1, \mathfrak{g}_2, \dots, \mathfrak{g}_N\},$$

which proves the second part.

For the first part, consider a second decomposition $\mathfrak{g} = \bigoplus_{i=1}^M \mathfrak{t}_i$. Each $\mathfrak{t}_i$ is an ideal and therefore is the direct sum of a subset of the $\mathfrak{g}_j$. $\mathfrak{t}_i$ is simple, so it must be the direct sum of exactly one $\mathfrak{g}_j$. Similarly, each $\mathfrak{g}_j$ is the direct sum of exactly one $\mathfrak{t}_i$. This proves the first part. $\square$

3、 There are some conclusions concerning radical and semisimple Lie algebra:

(1) if $\mathfrak{h}$ is an abelian ideal of $\mathfrak{g}$, then $\mathfrak{h} \subseteq \mathfrak{g}^\perp$.

(2) $\mathfrak{g}$ is a solvable Lie algebra iff $\mathfrak{g}^{(1)} \subseteq \mathfrak{g}^\perp$.

(3) $\mathfrak{g}$ is a semisimple Lie algebra, then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

sketch of proof. (1) Choose a basis of $\mathfrak{h}$ $x_1, \dots, x_k$, then extend them to a basis of $\mathfrak{g}$ $x_1, \dots, x_k, \dots, x_n$. To prove $\mathfrak{h} \subseteq \mathfrak{g}^\perp = \{x \in \mathfrak{g} \mid \kappa(x, y) = \text{tr}(\text{ad}_x \text{ad}_y) = 0, \forall y \in \mathfrak{g}\}$, we consider the matrix of $\mathfrak{h}$ on the basis $x_1, \dots, x_k, \dots, x_n$: $\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$.

(2) From Lie theorem, we know that $\mathfrak{g}$ is a solvable Lie algebra iff $\mathfrak{g}^{(1)}$ is nilpotent.

($\Rightarrow$) Similarly, the matrix of $\mathfrak{h}$ on the basis $x_1, \dots, x_k, \dots, x_n$ is: $\begin{pmatrix} A & * \\ 0 & 0 \end{pmatrix}$, where A is nilpotent.

($\Leftarrow$) Consider the matrix, then we can prove $\mathfrak{g}^{(1)}$ is nilpotent by using Engel's theorem.

We can prove (2) also by Cartan criterion: $\mathfrak{g}$ solvable iff $\kappa(\mathfrak{g}, \mathfrak{g}^{(1)}) = 0$ iff $\mathfrak{g}^{(1)} \subseteq \mathfrak{g}^\perp$.

(3) From 2, we know that $\mathfrak{g} = \bigoplus_{j=1}^N \mathfrak{g}_j$, then $[\mathfrak{g}, \mathfrak{g}] = [\bigoplus_{j=1}^N \mathfrak{g}_j, \mathfrak{g}] = \sum_{j=1}^N [\mathfrak{g}_j, \mathfrak{g}]$. For $\mathfrak{g}_i$ is simple ideal of $\mathfrak{g}$, thus $[\mathfrak{g}_i, \mathfrak{g}] = \mathfrak{g}_i$ or 0. If $[\mathfrak{g}_i, \mathfrak{g}] = 0$, then $\kappa(x, \mathfrak{g}) = 0, \forall x \in \mathfrak{g}_i$, contradicts to κ is non-degenerate. Thus $[\mathfrak{g}_i, \mathfrak{g}] = \mathfrak{g}_i$ and $\sum_{j=1}^N [\mathfrak{g}_j, \mathfrak{g}] = \sum_{j=1}^N \mathfrak{g}_j = \mathfrak{g}$. $\square$

9 Structure Theory of Semisimple Lie Algebras (I)

In this lecture and the few that follow, we will study the structure of finite dimensional semisimple Lie algebras with the aim of classifying them. This needs a detailed knowledge of root space decompositions. The first step is to use the Killing form to understand Cartan subalgebras and their actions under the adjoint representation.

Unless otherwise stated, we will assume throughout that our base field $\mathbb{F}$ is algebraically closed of characteristic zero.

9.1 Abstract Jordan decomposition

We begin by generalizing our notion of Jordan decomposition to an arbitrary Lie algebra.

Definition 9.1.1. An **abstract Jordan decomposition** of an element a of a Lie algebra $\mathfrak{g}$ is a decomposition of the form $a = a_s + a_n$, where

- (1) $\text{ad}a_s$ is a diagonalizable (equivalently semisimple) endomorphism of $\mathfrak{g}$.
- (2) $\text{ad}a_n$ is a nilpotent endomorphism.
- (3) $[a_s, a_n] = 0$.

Proposition 9.1.1. *Say the abstract Jordan decomposition in a Lie algebra $\mathfrak{g}$ exists, then it is unique iff $C(\mathfrak{g}) = 0$. (For the counterexample, one just need to consider $\mathfrak{t}(2, F)$.)*

proof. $(\Rightarrow)$ Suppose $c \in C(\mathfrak{g})$ is not 0, and that $a \in \mathfrak{g}$ has abstract Jordan decomposition $a = a_s + a_n$. Let $a'_s = a_s - c$ and $a'_n = a_n + c$. We note the following facts: $a = (a_s - c) + (a_n + c) = a'_s + a'_n$; $[a'_s, a'_n] = [a_s, a_n] = 0$; $\text{ad}a'_s = \text{ad}a_s$ and $\text{ad}a'_n = \text{ad}a_n$ as $C(\mathfrak{g}) = \ker \text{ad}$. Since $\text{ad}a_n$ is nilpotent, and $\text{ad}a_s$ is semisimple, we conclude that $a = a'_s + a'_n$ is an abstract Jordan decomposition. But $a'_s \neq a_s$ as $c \neq 0$. So the decomposition for a is not unique.

$(\Leftarrow)$ Suppose that $a = a_s + a_n = a'_s + a'_n$ are abstract Jordan decompositions for some $a \in \mathfrak{g}$. Since $\text{ad}(a_s + a_n)$ and $\text{ad}(a'_s + a'_n)$ are commuting operators (they are the same thing.), and $\text{ada} = \text{ad}a_s + \text{ad}a_n$ and $\text{ada} = \text{ad}a'_s + \text{ad}a'_n$ are Jordan decompositions, we know from Proposition 5.1.2., that

$$[\text{ad}a'_s, \text{ad}a_s + \text{ad}a_n] = 0, [\text{ad}a'_n, \text{ad}a_s + \text{ad}a_n] = 0$$

and using the proposition again:

$$[\text{ad}a'_s, \text{ad}a_s] = 0, [\text{ad}a'_n, \text{ad}a_n] = 0.$$

Since the sum of commuting semisimple (resp. nilpotent) operators is semisimple (resp. nilpotent),

$$\text{ad}a'_s - \text{ad}a_s = \text{ad}a_n - \text{ad}a'_n \Rightarrow \text{ad}a'_s - \text{ad}a_s = 0, \text{ad}a_n - \text{ad}a'_n = 0.$$

As $\ker \text{ad} = C(\mathfrak{g}) = 0$ we conclude that $a_n = a'_n$ and $a_s = a'_s$. □

Our first theorem in this section is to show the existence of abstract Jordan decompositions under the assumption of semisimplicity.

Theorem 9.1.1. *Let $\mathfrak{g}$ be a finite dimensional semisimple Lie algebra over $\mathbb{F}$.*

- (1) $C(\mathfrak{g}) = 0$.
- (2) All derivations of $\mathfrak{g}$ are inner. (The proof method by considering $\text{Der} \mathfrak{g}$ is worth learning.)
- (3) Any $a \in \mathfrak{g}$ admits a unique abstract Jordan decomposition. (which is exactly the classical Jordan decomposition of $\text{ad}a$).

proof. (1) $\mathfrak{g}$ has no nontrivial abelian ideals. Hence $C(\mathfrak{g}) = 0$.

(2) Consider homomorphism $\text{ad} : \mathfrak{g} \rightarrow \text{Der} \mathfrak{g}$ with kernel $C(\mathfrak{g})$. Since $C(\mathfrak{g}) = 0$, $\mathfrak{g} \cong \text{ad} \mathfrak{g}$.

Consider the trace form on Derg under the standard representation. The restriction to $\mathfrak{adg}$ is the Killing form on $\mathfrak{g}$, which is non-degenerate by semisimplicity. From Lemma 8.1.2.(2), we have that $\text{Derg} = \mathfrak{adg} \oplus (\mathfrak{adg})^\perp$ as a direct sum of Lie algebras.

We need to show that $(\mathfrak{adg})^\perp = 0$. In the contrary case, there exists a nonzero element $D \in (\mathfrak{adg})^\perp$. For $a \in \mathfrak{g}$, we have $[D, \mathfrak{ada}] = 0$. We recall that $[D, \mathfrak{ada}] = \mathfrak{ad}D(a)$. Hence $D(a) \in C(\mathfrak{g})$ for any $a \in \mathfrak{g}$. Since $C(\mathfrak{g}) = 0$, we conclude that $D = 0$ in contradiction to our assumption.

(3) Fix $a \in \mathfrak{g}$. Consider the classical Jordan decomposition of $\mathfrak{ada} = A_s + A_n$, where A_s is diagonalizable, A_n is nilpotent, and $[A_s, A_n] = 0$. Then we need to find a_s, a_n satisfying $A_s = \mathfrak{ada}_s, A_n = \mathfrak{ada}_n$. Then for $\mathfrak{ada} = \mathfrak{ada}_s + \mathfrak{ada}_n$, $\mathfrak{ad}(a - a_s - a_n) = 0$, we have $a = a_s + a_n$. For $0 = [A_s, A_n] = [\mathfrak{ada}_s, \mathfrak{ada}_n] = \mathfrak{ad}[a_s, a_n]$, $[a_s, a_n] \in \ker \mathfrak{ad} = C(\mathfrak{g})$. Thus $[a_s, a_n] = 0$.

Consider the generalized eigenspace decomposition of $\mathfrak{g}$ with respect to $\mathfrak{ada}$: $\mathfrak{g} = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_\lambda^a$. Recall that $\mathfrak{g}_\lambda^a = \{x \in \mathfrak{g} \mid (\mathfrak{ada} - \lambda I)^N x = 0 \text{ for some } N \in \mathbb{N}\}$ and $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$ with $A_s|_{\mathfrak{g}_\lambda} = \lambda I$. We want first to show that A_s is a derivation. By linearity, it is enough to check this for $x \in \mathfrak{g}_\lambda^a$ and $y \in \mathfrak{g}_\mu^a$. We have

$$A_s([x, y]) = (\lambda + \mu) \cdot [x, y] = [\lambda x, y] + [x, \mu y] = [A_s x, y] + [x, A_s y].$$

This means that $A_s \in \text{Derg}$. Then by part (2), all derivations are inner. So there exists $a_s \in \mathfrak{g}$ such that $\mathfrak{ada}_s = A_s$. Letting $a_n = a - a_s$, we have that $\mathfrak{ada}_n = A_n$. $\square$

9.2 Root

Theorem 9.2.1. Choose a Cartan subalgebra $\mathfrak{h}$ of $\mathfrak{g}$, and consider the root space decomposition $\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_\alpha$ with $\mathfrak{g}_0 = \mathfrak{h}$, where $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$. κ denote the Killing form on $\mathfrak{g}$.

(1) $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ if $\alpha + \beta \neq 0$.

(2)* $\kappa|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ is nondegenerate. Consequently $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate.

(3) $\mathfrak{h}$ is an abelian subalgebra of $\mathfrak{g}$.

(4)* $\mathfrak{h}$ consists of semisimple elements; namely, each $\mathfrak{adh}$ for $h \in \mathfrak{h}$ is diagonalizable. Consequently $\mathfrak{g}_\alpha = \{a \in \mathfrak{g} \mid [h, a] = \alpha(h)a, \forall h \in \mathfrak{h}\}$.

proof. (1) Let $a \in \mathfrak{g}_\alpha$ and $b \in \mathfrak{g}_\beta$ with $\alpha + \beta \neq 0$. Note that

$$(\mathfrak{adaadb})^N \mathfrak{g}_\gamma \subset \mathfrak{g}_{\gamma+N\alpha+N\beta} = 0$$

for $N \gg 0$ as $\mathfrak{g}$ is finite dimensional. So we can choose N sufficiently large so that the operator $(\mathfrak{adaadb})^N$ is zero on each summand of the decomposition. Therefore $\mathfrak{adaadb}$ is nilpotent. So the eigenvalues of $\mathfrak{adaadb}$ are zero, implying that $\kappa(a, b) = \text{tr}_{\mathfrak{g}}(\mathfrak{adaadb}) = 0$.

(2) By semisimplicity, we know that κ is non-degenerate on $\mathfrak{g}$. By part (1), we have $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ for $\alpha \neq -\beta$. So necessarily $\kappa|_{\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha}}$ is nondegenerate. Otherwise $\exists x \in \mathfrak{g}_\alpha, \kappa(x, \mathfrak{g}) = 0, x \in \mathfrak{g}^\perp$. Then take $\alpha = 0$, we get $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate.

(3) Consider $\mathfrak{adh}$, and note that $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is the trace form on $\mathfrak{adh}$ under the standard representation. As $\mathfrak{h}$ is solvable (indeed nilpotent¹), we know that $\mathfrak{adh}$ is solvable (which is given by the root space decomposition, i.e. every element in $\mathfrak{adh}$ is in $\mathfrak{t}(n, \mathbb{F})$ after a similar transformation.). So by Cartan's criterion, we find that $0 = \text{tr}_{\mathfrak{g}}(\mathfrak{adh}, [\mathfrak{adh}, \mathfrak{adh}]) = \kappa(\mathfrak{h}, [\mathfrak{h}, \mathfrak{h}])$. By part (2), however, $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate. Therefore we conclude that the derived subalgebra $[\mathfrak{h}, \mathfrak{h}]$ is zero.

(4) Consider $h \in \mathfrak{h}$. By Theorem 9.1.1. (3), h has an abstract Jordan decomposition of the form $h = h_s + h_n$, where $h_s, h_n \in \mathfrak{g}$ are such that $\mathfrak{adh}_s$ is diagonalizable, $\mathfrak{adh}_n$ is nilpotent, and $[h_s, h_n] = 0$.

By part (3), $[h, \mathfrak{h}] = 0$. Hence for $h' \in \mathfrak{h}$, we have $0 = \mathfrak{ad}([h', h]) = [\mathfrak{adh}', \mathfrak{adh}]$, i.e. $\mathfrak{adh}$ commutes with all $\mathfrak{adh}'$, $\forall h' \in \mathfrak{h}$. So we know that $[\mathfrak{adh}', (\mathfrak{adh})_s] = 0$, yielding $0 = [\mathfrak{adh}', (\mathfrak{adh})_s] = [\mathfrak{adh}', \mathfrak{adh}_s] =$

¹We should notice that $\mathfrak{ad}_{\mathfrak{h}}\mathfrak{h}$ is truly nilpotent, but $\mathfrak{ad}_{\mathfrak{g}}\mathfrak{h}$ is not necessarily nilpotent.

$\mathbf{ad}([h', h_s])$. By Theorem 9.1.1., $C_{\mathfrak{g}}(\mathfrak{h}) \subseteq C(\mathfrak{g}) = 0$, this implies $[h', h_s] = 0$ for $h' \in \mathfrak{h}$. But since $\mathfrak{h}$ is a Cartan subalgebra, we conclude that $h_s \in \mathfrak{g}_0^{\mathfrak{h}} = \mathfrak{h}$.

It remains to be shown that $h_n = 0$. Note that $h_n = h - h_s \in \mathfrak{h}$.

Since $\mathbf{ad}\mathfrak{h}$ is solvable, Lie's theorem implies that there exists a basis of $\mathfrak{g}$ such that the elements of $\mathbf{ad}\mathfrak{h}$ are upper triangular. As $\mathbf{ad}h_n$ is nilpotent, it is strictly upper triangular. Therefore $\mathrm{tr}(\mathbf{ad}h' \mathbf{ad}h_n) = 0$ for $h' \in \mathfrak{h}$; namely $\kappa(\mathfrak{h}, h_n) = 0$. By part (2), $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is nondegenerate. So $h_n = 0$, implying that $h = h_s$ is diagonalizable. $\square$

For arbitrary Lie algebras, Cartan subalgebras can have an unwieldy structure, and an unpredictable action under the adjoint representation. We see from the previous result, however, that with the assumption of semisimplicity the situation is much more transparent. We can rewrite the root space decomposition as

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha} \quad (9.1)$$

where Δ is the set of all $\alpha \neq 0$ such that $\mathfrak{g}_{\alpha} \neq 0$. Recall that $\mathfrak{g}_{\alpha} = \{a \in \mathfrak{g} \mid [h, a] = \alpha(h)a, \forall h \in \mathfrak{h}\}$.

Definition 9.2.1. We call $\alpha \in \Delta$ a **root** of $\mathfrak{g}$, and $\mathfrak{g}_{\alpha}$ the corresponding **root space**.

9.3 The killing form on root Δ

Having put the Killing form to good use in determining the structure of Cartan subalgebras, we would like to extend it to Δ . We have a canonical linear map

$$\nu : \mathfrak{h} \ni h \mapsto \kappa(h, \cdot) \in \mathfrak{h}^*.$$

Since $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is nondegenerate by Theorem 9.2.1 (2), this implies ν is injective. As $\mathfrak{h}$ and $\mathfrak{h}^*$ have the same dimension, ν determines a vector space isomorphism.

Definition 9.3.1. A bilinear form $\kappa : \mathfrak{h}^* \times \mathfrak{h}^* \rightarrow \mathbb{F}$ by the rule $\kappa(\gamma, \mu) = \kappa(\nu^{-1}(\gamma), \nu^{-1}(\mu))$. Note that $\kappa(\nu(h), \nu(h')) = \kappa(h, h') = \nu(h)(h') = \nu(h')(h)$.

Theorem 9.3.1. (1) If $\alpha \in \Delta$, $e \in \mathfrak{g}_{\alpha}$, $f \in \mathfrak{g}_{-\alpha}$, then $[e, f] = \kappa(e, f)\nu^{-1}(\alpha)$. Especially, $[\mathfrak{g}_{\alpha}, \mathfrak{g}_{-\alpha}] \neq 0$.

(2) If $\alpha \in \Delta$, then $\kappa(\alpha, \alpha) \neq 0$.

proof. (1) We know that $[e, f] \in \mathfrak{h}$. By Theorem 9.2.1 (2), $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is nondegenerate. So it is enough to show that $\kappa([e, f] - \kappa(e, f)\nu^{-1}(\alpha), h') = 0$ for any $h' \in \mathfrak{h}$. We have

$$\begin{aligned} \kappa([e, f] - \kappa(e, f)\nu^{-1}(\alpha), h') &= \kappa([e, f], h') - \kappa(e, f)\nu(\nu^{-1}(\alpha))(h') \\ &= -\kappa(e, [h', f]) - \kappa(e, f)\alpha(h') \\ &= \alpha(h')\kappa(e, f) - \kappa(e, f)\alpha(h') \\ &= 0. \end{aligned}$$

Note that we have used the invariance of κ , and the fact that $h' \in \mathfrak{h}$, $f \in \mathfrak{g}_{-\alpha}$. This gives the result.

(2) By Theorem 9.2.1. (2), $\kappa|_{\mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha}}$ is non-degenerate. So we can find $e \in \mathfrak{g}_{\alpha}$, $f \in \mathfrak{g}_{-\alpha}$ such that $\kappa(e, f) = 1$. By part (1), we know that $[e, f] = \nu^{-1}(\alpha)$. Hence

$$[\nu^{-1}(\alpha), e] = \alpha(\nu^{-1}(\alpha))e = \kappa(\alpha, \alpha)e$$

and similarly

$$[\nu^{-1}(\alpha), f] = -\alpha(\nu^{-1}(\alpha))f = -\kappa(\alpha, \alpha)f.$$

Suppose to the contrary that $\kappa(\alpha, \alpha) = 0$. By the above relations, we obtain a Lie algebra

$$\mathfrak{a} = \mathbb{F}e + \mathbb{F}f + \mathbb{F}\nu^{-1}(\alpha)$$

isomorphic to Heis_3 . Recall that Heis_3 is solvable with center $[\text{Heis}_3, \text{Heis}_3]$. So applying Lie's theorem to the restriction of the adjoint representation to $\mathfrak{a}$, we can find a basis of $\mathfrak{g}$ such that $\nu^{-1}(\alpha) = [\mathfrak{a}, \mathfrak{a}]$ is strictly upper triangular. But by Theorem 9.2.1. (4), we know that $\nu^{-1}(\alpha) \in \mathfrak{h}$ is diagonalizable. So we conclude that $\nu^{-1}(\alpha) = 0$, and $\alpha = 0$. This contradicts definition of root, and the fact that $\alpha \in \Delta$. $\square$

9.4 Some problems

1、 If $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ is a decomposition of a Lie algebra $\mathfrak{g}$ into a direct sum of ideals, then $\mathfrak{h}$ is a Cartan subalgebra of $\mathfrak{g}$ if and only if $\mathfrak{h} = \mathfrak{h} \cap \mathfrak{g}_1 \oplus \mathfrak{h} \cap \mathfrak{g}_2$, and $\mathfrak{h} \cap \mathfrak{g}_i$ is a Cartan subalgebra of $\mathfrak{g}_i$ ($i = 1, 2$).

proof. Denote $\mathfrak{h}_i = \mathfrak{h} \cap \mathfrak{g}_i, i = 1, 2$.

($\Rightarrow$) For $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, every element h in $\mathfrak{h}$ can be written as $h = h_1 + h_2, h_1 \in \mathfrak{h}_1, h_2 \in \mathfrak{h}_2$. Thus $\mathfrak{h} = \mathfrak{h}_1 + \mathfrak{h}_2$, for the intersection of $\mathfrak{g}_1$ and $\mathfrak{g}_2$ is 0, so the sum is a direct sum. For $\mathfrak{h}_i$ are subalgebras of $\mathfrak{h}$, $\mathfrak{h}_i$ are also nilpotent. For $x \in N_{\mathfrak{g}_1}(\mathfrak{h}_1), [x, \mathfrak{h}_1] \subseteq \mathfrak{h}_1$. For $x \in \mathfrak{g}_1, [x, \mathfrak{h}_2] \subseteq \mathfrak{g}_1 \cap \mathfrak{g}_2 = 0$. Thus $[x, \mathfrak{h}_2] = 0$. $[x, \mathfrak{h}] \subseteq [x, \mathfrak{h}_1] + [x, \mathfrak{h}_2] \subseteq \mathfrak{h}$, i.e. $x \in N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. Thus $x \in \mathfrak{h}_1$, i.e. $N_{\mathfrak{g}_1}(\mathfrak{h}_1) \subseteq \mathfrak{h}_1$. Thus conclude the proof.

($\Leftarrow$) For $x \in \mathfrak{h}, x = x_1 + x_2$, where $x_1 \in \mathfrak{h}_1, x_2 \in \mathfrak{h}_2$. $[x_2, \mathfrak{h}_1] \subseteq \mathfrak{g}_1 \cap \mathfrak{g}_2 = 0$, thus $[x, \mathfrak{h}_1] = [x_1, \mathfrak{h}_1] + [x_2, \mathfrak{h}_1] = [x_1, \mathfrak{h}_1] \subseteq \mathfrak{h}_1$. Thus $\mathfrak{h}_i$ are nilpotent ideals of $\mathfrak{h}$. So $\mathfrak{h}$ is nilpotent. We know that $N_{\mathfrak{g}}(\mathfrak{h}) \subseteq N_{\mathfrak{g}}(\mathfrak{h}_i)$, thus $N_{\mathfrak{g}}(\mathfrak{h}) \subseteq N_{\mathfrak{g}}(\mathfrak{h}_1) \cap N_{\mathfrak{g}}(\mathfrak{h}_2)$. For $x \in N_{\mathfrak{g}}(\mathfrak{h}_1), x = x_1 + x_2, x_i \in \mathfrak{g}_i$. For $[\mathfrak{g}_2, \mathfrak{h}_1] = 0$, we have $x_1 \in N_{\mathfrak{g}_1}(\mathfrak{h}_1) = \mathfrak{h}_1$ and $N_{\mathfrak{g}}(\mathfrak{h}_1) \subseteq \mathfrak{h}_1 \oplus \mathfrak{g}_2$. Obviously, $\mathfrak{h}_1 \oplus \mathfrak{g}_2 \subseteq N_{\mathfrak{g}}(\mathfrak{h}_1)$, thus $N_{\mathfrak{g}}(\mathfrak{h}_1) = \mathfrak{h}_1 \oplus \mathfrak{g}_2$. So $N_{\mathfrak{g}}(\mathfrak{h}) \subseteq (\mathfrak{h}_1 \oplus \mathfrak{g}_2) \cap (\mathfrak{g}_1 \oplus \mathfrak{h}_2) = \mathfrak{h}_1 \oplus \mathfrak{h}_2 = \mathfrak{h}$. Thus conclude the proof. $\square$

RMK: When $\mathfrak{h}$ is a CSA, we can get $\mathfrak{h}_i$ are ideals of $\mathfrak{h} = N_{\mathfrak{g}}(\mathfrak{h})$, for an element x in $\mathfrak{h}, [x, \mathfrak{h}_1] \subseteq [x, \mathfrak{h}] \cap [x, \mathfrak{g}_1] \subseteq \mathfrak{h} \cap \mathfrak{g}_1 = \mathfrak{h}_1$. Thus the sum of $[x, \mathfrak{h}_1]$ and $[x, \mathfrak{h}_2]$ is a direct sum. And $N_{\mathfrak{g}}(\mathfrak{h}) = N_{\mathfrak{g}}(\mathfrak{h}_1) \cap N_{\mathfrak{g}}(\mathfrak{h}_2)$.

RMK2: This problem is of highly importance, for it reveals the relation between the CSA of a semisimple Lie algebra $\mathfrak{g}$ and its simple ideal decomposition.

2、 Let $n > 1$. Let (X, Y) denote the Killing form on $\mathfrak{sl}(n, \mathbb{C})$.

(1) Write down the Killing form and use it to determine whether $\mathfrak{sl}(n, \mathbb{C})$ is semisimple.

(2) Find a Cartan subalgebra $\mathfrak{h}$ of $\mathfrak{sl}(n, \mathbb{C})$ and give the root space decomposition with respect to $\mathfrak{h}$.

solution. (1) From the example above, we know that $\kappa_{\mathfrak{sl}_n}(x, y) = 2n\text{tr}(xy)$. To prove $\mathfrak{sl}(n, \mathbb{C})$ is semisimple, we just need to prove the killing form on $\mathfrak{sl}(n, \mathbb{C})$ is non-degenerate. Consider the basis of $\mathfrak{sl}(n, \mathbb{C})$: $e_{ij}, i \neq j$ and $e_{ii} - e_{jj}, i \neq j$. For $x \in \mathfrak{sl}(n, \mathbb{C})$, if $\exists e_{ij}$, the coefficient of e_{ij} is $a_{ij} \neq 0$, then $\kappa(x, e_{ji}) = 2n\text{tr}(xe_{ji}) = 2na_{ij} \neq 0$. If $x = \sum a_i e_{ii}$, where $\sum a_i = 0$, then assume the first positive coefficient is a_1 and the first negative coefficient is a_2 , then $\kappa(x, a_1 e_{11} - a_2 e_{22}) > 0$. Thus $\mathfrak{sl}(n, \mathbb{C})$ is non-degenerate.

(2) Consider $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n) \mid a_i \in \mathbb{C}, \sum_{i=1}^n a_i = 0\}$, which is nilpotent and self-normalizing. Thus $\mathfrak{h}$ is a CSA of $\mathfrak{sl}(n, \mathbb{C})$. Let $h \in \mathfrak{h}$ act on $\mathfrak{g}$ by the adjoint representation: $\text{ad}_h(x) = [h, x]$. When $h = \text{diag}(a_1, \dots, a_n)$, we have $\text{ad}_h(e_{ij}) = (a_i - a_j)e_{ij}$. Thus e_{ij} is a root vector with the weight $\alpha_{ij}(h) = a_i - a_j$. Thus $\Delta = \{\alpha_{ij} = \varepsilon_i - \varepsilon_j \mid 1 \leq i \neq j \leq n\}$ and $\mathfrak{sl}(n, \mathbb{C}) = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$ is the root space decomposition. $\square$

3、 Find DerHeis_3 . Note that not all derivations are inner.¹

solution. Here the brackets are given by $[e_1, e_2] = -[e_2, e_1] = e_3$, where (e_1, e_2, e_3) denotes a basis. The

¹The solution is cited from mse.

inner derivations are of the form $\mathbf{ad}(x)$, and are linear combinations of

$$\mathbf{ad}(e_1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \mathbf{ad}(e_2) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \mathbf{ad}(e_3) = 0.$$

However, the Heisenberg Lie Algebra has many other derivations (outer derivations). In fact, all linear maps

of the form $D = \begin{pmatrix} d_1 & d_4 & 0 \\ d_2 & d_5 & 0 \\ d_3 & d_6 & d_1 + d_5 \end{pmatrix}$ are derivations of the Heisenberg Lie algebra. □

10 Structure Theory of Semisimple Lie Algebras (II)

10.1 Review and some further results of (I)

Throughout this lecture, let $\mathfrak{g}$ be a finite dimensional semisimple Lie Algebra over an algebraically closed field $\mathbb{F}$ of characteristic 0. So far, we have proved:

1. The Killing form κ of $\mathfrak{g}$ is non-degenerate.
2. The algebra $\mathfrak{g}$ contains a Cartan subalgebra $\mathfrak{h}$. Furthermore, $\mathfrak{h}$ is abelian and diagonalizable on $\mathfrak{g}$, and we have:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$$

where,

$$\mathfrak{g}_{\alpha} = \{a \in \mathfrak{g} \mid [h, a] = \alpha(h)a \text{ for all } h \in \mathfrak{h}\},$$

$$\Delta = \{\alpha \in \mathfrak{h}^* \mid \alpha \neq 0 \text{ and } \mathfrak{g}_{\alpha} \neq 0\},$$

$$[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subseteq \mathfrak{g}_{\alpha+\beta}, \text{ which is } 0 \text{ if } \alpha + \beta \notin \Delta \cup \{0\}, \text{ where } \mathfrak{g}_0 = \mathfrak{h}.$$

3. The restriction $\kappa|_{\mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha}}$, i.e. $\kappa(a, b)$ with $a \in \mathfrak{g}_{\alpha}$ and $b \in \mathfrak{g}_{-\alpha}$, is nondegenerate, induces a pairing between $\mathfrak{g}_{\alpha}$ and $\mathfrak{g}_{-\alpha}$. In particular, we have

$$\dim \mathfrak{g}_{\alpha} = \dim \mathfrak{g}_{-\alpha}.$$

One just need to consider $f : \mathfrak{g}_{\alpha} \rightarrow \mathfrak{g}_{-\alpha}^*$ with $f(x) = \kappa(x, \cdot)$ and prove f is an isomorphism. Note that $\mathfrak{g}_{\alpha}$ and $\mathfrak{g}_{-\alpha}^*$ are finitely dimensional, thus one only need to verify that f is a monomorphism using the fact that $\kappa|_{\mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha}}$ is non-degenerate.

4. The restriction $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate, hence we have an isomorphism $\nu : \mathfrak{h} \rightarrow \mathfrak{h}^*$ given by $\nu(h)(h') = \kappa(h, h')$ for all $h, h' \in \mathfrak{h}$. The map ν induces a bilinear form on $\mathfrak{h}^*$ by

$$\kappa(\alpha, \beta) = \beta(\nu^{-1}(\alpha)) = \alpha(\nu^{-1}(\beta))$$

for all $\alpha, \beta \in \mathfrak{h}^*$. We have proved that $\kappa(\alpha, \alpha) \neq 0$ if $\alpha \in \Delta$.

5. For all $\alpha \in \Delta, e \in \mathfrak{g}_{\alpha}$ and $f \in \mathfrak{g}_{-\alpha}$, we have:

$$[e, f] = \kappa(e, f) \nu^{-1}(\alpha) \tag{10.1}$$

Now, given $\alpha \in \Delta$, pick non-zero $E \in \mathfrak{g}_{\alpha}$ and $F \in \mathfrak{g}_{-\alpha}$ such that $\kappa(E, F) = \frac{2}{\kappa(\alpha, \alpha)}$. Let $H = \frac{2\nu^{-1}(\alpha)}{\kappa(\alpha, \alpha)}$. Then, we can check that:

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H.$$

The choice of E and F is possible by 3 and the last claim of 4. We only verify the first equality, the second being analogous and the third coming from (10.1). We have:

$$\left[\frac{2\nu^{-1}(\alpha)}{\kappa(\alpha, \alpha)}, E \right] = \frac{2\alpha(\nu^{-1}(\alpha))}{\kappa(\alpha, \alpha)} E = \frac{2\kappa(\nu^{-1}(\alpha), \nu^{-1}(\alpha))}{\kappa(\alpha, \alpha)} E = \frac{2\kappa(\alpha, \alpha)}{\kappa(\alpha, \alpha)} E = 2E,$$

where the first equality comes from $\nu^{-1}(\alpha) \in \mathfrak{h}$ and $E \in \mathfrak{g}_{\alpha}$. If we now let $\mathfrak{a}_{\alpha} = \mathbb{F}E + \mathbb{F}F + \mathbb{F}H$, then $\mathfrak{a}_{\alpha}$ is isomorphic to $\mathfrak{sl}(2, \mathbb{F})$ via:

$$E \rightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, F \rightarrow \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, H \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

10.2 Key lemma for $\mathfrak{sl}(2, \mathbb{F})$

Keeping $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$ and the representation image in mind.

Lemma 10.2.1. (Key Lemma for $\mathfrak{sl}(2, \mathbb{F})$). Let $\mathbb{F}$ be a field of characteristic 0. Let ϕ be a representation of $\mathfrak{sl}(2, \mathbb{F})$ in a vector space V over $\mathbb{F}$ and let $v \in V$ be a non-zero vector such that $\phi(E)v = 0$ and $\phi(H)v = \lambda v$ for some $\lambda \in \mathbb{F}$. Then:

- a) $\phi(H)\phi(F)^n v = (\lambda - 2n)\phi(F)^n v$ for any $n \in \mathbb{Z}_{\geq 0}$.
- b) $\phi(E)\phi(F)^n v = n(\lambda - n + 1)\phi(F)^{n-1} v$ for any $n \in \mathbb{Z}_{\geq 1}$.
- c) If $\dim V < \infty$, then $\lambda \in \mathbb{Z}_{\geq 0}$, the vectors $\phi(F)^j v$ for $0 \leq j \leq \lambda$ are linearly independent, and $\phi(F)^{\lambda+1} v = 0$.

proof. (a) We prove this by induction on n . For $n = 0$, the result is given to us. Suppose it holds for $n = k - 1$ for some $k > 0$. Then we complete the induction by:

$$\begin{aligned} \phi(H)\phi(F)^k v &= \phi(F)\phi(H)\phi(F)^{k-1} v + [\phi(H), \phi(F)]\phi(F)^{k-1} v \\ &= \phi(F)(\lambda - 2(k-1))\phi(F)^{k-1} v + \phi([H, F])\phi(F)^{k-1} v \\ &= (\lambda - 2(k-1))\phi(F)^k v + \phi(-2F)\phi(F)^{k-1} v \\ &= (\lambda - 2(k-1))\phi(F)^k v - 2\phi(F)^k v \\ &= (\lambda - 2k)\phi(F)^k v \end{aligned}$$

(b) Again, we use induction on n . For the case $n = 1$, we have $\phi(E)\phi(F)v = \phi(F)\phi(E)v + [\phi(E), \phi(F)]v = 0 + \phi([E, F])v = \phi(H)v = \lambda v$. Suppose the result holds for $n = k - 1$ for some $k > 1$. Then we complete the induction by:

$$\begin{aligned} \phi(E)\phi(F)^k v &= \phi(F)\phi(E)\phi(F)^{k-1} v + [\phi(E), \phi(F)]\phi(F)^{k-1} v \\ &= \phi(F)(k-1)(\lambda - (k-1) + 1)\phi(F)^{k-2} v + \phi([E, F])\phi(F)^{k-1} v \\ &= (k-1)(\lambda - (k-1) + 1)\phi(F)^{k-1} v + \phi(H)\phi(F)^{k-1} v \\ &= (k-1)(\lambda - (k-1) + 1)\phi(F)^{k-1} v + (\lambda - 2(k-1))\phi(F)^{k-1} v \\ &= k(\lambda - k + 1)\phi(F)^{k-1} v. \end{aligned}$$

(c) Suppose $\lambda \notin \mathbb{Z}_{\geq 0}$. Then, the term $n(\lambda - n + 1)$ is non-zero for all $n \geq 1$. Hence, by induction, we get $\phi(F)^n v \neq 0$ for all $n \in \mathbb{Z}_{\geq 0}$. But by a), this implies that all vectors $\phi(F)^n v$ are eigenvectors of $\phi(H)$ with distinct eigenvalues. This allows us to conclude $\dim V = \infty$, therefore proving the first claim.

If now, $\lambda \in \mathbb{Z}_{\geq 0}$, we see that the vectors $\phi(F)^n v$ are linearly independent for $0 \leq n \leq \lambda$, because they belong to different eigenvalues. Moreover, if $\phi(F)^{\lambda+1} v \neq 0$, then by induction we see that $\phi(F)^n v \neq 0$ for all $n > \lambda + 1$, and so there are infinitely many linearly independent vectors. Hence, if $\dim V < \infty$, then $\phi(F)^{\lambda+1} v = 0$, proving the last two claims. $\square$

Corollary 10.2.1. Using the notation of the Key Lemma for $\mathfrak{sl}(2, \mathbb{F})$, if v instead satisfies that $\phi(F)v = 0$ and $\phi(H)v = \lambda v$, then we have:

- a) $\phi(H)\phi(E)^n v = (\lambda + 2n)\phi(E)^n v$ for any $n \in \mathbb{Z}_{\geq 0}$.
- b) $\phi(F)\phi(E)^n v = -n(\lambda + n - 1)\phi(E)^{n-1} v$ for any $n \in \mathbb{Z}_{\geq 1}$.
- c) If $\dim V < \infty$, then $-\lambda \in \mathbb{Z}_{\geq 0}$, the vectors $\phi(E)^j v$ for $0 \leq j \leq -\lambda$ are linearly independent, and $\phi(E)^{-\lambda+1} v = 0$.

proof. It is enough to check that the function $\psi(E) = F, \psi(F) = E, \psi(H) = -H$, is an automorphism of $\mathfrak{a}_\alpha$. Indeed:

$$\begin{aligned} [\psi(H), \psi(E)] &= 2\psi(E), \\ [\psi(H), \psi(F)] &= -2\psi(F), \\ [\psi(E), \psi(F)] &= \psi(H). \end{aligned}$$

Thus, the Key Lemma shows that if $\phi(\psi(E))v = 0$ and $\phi(\psi(H))v = \lambda'v$ with $\lambda' \in \mathbb{F}$, then:

- a) $\phi(\psi(H))\phi(\psi(F))^nv = (\lambda' - 2n)\phi(\psi(F))^nv$ for any $n \in \mathbb{Z}_{\geq 0}$.
- b) $\phi(\psi(E))\phi(\psi(F))^nv = n(\lambda' - n + 1)\phi(\psi(F))^{n-1}v$ for any $n \in \mathbb{Z}_{\geq 1}$.
- c) If $\dim V < \infty$, then $\lambda' \in \mathbb{Z}_{\geq 0}$, the vectors $\phi(\psi(F))^jv$ for $0 \leq j \leq \lambda'$ are linearly independent, and $\phi(\psi(F))^{\lambda'+1}v = 0$.

Now, let $\lambda' = -\lambda$ and evaluate ψ to obtain the result. $\square$

RMK: The key point of the proof of the corollary is to view $\phi \circ \psi$ as a new representation. The lemma explains how the eigenvalue reduces when the power of $\phi(F)$ grows, whereas the corollary explains how the eigenvalue increases when the power of $\phi(F)$ grows.

10.3 Portray of Root Δ

Theorem 10.3.1. * The root space decomposition of $\mathfrak{g}$ with respect to a Cartan subalgebra $\mathfrak{h}$ and the set of roots Δ satisfy the following properties:

- a) $\dim \mathfrak{g}_\alpha = 1$ for all $\alpha \in \Delta$.
- b) If $\alpha, \beta \in \Delta$, then $\{\beta + n\alpha\}_{n \in \mathbb{Z}} \cap (\Delta \cup \{0\})$ is a finite connected string $\{\beta - p\alpha, \beta - (p-1)\alpha, \dots, \beta, \dots, \beta + (q-1)\alpha, \beta + q\alpha\}$, where $p, q \in \mathbb{Z}_{\geq 0}$ and $p - q = \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)}$.¹²
- c) If $\alpha, \beta, \alpha + \beta \in \Delta$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.
- d) If $\alpha \in \Delta$, then $n\alpha \in \Delta$ if and only if $n = 1$ or $n = -1$.

proof. a) Suppose that $\dim \mathfrak{g}_\alpha > 1$ for some $\alpha \in \Delta$, then $\dim \mathfrak{g}_{-\alpha} > 1$ by property 3 above. Consider the adjoint representation of the subalgebra $\mathfrak{a}_\alpha = \mathbb{F}E + \mathbb{F}F + \mathbb{F}H$ on $\mathfrak{g}$. In particular, recall $H = \frac{2\nu^{-1}(\alpha)}{\kappa(\alpha, \alpha)}$. Since $\dim \mathfrak{g}_{-\alpha} > 1$, there exists a non-zero vector $v \in \mathfrak{g}_{-\alpha}$ such that $\kappa(E, v) = 0$. (Consider the bilinearity of κ .) Hence, $(\mathbf{ad}E)v = [E, v] = \kappa(E, v)\nu^{-1}(\alpha) = 0$. But $(\mathbf{ad}H)v = [H, v] = \frac{2[\nu^{-1}(\alpha), v]}{\kappa(\alpha, \alpha)} = \frac{-2\alpha(\nu^{-1}(\alpha))v}{\kappa(\alpha, \alpha)} = \frac{-2\kappa(\alpha, \alpha)v}{\kappa(\alpha, \alpha)} = -2v$. Hence, $\dim \mathfrak{g} = \infty$ by the Key Lemma, which yields a contradiction.

b) Let q be the largest integer such that $\beta + q\alpha \in \Delta \cup \{0\}$. Notice $q \geq 0$. Pick a non-zero vector $v \in \mathfrak{g}_{\beta+q\alpha}$. Then, $(\mathbf{ad}E)v = 0$ since it lies in $\mathfrak{g}_{\beta+(q+1)\alpha}$. Also, $(\mathbf{ad}H)v = (\beta + q\alpha)(H)v = \left(\frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} + 2q\right)v$ (where H is the same as which in a). Hence, by the Key Lemma: $\lambda := \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} + 2q \in \mathbb{Z}_{\geq 0}$ and $(\mathbf{ad}F)^jv$ are non-zero vectors for $0 \leq j \leq \lambda$. But $(\mathbf{ad}F)^jv \in \mathfrak{g}_{\beta+(q-j)\alpha}$, so $\beta + q\alpha, \beta + (q-1)\alpha, \dots, \beta + (q-\lambda)\alpha \in \Delta \cup \{0\}$.

Define $p := -(q - \lambda) = q + \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)}$. Let p' be the largest integer for which $\beta - p'\alpha \in \Delta \cup \{0\}$. Again, notice $p' \geq 0$. Pick a non-zero vector $v' \in \mathfrak{g}_{\beta-p'\alpha}$. Then, $(\mathbf{ad}F)v' = 0$ and $(\mathbf{ad}H)v' = \left(\frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} - 2p'\right)v'$. By the corollary of the Key Lemma, we conclude that $-\lambda' := 2p' - \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} \in \mathbb{Z}_{\geq 0}$ and that $\beta - p'\alpha, \beta - (p'-1)\alpha, \dots, \beta - (p' + \lambda')\alpha \in \Delta \cup \{0\}$. Define $q' := -(p' + \lambda') = p' - \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)}$. Since q and p' are the largest integers for which $\beta + q\alpha \in \Delta \cup \{0\}$ (resp. $\beta - p'\alpha \in \Delta \cup \{0\}$), we conclude that $q \geq q'$ and $p' \geq p$. Hence, $\frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} = p - q \leq p' - q' = \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)}$, showing that $p = p', q = q', p, q \in \mathbb{Z}_{\geq 0}$.

c) Pick the largest integers p and q such that $\beta - p\alpha, \beta + q\alpha \in \Delta \cup \{0\}$. Pick a non-zero vector $v \in \mathfrak{g}_{\beta-p\alpha}$. Then, $(\mathbf{ad}F)v = 0$ and $(\mathbf{ad}H)v = \left(\frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} - 2p\right)v$. By the corollary of the Key Lemma, $(\mathbf{ad}E)^jv \neq 0$ for $0 \leq j \leq 2p - \frac{2\kappa(\alpha, \beta)}{\kappa(\alpha, \alpha)} = p + q$. But $q \geq 1$ since $\alpha + \beta \in \Delta$, so $(\mathbf{ad}E)^{p+1}v$ is a non-zero

¹Which is called Cartan integer, later we will find the value of Cartan integer can only be $0, \pm 1, \pm 2, \pm 3$.

²We should notice that (b) actually says that the string is made up of all roots $\beta + n\alpha$ and is connected.

vector. Its corresponding root is $\alpha + \beta$, and $(\mathbf{ad} E)^p v \in \mathfrak{g}_\beta$, so $[E, \mathfrak{g}_\beta] \neq 0$. Hence, $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$ since $\dim \mathfrak{g}_{\alpha+\beta} = 1$.

d) Let $\beta = n\alpha, n \neq 0$. Then, $\frac{2\kappa(\alpha, \beta)}{\kappa(\beta, \beta)} = \frac{2n}{n^2} = \frac{2}{n} \in \mathbb{Z}$. Hence, either $n = 2, 1, -1$ or -2 . However, $[\mathfrak{g}_\alpha, \mathfrak{g}_\alpha] = 0$ by a) (resp. $[\mathfrak{g}_{-\alpha}, \mathfrak{g}_{-\alpha}] = 0$), so 2α (resp. -2α) is not a root because otherwise, c) would imply that $[\mathfrak{g}_\alpha, \mathfrak{g}_\alpha] = \mathfrak{g}_{2\alpha}$ (resp. $[\mathfrak{g}_{-\alpha}, \mathfrak{g}_{-\alpha}] = \mathfrak{g}_{-2\alpha}$). $\square$

An immediate corollary of this theorem is:

Corollary 10.3.1. *If $\alpha, \beta \in \Delta$, $\beta \neq \pm\alpha$. Then an α string through β has at most 4 roots.*

proof. Say there are five roots in the string, then we can always assume the string is symmetric by replacing the middle one by γ , thus we can get:

$$\gamma - 2\alpha, \gamma - \alpha, \gamma, \gamma + \alpha, \gamma + 2\alpha$$

Thus we have $\kappa(\gamma, \alpha) = 0$. Then we consider the γ string through $\gamma + 2\alpha$, we get $2(\gamma + \alpha)$ and 2α . By d), we know the γ string through $\gamma + 2\alpha$ has only one root. Thus $\kappa(\gamma, \gamma) + 2\kappa(\gamma, \alpha) = 0$. Then we have $\kappa(\gamma, \gamma) = 0$. However, we know that for $\gamma \in \Delta$, $\kappa(\gamma, \gamma) \neq 0$ from 10.1.4. Thus we conclude the proof. $\square$

Example 10.3.1. Let $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{F})$. We know κ is non-degenerate, so $\mathfrak{g}$ is semisimple. We will find all possibilities for p and q in the proof above.

To construct $\mathfrak{h}$, take a diagonal matrix $a = \text{diag}(a_1, a_2, \dots, a_n) \in \mathfrak{g}$ all of whose diagonal entries are distinct. Thus it's regular in $\mathfrak{g}$. Hence, $\mathfrak{g}_0^a$ is a Cartan subalgebra of $\mathfrak{g}$, so let $\mathfrak{h} = \mathfrak{g}_0^a$. As $(\mathbf{ad} a)^N e_{ij} = (a_i - a_j)^N e_{ij}$ for all $N \geq 0$, we see that $\mathfrak{h}$ is precisely the set of diagonal matrices of $\mathfrak{g}$.

A basis for $\mathfrak{h}^*$ is given by $\{\varepsilon_1 - \varepsilon_2, \varepsilon_2 - \varepsilon_3, \dots, \varepsilon_{n-1} - \varepsilon_n\}$, $i \in \{1, \dots, n\}$. We can check that:

$$\mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{F}e_{ij} \text{ for all } i, j \in \{1, \dots, n\}, i \neq j.$$

Hence, the set $\Delta := \{\varepsilon_i - \varepsilon_j \mid i, j \in \{1, \dots, n\}, i \neq j\}$ is a root system for $\mathfrak{g}$.

For no pair of roots $\alpha, \beta \in \Delta$ it is true that $\beta \pm 3\alpha \in \Delta \cup \{0\}$. Thus, the only possibilities for (q, p) are $(2, 0), (1, 1), (0, 2), (1, 0), (0, 1), (0, 0)$.

When $n = 2$, we can only have $(q, p) = (2, 0), (0, 2)$. Let $\alpha := \varepsilon_1 - \varepsilon_2$. Then $\Delta = \{\pm\alpha\}$, so $\alpha = \alpha + (0)\alpha, 0 = \alpha - (1)\alpha, -\alpha = \alpha - (2)\alpha$ and we have $\alpha, 0, -\alpha \in \Delta \cup \{0\}$, giving all possible values for p and q .

If $n = 3$, we can only have $(q, p) = (2, 0), (0, 2), (1, 0), (0, 1)$. We have the pairs $(2, 0)$ and $(0, 2)$ by the previous case. Now, letting $\beta := \varepsilon_1 - \varepsilon_3$ and $\gamma := \varepsilon_2 - \varepsilon_3$ so that $\Delta = \{\pm\alpha, \pm\beta, \pm\gamma\}$, we see that $\alpha - \beta, \alpha \in \Delta \cup \{0\}$ but $\alpha - 2\beta, \alpha + \beta \notin \Delta \cup \{0\}$, and similar relations hold among α, γ and β, γ by symmetry.

If $n \geq 4$, we can only have $(q, p) = (2, 0), (0, 2), (1, 0), (0, 1), (0, 0)$. The first four pairs come from the previous two cases. The fifth pair $(0, 0)$ occurs if we let $\delta = \varepsilon_3 - \varepsilon_4$ and then notice that $\alpha - \delta, \alpha + \delta \notin \Delta \cup \{0\}$.

In general, let $\alpha_{ij} := \varepsilon_i - \varepsilon_j$ for all $i, j \in \{1, \dots, n\}$ with $i \neq j$. Then, for all multisets $\{i, j, k, l\} \subseteq \{1, \dots, n\}$:

if $|\{i, j\} \cap \{k, l\}| = 2$, then α_{ij} and α_{kl} are related by pairs $(2, 0), (0, 2)$;

if $|\{i, j\} \cap \{k, l\}| = 1$, then α_{ij} and α_{kl} are related by pairs $(1, 0), (0, 1)$;

if $|\{i, j\} \cap \{k, l\}| = 0$, then α_{ij} and α_{kl} are related by the pair $(0, 0)$.

10.4 Some problems

1. Show that $h = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} \in \mathfrak{sl}(n, \mathbb{F})$ is a regular element if and only if $\lambda_i \neq \lambda_j$ when $i \neq j$.

sketch of proof. ($\Rightarrow$) Let $h = \text{diag}(\lambda_1, \dots, \lambda_n)$ be a diagonal matrix in $\mathfrak{sl}(n, \mathbb{F})$. Let us compute its centralizer in $\mathfrak{sl}(n, \mathbb{F})$: Take $X = (x_{ij}) \in \mathfrak{sl}(n, \mathbb{F})$, and suppose $[h, X] = hX - Xh = 0$. The (i, j) -th

entry of this commutator is: $([h, X])_{ij} = (\lambda_i - \lambda_j)x_{ij}$. Therefore, $[h, X] = 0$ if and only if $(\lambda_i - \lambda_j)x_{ij} = 0$ for all i, j .

This implies:

- (1) For all $i \neq j$, if $\lambda_i \neq \lambda_j$, then $x_{ij} = 0$.
- (2) For $i = j$, there is no condition beyond the trace-zero constraint.
- (3) If $\lambda_i = \lambda_j$ for some $i \neq j$, then x_{ij} is allowed to be arbitrary.

Hence, the centralizer $\mathfrak{g}_h$ consists of matrices $X = (x_{ij})$ where:

- (1) $x_{ij} = 0$ for $i \neq j$ unless $\lambda_i = \lambda_j$.
- (2) The trace of X is zero.

□

11 Structure Theory of Semisimple Lie Algebras (III)

11.1 Further portray on Root Δ

Proposition 11.1.1. Choose a Cartan subalgebra $\mathfrak{h}$, for $h_1, h_2 \in \mathfrak{h}$, we have:

$$\kappa(h_1, h_2) = \text{tr}_{\mathfrak{g}}(\text{ad}h_1 \text{ad}h_2) = \sum_{\alpha \in \Delta} \alpha(h_1)\alpha(h_2) \quad (11.1)$$

sketch of proof. Consider the root space decomposition concerning $\mathfrak{h}$: $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$, $[\mathfrak{h}, \mathfrak{h}] = 0$, $\dim \mathfrak{g}_{\alpha} = 1$. $\text{tr}_{\mathfrak{g}}(\text{ad}h_1 \text{ad}h_2) = \text{tr}_{\mathfrak{h}}(\text{ad}h_1 \text{ad}h_2) + \sum_{\alpha \in \Delta} \text{tr}_{\mathfrak{g}_{\alpha}}(\text{ad}h_1 \text{ad}h_2) = 0 + \sum_{\alpha \in \Delta} \alpha(h_1)\alpha(h_2)$. $\square$

Thus using the isomorphism $\nu : \mathfrak{h} \rightarrow \mathfrak{h}^*$ by $\nu(h)(h') = \kappa(h, h')$, we can define a bilinear form κ on $\mathfrak{h}^*$:

$$\kappa(\lambda_1, \lambda_2) = \sum_{\alpha \in \Delta} \alpha(\nu^{-1}(\lambda_1))\alpha(\nu^{-1}(\lambda_2)) = \sum_{\alpha \in \Delta} \kappa(\lambda_1, \alpha)\kappa(\lambda_2, \alpha) \quad (11.2)$$

Definition 11.1.1. $\mathfrak{h}_{\mathbb{Q}}^* \subset \mathfrak{h}^*$ is the $\mathbb{Q}$ -span of Δ . (Note that $\mathbb{Q} \subset \mathbb{F}$ since $\text{char} \mathbb{F} = 0$.)

Theorem 11.1.1. For $\mathfrak{g}$ as above, the following are true:

- 1) Δ spans $\mathfrak{h}^*$ over $\mathbb{F}$.
- 2) $\kappa(\alpha, \beta) \in \mathbb{Q}$ for all $\alpha, \beta \in \Delta$.
- 3) $\kappa|_{\mathfrak{h}_{\mathbb{Q}}^* \times \mathfrak{h}_{\mathbb{Q}}^*}$ is a positive definite symmetric bilinear form with values in $\mathbb{Q}$.

proof. 1) Suppose Δ does not span $\mathfrak{h}^*$, then there exists a non-zero $h \in \mathfrak{h}$ such that $\alpha(h) = 0$ for all $\alpha \in \Delta$.¹ This implies that $[h, \mathfrak{g}_{\alpha}] = 0$ for all α from the definition of root space. It is also true that $[h, \mathfrak{h}] = 0$, as the Cartan subalgebra $\mathfrak{h}$ is abelian. Therefore, h is in $C(\mathfrak{g})$. But $\mathfrak{g}$ is semisimple, so $C(\mathfrak{g}) = 0$, which is a contradiction.

2) From equation (11.2) we have:

$$\kappa(\lambda, \lambda) = \sum_{\alpha \in \Delta} \kappa(\lambda, \alpha)^2, \quad \forall \lambda \in \mathfrak{h}^* \quad (11.3)$$

For $\beta \in \Delta$, $\kappa(\beta, \beta) \neq 0$, so:

$$\frac{4}{\kappa(\beta, \beta)} = \sum_{\alpha \in \Delta} \left(\frac{2\kappa(\beta, \alpha)}{\kappa(\beta, \beta)} \right)^2, \quad \beta \in \Delta \quad (11.4)$$

It follows that $\frac{4}{\kappa(\beta, \beta)} \in \mathbb{Z}$ for $\beta \in \Delta$ because by the string condition $\frac{2\kappa(\beta, \alpha)}{\kappa(\beta, \beta)} = p - q \in \mathbb{Z}$. This implies that $\kappa(\beta, \beta) \in \mathbb{Q}$ for $\beta \in \Delta$.

But since $\frac{2\kappa(\alpha, \beta)}{\kappa(\beta, \beta)} \in \mathbb{Z}$ for $\alpha, \beta \in \Delta$ and $\kappa(\beta, \beta) \in \mathbb{Q}$ we conclude that $\kappa(\alpha, \beta) \in \mathbb{Q}$.

3) From part 2), $\kappa(\alpha, \beta) \in \mathbb{Q}$, hence $\kappa(\lambda, \alpha) \in \mathbb{Q}$ for $\lambda \in \mathfrak{h}_{\mathbb{Q}}^*$, $\alpha \in \Delta$. Since by part 1), Δ spans $\mathfrak{h}^*$ and the Killing form is non-degenerate on $\mathfrak{h}$, its restriction to $\mathfrak{h}_{\mathbb{Q}}^*$ is non-degenerate as well.

Hence by equation (11.3), $\kappa(\lambda, \lambda) \geq 0$ for $\lambda \in \mathfrak{h}_{\mathbb{Q}}^*$ since it is the sum of rational squares. This proves that κ is positive and semi-definite. And from linear algebra we know any non-degenerate positive semi-definite symmetric bilinear form is positive definite. This proves part 3). $\square$

We will now examine how the decomposition of a Lie algebra corresponds to a decomposition of its root space.

Let $\mathfrak{g}$ be the direct sum of two ideals, $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, where both $\mathfrak{g}_i$ are semisimple. Consider for each of them the root space decomposition. To do this we choose a Cartan subalgebra $\mathfrak{h}_i$ in each $\mathfrak{g}_i$.

$$\mathfrak{g}_i = \mathfrak{h}_i \oplus \bigoplus_{\alpha \in \Delta_i} (\mathfrak{g}_i)_{\alpha}, \quad \mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha},$$

¹There are some gaps, which is illustrated in the supplement.

where $\mathfrak{h}$ is $\mathfrak{h}_1 \oplus \mathfrak{h}_2$ and $\Delta = \Delta_1 \sqcup \Delta_2$. We can verify that $\mathfrak{h}$ is nilpotent, for $\mathfrak{g}_1, \mathfrak{g}_2$ are ideals, $[\mathfrak{h}_1, \mathfrak{h}_2] \subseteq \mathfrak{h}_1 \cap \mathfrak{h}_2 = 0$. If $x = g_1 + g_2$, $g_i \in \mathfrak{g}_i$ and $[x, \mathfrak{h}] \subseteq \mathfrak{h}$, we have $[g_1 + g_2, \mathfrak{h}_1 + \mathfrak{h}_2] = [g_1, \mathfrak{h}_1] + [g_2, \mathfrak{h}_2] \subseteq \mathfrak{h}_1 + \mathfrak{h}_2$, whereas $[g_i, \mathfrak{h}_i] \subseteq \mathfrak{g}_i$. Thus $g_i \in \mathfrak{h}_i$. Thus $x \in \mathfrak{h}$ and $\mathfrak{h}$ is a CSA of $\mathfrak{g}$. Moreover, we can show that $(\mathfrak{g}_i)_\alpha$ is a root space of $\mathfrak{h}$. For $\alpha \in \Delta_1$, we just need to consider $\tilde{\alpha}(h_1 + h_2) = \alpha(h_1)$.

If $\alpha \in \Delta_1$ and $\beta \in \Delta_2$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = 0$, then we have that $\alpha + \beta \notin \Delta$ and $\alpha + \beta \neq 0$. In other words,

$$\Delta = \Delta_1 \sqcup \Delta_2, \quad \alpha + \beta \notin \Delta \cup \{0\} \text{ if } \alpha \in \Delta_1, \beta \in \Delta_2. \quad (11.5)$$

Moreover, we can prove that equation (11.5) is equal to (where $(\cdot, \cdot)$ is the killing form):

$$\Delta = \Delta_1 \cup \Delta_2; (\Delta_1, \Delta_2) = 0 \quad (11.6)$$

proof. ($\Rightarrow$) If $\beta \in \Delta_2$, then $-\beta \in \Delta_2$. Thus consider the α string through β , we can get $p = q = 0$. Thus $\kappa(\alpha, \beta) = 0, \forall \alpha \in \Delta_1, \beta \in \Delta_2$.

($\Leftarrow$) If $\alpha \in \Delta_1 \cap \Delta_2$, then $\kappa(\alpha, \alpha) = 0$, which is a contradiction. Thus the union is disjoint. If for $\alpha \in \Delta_1, \beta \in \Delta_2, \alpha + \beta \in \Delta$, say it's in Δ_1 . Then $\kappa(\alpha + \beta, \beta) = \kappa(\alpha, \beta) = 0$. Thus $\kappa(\beta, \beta) = 0$. If $\alpha + \beta = 0$, i.e. $\beta = -\alpha$. Then $\kappa(\alpha, -\alpha) = 0 = \kappa(\alpha, \alpha)$. $\square$

RMK: Moreover, we can deduce that if $\alpha \in \Delta_1$, then $-\alpha \in \Delta_1$. Thus if Δ is indecomposable, $\alpha \in \Delta$, then $-\alpha \in \Delta$.

Definition 11.1.2. The set Δ in a vector space V is called **indecomposable** if it cannot be decomposed into a disjoint union of non-empty subsets Δ_1 and Δ_2 such that equation (11.5) holds.

With this notion, we can give a simple way to check if a semisimple Lie algebra is simple.

11.2 Criterion of Semisimple Lie algebra

Lemma 11.2.1. (The inheritance of direct sum decomposition) Let $\mathfrak{h}$ be an abelian Lie algebra and ϕ its representation in a vector space V such that V has a weight space decomposition: $V = \bigoplus_{\lambda \in \mathfrak{h}^*} V_\lambda$, where $V_\lambda = \{v \in V \mid \phi(h)v = \lambda(h)v, \forall h \in \mathfrak{h}\}$. If $U \subset V$ is a $\phi(\mathfrak{h})$ -invariant subspace, then $U = \bigoplus_{\lambda \in \mathfrak{h}^*} (U \cap V_\lambda)$.

proof. For $u \in U$, let: $u = \sum_{i=1}^n v_{\lambda_i}, v_{\lambda_i} \in V_{\lambda_i}, \lambda_i \neq \lambda_j$. We will prove that all v_{λ_i} are in U by induction on n . For the case $n = 1$, $v_{\lambda_1} = u \in U$.

For the case, $n > 1$ we apply $\phi(h)$ to both sides:

$$\phi(h)u = \sum_{i=1}^n \lambda_i(h)v_{\lambda_i}$$

where $h \in \mathfrak{h}$ is chosen such that $\lambda_i(h) \neq \lambda_j(h)$ for $i \neq j$. The choose is always exists. We can consider the subspace of $\mathfrak{h}$: $\mathfrak{h}_{ij} = \{h \in \mathfrak{h} \mid \lambda_i(h) = \lambda_j(h)\}$, which is always a proper subspace of $\mathfrak{h}$. Thus for i, j are finite, $\mathfrak{h}_{ij}$ are finitely many proper subspaces of $\mathfrak{h}$, whose union can not cover $\mathfrak{h}$. For $u = \sum_{i=1}^n v_{\lambda_i}$, thus we have $\lambda_1 u = \sum_{i=1}^n \lambda_1 v_{\lambda_i}$. Then using a subtraction, we get $\phi(h)u - \lambda_1(h)u = \sum_{i=2}^n (\lambda_i(h) - \lambda_1(h))v_{\lambda_i}$, where each term is not 0 by assumption. By the inductive assumption, $v_{\lambda_i} \in U$ for $i \geq 2$, and we have $\phi(h)u - \lambda_1(h)u \in U$, hence also $v_{\lambda_1} \in U$. $\square$

RMK: We can also using the Vandermonde determinant to solve the equations.

Theorem 11.2.1. Let $\mathfrak{g} = \mathfrak{h} \oplus (\bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha)$ be a decomposition of a finite dimensional Lie algebra $\mathfrak{g}$ into a direct sum of subspaces such that the following properties hold:

- 1) $\mathfrak{h}$ is an abelian subalgebra and $\dim \mathfrak{g}_\alpha = 1$ for all $\alpha \in \Delta$, where $\mathfrak{g}_\alpha = \{a \in \mathfrak{g} \mid [h, a] = \alpha(h)a, \forall h \in \mathfrak{h}\}$,
- 2) $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{F}h_\alpha$, where $h_\alpha \in \mathfrak{h}$ is such that $\alpha(h_\alpha) \neq 0$,
- 3) $\mathfrak{h}^*$ is spanned by Δ .

Then $\mathfrak{g}$ is a semisimple Lie algebra. Moreover if the set Δ is indecomposable, then $\mathfrak{g}$ is simple.

proof. We want to prove that if $\mathfrak{a}$ is an abelian ideal of $\mathfrak{g}$, then $\mathfrak{a} = 0$. Note that since $\mathfrak{a}$ is an ideal, it is invariant with respect to $\text{ad } h$ on $\mathfrak{g}$. Hence by the lemma, either $\mathfrak{g}_\alpha$ is in $\mathfrak{a}$ for some α or $\mathfrak{h} \cap \mathfrak{a}$ is non-zero. This uses the fact that $\dim \mathfrak{g}_\alpha = 1$.

In the first case, $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{F}h_\alpha \subset \mathfrak{a}$, but $[h_\alpha, \mathfrak{g}_\alpha] \neq 0$ since $\alpha(h_\alpha) \neq 0$. So $\mathfrak{a}$ contains the non-abelian subalgebra $\mathbb{F}h_\alpha \oplus \mathfrak{g}_\alpha$, which is impossible since $\mathfrak{a}$ is abelian.

In the second case, where $\mathfrak{h} \cap \mathfrak{a}$ is non-zero, let $h \in \mathfrak{h} \cap \mathfrak{a}$, $h \neq 0$. By condition 3), $\mathfrak{h}^*$ is spanned by Δ , so $\alpha(h) \neq 0$ for some $\alpha \in \Delta$. Hence $[h, \mathfrak{g}_\alpha] = \mathfrak{g}_\alpha \subset \mathfrak{a}$, which again contradicts that $\mathfrak{a}$ is abelian. So $\mathfrak{a} = 0$. This proves that $\mathfrak{g}$ is semisimple.

Now the proof of simplicity, given that Δ is indecomposable. Since $\mathfrak{g}$ is semisimple, we have that $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, where the $\mathfrak{g}_i$ are non-zero ideals. In the contrary case, by our discussion before definition 11.1.2., this implies that Δ is decomposable. This is a contradiction, so $\mathfrak{g}$ is simple. $\square$

RMK: In fact, $\mathfrak{g}$ is simple iff its root Δ is indecomposable.

The following is an easy method to determine if a set Δ is indecomposable.

Proposition 11.2.1. *Prove that a finite subset: $\Delta \subset V \setminus \{0\}$ is an indecomposable set iff for any $\alpha, \beta \in \Delta$, $\exists$ a sequence $\gamma_1, \gamma_2, \dots, \gamma_s$ such that $\alpha = \gamma_1, \beta = \gamma_s$ and $\gamma_i + \gamma_{i+1} \in \Delta$ for $i = 1, \dots, s-1$.*

Also for any $\Delta \subset V \setminus \{0\}$, it has a canonical decomposition into a disjoint union of indecomposable sets.

proof. Consider the contrapositive. Suppose Δ is decomposable. Therefore, $\Delta = \Delta_1 \cup \Delta_2$. Take some $\alpha \in \Delta_1, \beta \in \Delta_2$. No sequence can exist with $\gamma_1 = \alpha$ and $\gamma_s = \beta$. For at some step in any such sequence $\gamma_i \in \Delta_1$ and $\gamma_{i+1} \in \Delta_2$. But $\gamma_i + \gamma_{i+1} \in \Delta$ contradicts the definition of a decomposition.

Suppose for some $\alpha, \beta \in \Delta$ no sequence exists. Let Δ_1 be the set of roots for which a sequence with α exists. Let Δ_2 be the set of roots for which a sequence with α does not exist. This is clearly a disjoint partition of Δ . Further Δ_1 and Δ_2 are non-empty as $\alpha \in \Delta_1, \beta \in \Delta_2$. For $\alpha' \in \Delta_1, \beta' \in \Delta_2, \alpha' + \beta' \notin \Delta$ as otherwise one can concatenate the sequence from $\alpha = \gamma_1$ to $\alpha' = \gamma_s$ and from $\alpha' = \gamma_s$ to $\beta' = \gamma_{s+1}$. Thus, $\Delta_1 \sqcup \Delta_2$, where Δ_1 is indecomposable. Now apply the same argument to Δ_2 , etc. Since Δ is a finite set, we obtain the decomposition of Δ in a finite number of steps. $\square$

RMK: When we use this proposition to judge whether a set is indecomposable, we ignore the case when $\beta = \pm\alpha$, for $\pm\alpha$ is always in the same root of α .

Corollary 11.2.1. *Similarly, we can prove: a finite subset $\Delta \subset V \setminus \{0\}$ is an indecomposable set iff for any $\alpha, \beta \in \Delta, \exists$ a sequence $\gamma_1, \gamma_2, \dots, \gamma_s$ such that $\alpha = \gamma_1, \beta = \gamma_s$ and $\kappa(\gamma_i, \gamma_{i+1}) \neq 0$ for $i = 1, \dots, s-1$.*

proof. $(\Rightarrow)$ We can get a disjoint partition of Δ similarly as in the proposition. We only need to prove (11.6), i.e. $\forall \alpha' \in \Delta_1, \beta' \in \Delta_2, \kappa(\alpha', \beta') = 0$. Otherwise, we can get the contradiction samely, by concatenating the strings.

$(\Leftarrow)$ We should use the equivalent definition (11.6). And we can get the contradiction samely. $\square$

11.3 Supplement: Dual space

Let V be an n -dimensional vector space over a field $\mathbb{F}$.

Definition 11.3.1. The *dual space* of V , denoted V^* , is by definition the set of all linear maps from V to $\mathbb{F}$. Thus, if $f, g \in V^*$, then $f + g$ is defined by $(f + g)(v) = f(v) + g(v)$ for $v \in V$, and if $\lambda \in \mathbb{F}$, then λf is defined by $(\lambda f)(v) = \lambda f(v)$. Moreover, we can define the dual space of V^* in the same way.

Proposition 11.3.1. *The basis $\{v_1, \dots, v_n\}$ of vector space V can induce a basis of the dual space, which is called **dual basis** of V . In particular $\dim V = \dim V^*$.*

sketch of proof. Given the basis $\{v_1, \dots, v_n\}$ of V , consider $f_i : V \rightarrow \mathbb{F}$ be the linear map defined on basis elements by

$$f_i(v_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

It is not hard to check that $f_1, \dots, f_n$ is a basis for V^* . □

Proposition 11.3.2. The dual space of V^* can be identified with V in a natural way, using the **evaluation map**. And thus we have $\dim V = \dim V^* = \dim V^{**}$.

sketch of proof. Given $v \in V$, we may define an evaluation map $\varepsilon_v : V^* \rightarrow \mathbb{F}$ by

$$\varepsilon_v(f) := f(v), \text{ for all } f \in V^*.$$

It is straightforward to check that ε_v is linear and so belongs to the dual space of V^* ; that is, to V^{**} . Moreover, the map $v \rightarrow \varepsilon_v$ (which we might call ε) from V to V^{**} is itself linear. If $\varepsilon(v) = \varepsilon_v = 0$, which is a 0-function, then for all $f \in V^*$, $f(v) = 0$. However, for $v \neq 0$, we can find a dual element v^* in V^* , $v^*(u) = \delta_{uv}$. Thus $\ker \varepsilon = 0$, ε is a monomorphism and thus $\varepsilon : V \rightarrow V^{**}$ is an isomorphism. □

Proposition 11.3.3. If U is a subspace of V , we let $U^0 = \{f \in V^* \mid f(u) = 0 \text{ for all } u \in U\}$ be the **annihilator** of U in V^* . One can show that U^0 is a subspace of V^* and that

$$\dim U + \dim U^0 = \dim V.$$

proof. Say the basis of U is $e_1, \dots, e_m$, and we span them to a basis of V $e_1, \dots, e_m, e_{m+1}, \dots, e_n$. An element in V^* can be presented as $f = \sum_{i=1}^n a_i e_i^*$. Thus from $f(u) = 0 \forall u \in U$ we have the equation set:

$$\begin{cases} a_1 e_1^*(e_1) + \dots + a_m e_m^*(e_1) + a_{m+1} e_{m+1}^*(e_1) + \dots + a_n e_n^*(e_1) = 0 \\ a_1 e_1^*(e_2) + \dots + a_m e_m^*(e_2) + a_{m+1} e_{m+1}^*(e_2) + \dots + a_n e_n^*(e_2) = 0 \\ \vdots \\ a_1 e_1^*(e_m) + \dots + a_m e_m^*(e_m) + a_{m+1} e_{m+1}^*(e_m) + \dots + a_n e_n^*(e_m) = 0 \end{cases}$$

Obviously the solution space is U^0 and we can complete the proof from the theory of solving systems of linear equations. □

anotherproof. We can use the language of map to simplify the proof. Consider the restriction map $r : V^* \rightarrow U^*$, which takes a linear map $f : V \rightarrow \mathbb{F}$ and regards it just as a map on U . We know that $\text{Ker } r = U^0$ and $\text{Im } r = U^*$. Hence prove that $\dim U + \dim U^0 = \dim V$. □

RMK: Given a subspace W of V^* , we can similarly define the annihilator of W in V^{**} . Under the identification of V^{**} with V , the annihilator of W becomes $W^0 = \{v \in V \mid f(v) = 0 \text{ for all } f \in W\}$.

In particular, we have $\dim W + \dim W^0 = \dim V$. (Which is precisely the gap in the proof of Theorem 11.1.1.)

11.4 Some problems

1、 Suppose that $\mathfrak{g}$ is a complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$ and root system Δ . Use the results of this lecture to prove that $\dim \mathfrak{g} = \dim \mathfrak{h} + |\Delta|$. Hence show that there are no semisimple Lie algebras of dimensions 4, 5, or 7.

proof. The equation is obvious, for every $\mathfrak{g}_\alpha$, $\alpha \in \Delta$, $\dim \mathfrak{g}_\alpha = 1$. Moreover, for $\alpha, -\alpha \in \Delta$, $|\Delta|$ is always even.

When $\dim \mathfrak{g} = 4$, then $|\Delta| = 2$ and $\dim \mathfrak{h} = 2$. Thus There are only $\alpha, -\alpha \in \Delta$, thus the elements in Δ can only span a 1-dimensional vector space, whereas $\dim \mathfrak{h}^* = \dim \mathfrak{h} = 2$. Thus we have a contradiction.

When $\dim \mathfrak{g} = 5$ with $|\Delta| = 2$ and $\dim \mathfrak{h} = 3$, we can get a contradiction samely. Thus we will only consider $|\Delta| = 4$ and $\dim \mathfrak{h} = 1$. For $\Delta \subseteq \mathfrak{h}^*$, which is 1-dimensional, we can assume the elements in Δ are $\alpha, -\alpha, k\alpha, -k\alpha$, for some $k \in \mathbb{Q}_{>0}, k \neq 1$. For $\frac{2\kappa(\alpha, k\alpha)}{\kappa(\alpha, \alpha)} = 2k \in \mathbb{Z}$, k is a multiple of $\frac{1}{2}$, say $k = \frac{m}{2}$.

$\frac{2\kappa(\alpha, k\alpha)}{\kappa(k\alpha, k\alpha)} = \frac{2}{k} = \frac{4}{m} \in \mathbb{Z}$. Thus k can only be $\frac{1}{2}$ or 2. Thus we have a contradiction.

When $\dim \mathfrak{g} = 7$ with $|\Delta| = 2$ and $\dim \mathfrak{h} = 5$ or $|\Delta| = 4$ and $\dim \mathfrak{h} = 3$, we can get a contradiction similar with the case $\dim \mathfrak{g} = 4$. Thus we only need to analyze when $|\Delta| = 6$ and $\dim \mathfrak{h} = 1$, in which case $\Delta = \{\pm\alpha, \pm k_1\alpha, \pm k_2\alpha\}$. Then we can get the contradiction as in the case $\dim \mathfrak{g} = 5$. $\square$

2、Show that the set of diagonal matrices in $\mathfrak{so}(4, \mathbb{C})$ (which is D_l in subsection 1.2.) forms a Cartan subalgebra of $\mathfrak{so}(4, \mathbb{C})$ and determine the corresponding root space decomposition. Hence show that $\mathfrak{so}(4, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{sl}(2, \mathbb{C})$.

proof. Say $e_{ij} = E_{ij} - E_{5-j, 5-i}, i+j < 5$. If $i+j > 5$, denote $e_{ij} = -e_{5-j, 5-i}$. For quantitative relation, we know that $\dim \mathfrak{h} = 2$ and $|\Delta| = 4$. $[e_{11}, e_{22}] = [E_{11} - E_{44}, E_{22} - E_{33}] = 0$, thus the subalgebra generated by e_{11} and e_{22} is nilpotent. Denote $\mathfrak{h}$ as the subalgebra generated by e_{11} and e_{22} . We just need to verify that e_{11} is a regular element, i.e. $\dim \mathfrak{g}_0^{e_{11}} = 0$. Clearly, $\mathfrak{h} \subseteq \mathfrak{g}_0^{e_{11}}$, for $e_{ij}, i \neq j$, we know that $[e_{11}, e_{ij}] \notin \mathfrak{h}$. Thus $\mathfrak{h} = \mathfrak{g}_0^{e_{11}}$ and e_{11} is a regular element. Thus $\mathfrak{h} = \mathfrak{g}_0^{e_{11}}$ is a CSA.

We can show that $[e_{ij}, e_{kl}] = [E_{ij} - E_{5-j, 5-i}, E_{kl} - E_{5-l, 5-k}] = \delta_{jk}E_{il} - \delta_{il}E_{kj} - \delta_{5-i, k}E_{5-j, l} + \delta_{5-j, l}E_{k, 5-i} - \delta_{j, 5-l}E_{i, 5-k} + \delta_{i, 5-k}E_{5-l, j} + \delta_{il}E_{5-j, 5-k} - \delta_{kj}E_{5-l, 5-i} = \delta_{jk}e_{il} - \delta_{il}e_{kj} - \delta_{5-i, k}e_{5-j, l} + \delta_{5-j, l}e_{k, 5-i}$. Thus $[e_{11}, e_{12}] = e_{12}, [e_{22}, e_{12}] = -e_{12}, e_{12}$ belongs to root $\varepsilon_1 - \varepsilon_2$. $[e_{11}, e_{13}] = e_{13}, [e_{22}, e_{13}] = e_{13}, e_{13}$ belongs to root $\varepsilon_1 + \varepsilon_2$. $[e_{11}, e_{21}] = -e_{21}, [e_{22}, e_{21}] = e_{21}, e_{21}$ belongs to root $-\varepsilon_1 + \varepsilon_2$. $[e_{11}, e_{31}] = -e_{31}, [e_{22}, e_{31}] = -e_{31}, e_{31}$ belongs to root $-\varepsilon_1 - \varepsilon_2$. Thus $\Delta = \{\varepsilon_1 - \varepsilon_2, \varepsilon_2 - \varepsilon_1, \varepsilon_1 + \varepsilon_2, -\varepsilon_1 - \varepsilon_2\}$ and the discussion gives the root space decomposition.

Let $x = e_{11} + e_{22}, y = e_{11} - e_{22}$. Then we have:

$$[x, e_{12}] = 0, [x, e_{13}] = 2e_{13}, [x, e_{31}] = -2e_{31}, [x, e_{21}] = 0, [e_{13}, e_{31}] = x$$

$$[y, e_{12}] = 2e_{12}, [y, e_{13}] = 0, [y, e_{31}] = 0, [y, e_{21}] = -2e_{21}, [e_{12}, e_{21}] = y$$

Thus, $\{e_{13}, e_{31}, x\}$ and $\{e_{12}, e_{21}, y\}$ are isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. Thus we complete the proof. $\square$

12 Classical (Semi) Simple Lie Algebras

Let V be a vector space over $\mathbb{F}$ and B a bilinear form: $V \times V \rightarrow \mathbb{F}$.

Set

$$O_{V,B}(\mathbb{F}) = \{a \in \mathfrak{gl}(V, \mathbb{F}) \mid B(a(u), v) + B(u, a(v)) = 0, \forall u, v \in V\} \subseteq \mathfrak{gl}(V, \mathbb{F}).$$

Proposition 12.0.1. $O_{V,B}(\mathbb{F})$ is a subalgebra of the Lie algebra $\mathfrak{gl}(V, \mathbb{F})$.

proof. We only need to verify $O_{V,B}$ is closed under bracket, as $O_{V,B}$ is trivially a subspace of $\mathfrak{gl}(V, \mathbb{F})$. Consider $a, b \in O_{V,B}$. Then

$$B(ab(u), v) = -B(b(u), a(v)) = B(u, ba(v))$$

from the property of a and b . Similarly,

$$B(ba(u), v) = -B(a(u), b(v)) = B(u, ab(v))$$

subtracting the second of these from the first,

$$B([a, b](u), v) = B(u, -[a, b](v)) = -B(u, [a, b](v))$$

by the bilinearity of B . Hence, $O_{V,B}$ is closed under the bracket. $\square$

Choosing the standard basis of $V = \mathbb{F}^n$ and denoting by B the matrix of the bilinear form in this basis, we can prove we get the subalgebra

$$\begin{aligned} O_{n,B}(\mathbb{F}) &= \{a \in \mathfrak{gl}(V, \mathbb{F}) \mid u^T a^T B v + u^T B a v = u^T (a^T B + B a) v = 0, \forall u, v \in V\} \\ &= \{a \in \mathfrak{gl}(n, \mathbb{F}) \mid a^T B + B a = 0\} \subseteq \mathfrak{gl}(n, \mathbb{F}), \end{aligned}$$

where a^T denotes the transpose of matrix a . We also write $O_{n,B}(\mathbb{F})$ as $\mathfrak{g}(n, B, \mathbb{F})$.

For different choices of basis, we get isomorphic Lie algebras $O_{n,B}(\mathbb{F})$. In the rest of this section, we will consider symmetric non-degenerate and skew-symmetric non-degenerate bilinear forms, which will give B_l , D_l and C_l separately. One should notice that the CSA of $\mathfrak{g}$ may vary between the choice of B , but the root will not. However, we should look at A_l first.

12.1 $A_l : \mathfrak{sl}(l+1, \mathbb{F})$

$\mathfrak{g} = \mathfrak{sl}(n, \mathbb{F})$, where $\mathbb{F}$ is an algebraically closed field with characteristic 0.

Proposition 12.1.1. $\mathfrak{g}$ is semisimple and simple when $\mathbb{F}$ is not characteristic 2 and $\text{char } \mathbb{F} \nmid n$.

proof. Let the Cartan subalgebra $\mathfrak{h}$ be the set of all traceless diagonal matrices. $\mathfrak{h}$ lies in $\mathfrak{d}_n$, the space of all diagonal matrices. Denote by ε_i the following linear function on $\mathfrak{d}_n$: $\varepsilon_i(A) = a_i$, the i -th coordinate of the diagonal of matrix A . The set $\{\varepsilon_i \mid i = 1, \dots, n\}$ is clearly a basis of $\mathfrak{d}_n^*$.

If $\text{char } \mathbb{F} \nmid n$, another basis is $\{\varepsilon_i - \varepsilon_{i+1}, \varepsilon_1 + \dots + \varepsilon_n \mid i = 1, \dots, n-1\}$, since it generates the first basis and has the same size. $\mathfrak{h} = \{a \in \mathfrak{d}_n \mid (\varepsilon_1 + \dots + \varepsilon_n)a = 0\}$. Therefore, $\mathfrak{h}^* = \mathfrak{d}_n^*/(\varepsilon_1 + \dots + \varepsilon_n)$, and $\{\varepsilon_i - \varepsilon_{i+1} \mid 1 \leq i \leq n-1\}$ is a basis for $\mathfrak{h}^*$. The root space decomposition of $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{F})$ is $\mathfrak{g} = \mathfrak{h} \oplus (\oplus_{\alpha \in \Delta} \mathfrak{g}_\alpha)$, where $\Delta = \{\varepsilon_i - \varepsilon_j \mid i \neq j\}$ and $\mathfrak{g}_{\varepsilon_i - \varepsilon_j} = \mathbb{F}e_{ij}$.

To prove $\mathfrak{sl}(n, \mathbb{F})$ is semisimple, we check the conditions of theorem 11.2.1. 1). is clearly true. 2). $[e_{ij}, e_{ji}] = e_{ii} - e_{jj}$, $(\varepsilon_i - \varepsilon_j)(e_{ii} - e_{jj}) = 2 \neq 0$. This is why $\mathbb{F}$ can not be characteristic 2. 3). When $\text{char } \mathbb{F} \nmid n$, the $\varepsilon_i - \varepsilon_j$ span $\mathfrak{h}^*$, so Δ spans $\mathfrak{h}^*$. This implies $\mathfrak{sl}(n, \mathbb{F})$ is semisimple for any $n \neq 2$ and $\text{char } \mathbb{F} \nmid n$.

To show it is simple, we prove that Δ is indecomposable. Let $\alpha = \varepsilon_i - \varepsilon_j$, $\beta = \varepsilon_s - \varepsilon_t$. Using proposition 11.2.1., let $\gamma_1 = \alpha$, $\gamma_2 = \varepsilon_j - \varepsilon_s$, $\gamma_3 = \beta$. This is a string from α to β as $\gamma_1 + \gamma_2 = \varepsilon_i - \varepsilon_s \in \Delta$ and $\gamma_2 + \gamma_3 = \varepsilon_j - \varepsilon_t \in \Delta$. This proves Δ is indecomposable and that $\mathfrak{sl}(n, \mathbb{F})$ is simple. $\square$

RMK: The above argument fails if $\text{char } \mathbb{F} \mid n$. As $\mathfrak{sl}(n, \mathbb{F})$ contains a non-trivial abelian ideal $C(\mathfrak{sl}(n, \mathbb{F}))$, since $I_n \in C(\mathfrak{sl}(n, \mathbb{F}))$. Moreover, the argument fails because Δ does not span $\mathfrak{h}^*$. When $\text{char } \mathbb{F} \mid n$, $I_n \in \mathfrak{h}$ and Δ is still $\{\varepsilon_i - \varepsilon_j \mid i \neq j\}$. Since $(\varepsilon_i - \varepsilon_j)(I_n) = 0$, Δ can't span $\mathfrak{h}^*$ for I_n^* is not in Δ .

12.2 B_l and D_l : $\mathfrak{so}(2n+1, \mathbb{F})$ and $\mathfrak{so}(2n, \mathbb{F})$

Now, consider the case where B is a symmetric non-degenerate bilinear form. If $\mathbb{F}$ is algebraically closed and $\text{char } \mathbb{F} \neq 2$, one can choose a basis in which the matrix of B is any symmetric non-degenerate matrix.

Write $\dim V = N$. We will choose a basis such that

$$B = \begin{pmatrix} 0 & 0 & 0 & \cdots & & 1 \\ 0 & 0 & & & \ddots & \\ 0 & & \ddots & 1 & & \vdots \\ \vdots & & 1 & \ddots & & 0 \\ & \ddots & & & 0 & 0 \\ 1 & & \cdots & 0 & 0 & 0 \end{pmatrix}$$

and denote by $\mathfrak{so}(N, \mathbb{F})$ the corresponding Lie algebra $O_{N,B}(\mathbb{F})$.

Proposition 12.2.1. $\mathfrak{so}(N, \mathbb{F}) = \{a \in \mathfrak{gl}(N, \mathbb{F}) \mid a + a' = 0\}$ where a' is the transposition of a with respect to the anti-diagonal.

proof. $\mathfrak{so}(N, \mathbb{F}) = \{a \in \mathfrak{gl}(N, \mathbb{F}) \mid a^T B + B a = 0\}$ where B is the matrix consisting of ones along the anti-diagonal.

As $B = B^T$, we have $a^T B = a^T B^T = (Ba)^T$. Viewing B as a permutation matrix, we get Ba permutes the rows by $\text{row}_i \mapsto \text{row}_{N+1-i}$. Transposing and reapplying B , we get $B(Ba)^T = a'$ and $BBa = a$. Hence we obtain the following sequence of implications

$$\begin{aligned} a^T B + B a &= 0 \\ (Ba)^T + B a &= 0 \\ B(Ba)^T + a &= 0. \end{aligned}$$

and finally $a' + a = 0$. Therefore, $\mathfrak{so}(N, \mathbb{F}) = \{a \in \mathfrak{gl}(N, \mathbb{F}) \mid a + a' = 0\}$. $\square$

Corollary 12.2.1. $\mathfrak{so}(2, \mathbb{F}) = \left\{ \begin{pmatrix} a_{11} & 0 \\ 0 & -a_{11} \end{pmatrix}, a_{11} \in \mathbb{F} \right\}$, which is one-dimensional abelian, hence not semisimple. We can easily find the basis of $\mathfrak{so}(2n+1, \mathbb{F})$ is $e_{ij} - e_{2n+2-j, 2n+2-i}$, $i+j < 2n+2$. Thus $\dim \mathfrak{so}(2n+1, \mathbb{F}) = 2n^2 + n$. The basis of $\mathfrak{so}(2n, \mathbb{F})$ is $e_{ij} - e_{2n+1-j, 2n+1-i}$, $i+j < 2n+1$. Thus $\dim \mathfrak{so}(2n, \mathbb{F}) = 2n^2 - n$. We denote: $\mathfrak{so}(2n+1, \mathbb{F})$ for $n \geq 1$ (type B) and $\mathfrak{so}(2n, \mathbb{F})$ for $n \geq 3$ (type D).

Proposition 12.2.2. Assume $N \geq 3$, then $\mathfrak{so}(N, \mathbb{F})$ is semisimple.

proof. We show this by the study of the root space decomposition. We denote $h_{ij} = e_{ij} - e_{2n+2-j, 2n+2-i}$, $i+j < 2n+2$ for $\mathfrak{so}(2n+1, \mathbb{F})$, and $h_{ij} = e_{ij} - e_{2n+1-j, 2n+1-i}$, $i+j < 2n+1$ for $\mathfrak{so}(2n, \mathbb{F})$. And denote $h_{ij} = -h_{ji}$, when $i+j \geq 2n+2$ ($2n+1$ for $\mathfrak{so}(2n, \mathbb{F})$).

Case 1: $N = 2n + 1$ (odd). Let

$$\mathfrak{h} = \left\{ \begin{pmatrix} a_1 & & & & & \\ & \ddots & & & & \\ & & a_n & & & \\ & & & 0 & & \\ & & & & -a_n & \\ & & & & & \ddots \\ & & & & & & -a_1 \end{pmatrix} \right\} \subseteq \mathfrak{so}(2n+1, \mathbb{F}),$$

This is a Cartan subalgebra since it contains a diagonal matrix with distinct entries. Clearly it's abelian, thus nilpotent. We only need to verify that if $[x, \mathfrak{h}] \subseteq \mathfrak{h}$, then $x \in \mathfrak{h}$. Say $h = \sum_{k=1}^n a_k h_{kk}$, we compute: (where the last step is an abuse of notation, if a_l doesn't exist, then $a_l = 0$, $l = i, j, 2n+2-i, 2n+2-j$.)

$$\begin{aligned} \text{adh}(h_{ij}) &= \left[\sum_{k=1}^n a_k h_{kk}, e_{ij} - e_{2n+2-j, 2n+2-i} \right] = \sum_{k=1}^n a_k [e_{kk} - e_{2n+2-k, 2n+2-k}, e_{ij} - e_{2n+2-j, 2n+2-i}] \\ &= \sum_{k=1}^n a_k (\delta_{ki} e_{kj} - \delta_{jk} e_{ik} - \delta_{2n+2-k, i} e_{2n+2-k, j} + \delta_{2n+2-k, j} e_{i, 2n+2-k} \\ &\quad - \delta_{k, 2n+2-j} e_{k, 2n+2-i} + \delta_{2n+2-i, k} e_{2n+2-j, k} + \delta_{kj} e_{2n+2-k, 2n+2-i} - \delta_{ik} e_{2n+2-j, 2n+2-k}) \\ &= \sum_{k=1}^n a_k (\delta_{ik} h_{kj} - \delta_{jk} h_{ik} - \delta_{2n+2-k, i} h_{2n+2-k, j} + \delta_{2n+2-k, j} h_{i, 2n+2-k}) \\ &= (a_i - a_j - a_{2n+2-i} + a_{2n+2-j}) h_{ij} \end{aligned}$$

Consider $h \in \mathfrak{h}$ with $a_i \neq a_j$, $i \neq j$. Then we can verify $\mathfrak{h}$ is a Cartan subalgebra. We should notice that $a_i, a_j, a_{2n+2-i}, a_{2n+2-j}$ is not always exists, for $1 \leq k \leq n$. Thus they exists iff $1 \leq l \leq n$, $l = i, j, 2n+2-i, 2n+2-j$.

Case 2: $N = 2n$ (even). Let

$$\mathfrak{h} = \left\{ \begin{pmatrix} a_1 & & & & \\ & \ddots & & & \\ & & a_n & & \\ & & & -a_n & \\ & & & & \ddots \\ & & & & & -a_1 \end{pmatrix} \right\} \subseteq \mathfrak{so}(2n, \mathbb{F}),$$

This is a Cartan subalgebra for the same reason. We can compute similarly: $\text{adh}(h_{ij}) = (a_i - a_j - a_{2n+1-i} + a_{2n+1-j}) h_{ij}$.

In both cases, $\dim \mathfrak{h} = n$ and $\varepsilon_1, \dots, \varepsilon_n$ form a basis $\mathfrak{h}^*$. Note that $\varepsilon_{N+1-j}|_{\mathfrak{h}} = -\varepsilon_j|_{\mathfrak{h}}$ and $\varepsilon_{\frac{N+1}{2}}|_{\mathfrak{h}} = 0$ if N is odd. All eigenvectors for $\text{ad} \mathfrak{h}$ are elements $h_{ij}, i, j \in \{1, 2, \dots, N\}$. For $\mathfrak{so}(2n+1, \mathbb{F})$, when $i, j \in \{1, \dots, n\}$, h_{ij} belongs to root $\varepsilon_i - \varepsilon_j$; when $i \in \{n+2, \dots, 2n\}$, $j \in \{1, \dots, n\}$, h_{ij} belongs to root $-\varepsilon_i - \varepsilon_j$; when $i \in \{1, \dots, n\}, j \in \{n+2, \dots, 2n\}$, h_{ij} belongs to root $\varepsilon_i + \varepsilon_j$; when $i = n+1, j \in \{1, \dots, n\}$, h_{ij} belongs to root $-\varepsilon_j$; when $j = n+1, i \in \{1, \dots, n\}$, h_{ij} belongs to root ε_i . The situation for $\mathfrak{so}(2n, \mathbb{F})$ is similar with the first three cases.¹

Hence the set of roots is: (For ε_i , we should notice that its inverse is in $\{\varepsilon_j | j \in \{1, \dots, n\}\}$.)

$$N = 2n + 1 : \Delta_{\mathfrak{so}(2n+1, \mathbb{F})} = \{\varepsilon_i - \varepsilon_j, \varepsilon_i, -\varepsilon_j, \varepsilon_i + \varepsilon_j, -\varepsilon_i - \varepsilon_j | i, j \in \{1, \dots, n\}, i \neq j\},$$

$$N = 2n : \Delta_{\mathfrak{so}(2n, \mathbb{F})} = \{\varepsilon_i - \varepsilon_j, \varepsilon_i + \varepsilon_j, -\varepsilon_i - \varepsilon_j | i, j \in \{1, \dots, n\}, i \neq j\}$$

¹One should always keep the blocks in the matrix in mind. We always assume $i + j < N + 1$ in all the analysis.

Then we can verify $\mathfrak{so}(N, \mathbb{F})$ is semisimple using the theorem 11.2.1. : (1), $\mathfrak{h}$ is obviously abelian, and $\dim \mathfrak{g}_\alpha = 1$, for all $\alpha \in \Delta$, which is clear; (2), $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{F}h_\alpha \subseteq \mathfrak{h}$ and $\alpha(h_\alpha) \neq 0$. We compute $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = [e_{ij} - e_{N+1-j, N+1-i}, e_{ji} - e_{N+1-i, N+1-j}] = h_{ii} - h_{jj} = h_\alpha \in \mathfrak{h}$. As $\alpha(h_\alpha) \neq 0$, $\mathfrak{so}(N, \mathbb{F})$ is semisimple if $N \geq 3$. (3), $\mathfrak{h}^*$ is spanned by Δ . It is clear for type B. For (3) in case D, we have roots $\varepsilon_i - \varepsilon_j$ and $\varepsilon_i + \varepsilon_j$, adding and dividing by 2 (as $\text{char } \mathbb{F} \neq 2$) gives us ε_i , hence (3) holds. $\square$

Proposition 12.2.3. Show $\mathfrak{so}(N, \mathbb{F})$ is simple if $N = 3$ or $N \geq 5$ by showing that Δ is indecomposable.

proof. To show simple for $N = 3$ and $N \geq 5$, we show that Δ is indecomposable. This is clear for $N = 3$. We list pairs and corresponding paths for $N \geq 3$. $N = 5$ is done separately. For ease of notation, we write ε_i as i and remark that any root is connected to its negative by the path of length one.

- $(i + j) \rightarrow (j + k)$ via $(i + j, -k - i, j + k)$
- $(i + j) \rightarrow (i - j)$ via $(i + j, k - i, j - k, i - j)$
- $(i + j) \rightarrow (i)$ via $(i + j, -j, i)$

Therefore when $N > 5$, we may concatenate and find paths through $(i + j)$. When $N = 5$, we had the issue of connecting $(i + j)$ to $(i - j)$. As N is odd in this case, we may use the path $(i + j, -i, i - j)$. Therefore, $\mathfrak{so}(N, \mathbb{F})$ is simple for $N = 3$ and $N \geq 5$. $\square$

12.3 C_l : $\mathfrak{sp}(2l, \mathbb{F})$

Now consider the case where B is a skew-symmetric non-degenerate bilinear form. If $\mathbb{F}$ is of characteristic $\neq 2$, one can choose a basis in which the matrix of B is any skew-symmetric non-degenerate matrix where $N = \dim V = 2n$ (even). We get

$$\mathfrak{sp}(2n, \mathbb{F}) = \{a \in \mathfrak{gl}(2n, \mathbb{F}) \mid a^T B + Ba = 0\} \subseteq \mathfrak{gl}(2n, \mathbb{F}).$$

The best choice of B is

$$B = \begin{pmatrix} 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & & \ddots & \\ 0 & & \ddots & 1 & \\ \vdots & & -1 & \ddots & 0 \\ & \ddots & & & 0 & 0 \\ -1 & \cdots & 0 & 0 & 0 \end{pmatrix}$$

Proposition 12.3.1. $\mathfrak{sp}(2n, \mathbb{F}) = \left\{ \begin{pmatrix} m & p \\ q & t \end{pmatrix} \mid \text{all } m, p, q, t \text{ are } n \times n \text{ such that } p = p', q = q', t = -m' \right\}$.

Moreover, the anti-diagonal of elements in $\mathfrak{sp}(2n, \mathbb{F})$ is arbitrary and the dimension of $\mathfrak{sp}(2n, \mathbb{F})$ is $2n^2 + n$.

proof. Denote $B = \begin{pmatrix} 0 & j \\ -j & 0 \end{pmatrix}$. And we find that the inverse of j is j thus the inverse of B is B^T . Thus from $a^T B + Ba = 0$, we get

$$\begin{pmatrix} m^T & q^T \\ p^T & t^T \end{pmatrix} \begin{pmatrix} 0 & j \\ -j & 0 \end{pmatrix} + \begin{pmatrix} 0 & j \\ -j & 0 \end{pmatrix} \begin{pmatrix} m & p \\ q & t \end{pmatrix} = \begin{pmatrix} -q^T j + j q & m^T j + j t \\ -t^T j - j m & p^T j - j p \end{pmatrix} = 0.$$

Thus we have $p^T j = jp$, $q^T j = jq$, $m^T j = -jt$. Say $p = (p_{ij})$, thus $p^T = (p_{ji}) = jpj = (p_{n+1-i, n+1-j})$. Thus we know that $p = p'$. Similarly, we can prove $q = q'$, $t = -m'$. For $p = p'$, $q = q'$, we know that the anti-diagonal of elements in $\mathfrak{sp}(2n, \mathbb{F})$ is arbitrary. Consider $p = p'$, thus its dimension is $1 + 2 + \cdots + n =$

$\frac{n(n+1)}{2}$. Similarly, the dimension of q is $\frac{n(n+1)}{2}$. t is determined by m , thus the dimension provided by m and t is the dimension of m , i.e. n^2 dimensions. Thus the dimension of $\mathfrak{sp}(2n, \mathbb{F})$ is $2n^2 + n$. $\square$

Corollary 12.3.1. We denote $N_1 = \{1, \dots, n\}$, $N_2 = \{n+1, \dots, 2n\}$. From $p = p', q = q', t = -m'$, we know that the basis of $\mathfrak{sp}(2n, \mathbb{F})$ is:

$$\begin{cases} e_{ij} + e_{2n+1-j, 2n+1-i}, & i \in N_2, j \in N_1, i+j < 2n+1 \text{ or } j \in N_2, i \in N_1, i+j < 2n+1 \\ e_{ij} - e_{2n+1-j, 2n+1-i}, & i, j \in N_1 \\ e_{ij}, & i+j = 2n+1 \end{cases}$$

We denote f_{ij} for the basis of the first case, h_{ij} for the basis of the second case and e_{ij} for the last case.

Proposition 12.3.2. $\mathfrak{h} = \{\text{diag}(a_1, \dots, a_n, -a_n, \dots, -a_1) \mid a_i \in \mathbb{F}\}$, the set of all diagonal matrices in $\mathfrak{sp}(2n, \mathbb{F})$, is the CSA of $\mathfrak{sp}(2n, \mathbb{F})$. All eigenvectors for $\text{ad}\mathfrak{h}$ are f_{ij}, h_{ij}, e_{ij} . And the set of roots is

$$\Delta_{\mathfrak{sp}(2n, \mathbb{F})} = \{\varepsilon_i - \varepsilon_j, 2\varepsilon_i, -2\varepsilon_i, \varepsilon_i + \varepsilon_j, -\varepsilon_i - \varepsilon_j \mid i, j \in \{1, \dots, n\}, i \neq j\}.$$

proof. We assume $\mathfrak{h}$ is a CSA and do the computation first. For $h = \sum_{k=1}^n a_k h_{kk}$, we compute the action of $\text{ad}h$ on three kinds of basis. After the computation, we give the root for every eigenvectors (We claim the basis are eigenvectors here and verify during the computation.).

$$\begin{aligned} \text{ad}h(f_{ij}) &= \left[\sum_{k=1}^n a_k h_{kk}, e_{ij} + e_{2n+1-j, 2n+1-i} \right] = \sum_{k=1}^n a_k [e_{kk} - e_{2n+1-k, 2n+1-k}, e_{ij} + e_{2n+1-j, 2n+1-i}] \\ &= \sum_{k=1}^n a_k (\delta_{ki} e_{kj} - \delta_{jk} e_{ik} - \delta_{2n+1-k, i} e_{2n+1-k, j} + \delta_{2n+1-k, j} e_{i, 2n+1-k} \\ &\quad + \delta_{k, 2n+1-j} e_{k, 2n+1-i} - \delta_{2n+1-i, k} e_{2n+1-j, k} - \delta_{kj} e_{2n+1-k, 2n+1-i} + \delta_{ik} e_{2n+1-j, 2n+1-k}) \\ &= \sum_{k=1}^n a_k (\delta_{ik} f_{kj} - \delta_{jk} f_{ik} - \delta_{2n+1-k, i} f_{2n+1-k, j} + \delta_{2n+1-k, j} f_{i, 2n+1-k}) \\ &= (a_i - a_j - a_{2n+1-i} + a_{2n+1-j}) f_{ij} \end{aligned}$$

For a_k is from a_1 to a_n , thus the root for f_{ij} , $i \in N_2, j \in N_1, i+j < 2n+1$ is $-\varepsilon_i - \varepsilon_j$ and the root for f_{ij} , $j \in N_2, i \in N_1, i+j < 2n+1$ is $\varepsilon_i + \varepsilon_j$.

The computation for h_{ij} is exactly the same as which in $\mathfrak{so}(N+1, \mathbb{F})$, thus $\text{ad}h(h_{ij}) = (a_i - a_j)h_{ij}$ (We using the condition $i, j \in N_1$). Thus h_{ij} are eigenvectors and the root of h_{ij} is $\varepsilon_i - \varepsilon_j$.

$$\begin{aligned} \text{ad}h(e_{i, 2n+1-i}) &= \left[\sum_{k=1}^n a_k h_{kk}, e_{i, 2n+1-i} \right] = \sum_{k=1}^n a_k [e_{kk} - e_{2n+1-k, 2n+1-k}, e_{i, 2n+1-i}] \\ &= \sum_{k=1}^n a_k (\delta_{ki} e_{k, 2n+1-i} - \delta_{2n+1-i, k} e_{ik} - \delta_{2n+1-k, i} e_{2n+1-k, 2n+1-i} + \delta_{ik} e_{i, 2n+1-k}) \\ &= \sum_{k=1}^n 2a_k (\delta_{ki} e_{k, 2n+1-i} - \delta_{2n+1-i, k} e_{ik}) = 2(a_i - a_{2n+1-i}) e_{ij} \end{aligned}$$

Thus e_{ij} are eigenvectors of h and its root is 2ε , if $i \in N_1$ and -2ε if $i \in N_2$. Thus we have proved All eigenvectors for $\text{ad}\mathfrak{h}$ are f_{ij}, h_{ij}, e_{ij} . And the set of roots is $\Delta_{\mathfrak{sp}(2n, \mathbb{F})} = \{\varepsilon_i - \varepsilon_j, 2\varepsilon_i, -2\varepsilon_i, \varepsilon_i + \varepsilon_j, -\varepsilon_i - \varepsilon_j \mid i, j \in \{1, \dots, n\}, i \neq j\}$. We will prove $\mathfrak{h}$ is a CSA to finish the proof.

Clearly, $\mathfrak{h}$ is abelian, thus nilpotent. We have shown that all basis are eigenvectors, thus $\mathfrak{h}$ is self-normalizing, thus is a CSA. $\square$

Proposition 12.3.3. $\mathfrak{sp}(2n, \mathbb{F})$ is always semisimple. Moreover, $\Delta_{\mathfrak{sp}(2n, \mathbb{F})}$ always indecomposable and deduce that $\mathfrak{sp}(2n, \mathbb{F})$ simple for all $n \geq 2$.

proof. First, let's prove $\mathfrak{g}$ is semisimple. Clearly, $\mathfrak{h}$ is abelian and $\dim \mathfrak{g}_\alpha = 1$, for all $\alpha \in \Delta$. Then we should consider $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$. If $\alpha = \varepsilon_i - \varepsilon_j$, the case is the same as which in $\mathfrak{so}(N+1, \mathbb{F})$; if $\alpha = 2\varepsilon_i$, then $\mathfrak{g}_\alpha = \mathbb{F}e_{i, 2n+1-i}$, $i \in N_1$ and $\mathfrak{g}_{-\alpha} = \mathbb{F}e_{2n+1-i, i}$, thus $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{F}[e_{i, 2n+1-i}, e_{2n+1-i, i}] = \mathbb{F}(e_{ii} - e_{2n+1-i, 2n+1-i})$, which is in $\mathfrak{h}$ and $\alpha(e_{ii} - e_{2n+1-i, 2n+1-i}) = 2 \neq 0$; if $\alpha = \varepsilon_i + \varepsilon_j$, the eigenvector is f_{ij} , $j \in N_2, i \in N_1, i+j < 2n+1$ and the eigenvector of $-\varepsilon_i - \varepsilon_j$ is f_{ji} , thus $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{F}[f_{ij}, f_{ji}] = \mathbb{F}[e_{ij} + e_{2n+1-j, 2n+1-i}, e_{ji} + e_{2n+1-i, 2n+1-j}] = \mathbb{F}(h_{ii} + h_{2n+1-j, 2n+1-j}) \subseteq \mathfrak{h}$, $\alpha(h_{ii} + h_{2n+1-j, 2n+1-j}) = 2 \neq 0$. Thus the condition (2) is satisfied. (3) is satisfied obviously. Thus $\mathfrak{g}$ is semisimple.

The proof of simplicity is almost the same as $\mathfrak{so}(N+1, \mathbb{F})$, we will not repeat. $\square$

RMK: Thus we get four series of simple Lie algebras $A_n = \mathfrak{sl}(n, \mathbb{F}) (n \geq 1)$, $B_n = \mathfrak{so}(2n+1, \mathbb{F}) (n \geq 1)$, $C_n = \mathfrak{sp}(2n, \mathbb{F}) (n \geq 1)$, $D_n = \mathfrak{so}(2n, \mathbb{F}) (n \geq 4)$ called the classical simple Lie algebras.¹

12.4 The properties of bilinear forms on simple Lie algebra

Proposition 12.4.1. Let $\mathfrak{g}$ be a finite dimensional simple Lie algebra. Then

- a) Any symmetric invariant bilinear form is either non-degenerate or identically zero.
- b) Any two non-degenerate such bilinear forms are proportional: $(a, b)_1 = \lambda(a, b)_2$.

proof. a) If $(\cdot, \cdot)$ is an invariant bilinear form and I is its kernel, then I is an ideal, hence $\mathfrak{g}$ simple implies that either $I = 0$ or $I = \mathfrak{g}$.

b) Choose a basis of $\mathfrak{g}$ and let B_i be the matrix of $(\cdot, \cdot)_i$ in the basis. $\det(B_i) \neq 0$. Consider $\det(B_1 - \lambda B_2) = \det(B_2) \det(B_1 B_2^{-1} - \lambda I) = 0$ iff λ is an eigenvalue of $B_1 B_2^{-1}$. Hence the form $(a, b)_1 - \lambda(a, b)_2$ ² is a degenerate, invariant, bilinear form as $\det$ is 0. Hence the form $(a, b)_1 - \lambda(a, b)_2$ is identically zero by (a), which implies $(a, b)_1 = \lambda(a, b)_2$. $\square$

Corollary 12.4.1. If $\mathfrak{g} \subseteq \mathfrak{gl}(N, \mathbb{F})$ is a simple Lie algebra, then the Killing form on $\mathfrak{g}$ is proportional to the trace form $(a, b) = \text{tr}(ab)$ on $\mathfrak{g}$, here we should consider the canonical representation.

Example 12.4.1. On $\mathfrak{gl}(N, \mathbb{F})$: (1) $\text{tr}(e_{ii}e_{ij}) = \delta_{ij}$ (with e_{ii} basis of $\mathfrak{d}_n$), hence the induced bilinear form on $\mathfrak{d}_n^* = (\varepsilon_i, \varepsilon_j) = \delta_{ij}$ (2).

Hence for all classical simple Lie algebras A, B, C, D , the Killing form is a positive constant multiple of (1) and on $\mathfrak{h}^*$ is a positive constant multiple of (2).

¹We require $n \geq 4$ for D_n for D_3 is the same as A_3 , which we will see later.

²The λ is set to be a nonzero eigenvalue of $B_1 B_2^{-1}$. The eigenvalue is always exists, for $B_1 B_2^{-1}$ is invertible not nilpotent.

13 Root Systems and Root Lattices

13.1 Root system

Definition 13.1.1. Let V be a finite dimensional real Euclidean space, i.e. V finite dimensional vector space over $\mathbb{R}$ with symmetric positive definite bilinear form $(\cdot, \cdot)$. Let $\Delta \subset V$ be a subset of V . Then the pair (V, Δ) is called a **root system** if:

- i) Δ finite, $0 \notin \Delta$, Δ spans V over $\mathbb{R}$;
- ii) (String Condition) For any $\alpha, \beta \in \Delta$, the set $\{\beta + j\alpha \mid j \in \mathbb{Z}\} \cap (\Delta \cup \{0\})$ is a string $\beta - p\alpha, \beta - (p-1)\alpha, \dots, \beta + q\alpha$ where $p, q \in \mathbb{Z}_{\geq 0}$, and $p - q = 2\frac{(\alpha, \beta)}{(\alpha, \alpha)}$;
- iii) For all $\alpha \in \Delta$, we have $k\alpha \in \Delta$ if and only if $k = 1$ or $k = -1$.

Example 13.1.1. The basic example: Let $\mathfrak{g}$ be a finite dimensional semisimple Lie algebra over an algebraically closed field $\mathbb{F}$ of characteristic 0, $\mathfrak{h}$ a Cartan subalgebra, $\Delta \subset \mathfrak{h}_{\mathbb{Q}}^*$ the set of roots, $(\cdot, \cdot)$ the Killing form on $\mathfrak{h}_{\mathbb{Q}}^*$ which is $\mathbb{Q}$ -valued and positive definite.

Let $V = \mathbb{R} \otimes \mathfrak{h}_{\mathbb{Q}}^*$, i.e. linear combinations of roots with real coefficients and extend the Killing form by bilinearity. Then the pair (V, Δ) is a root system, called root system of $\mathfrak{g}$.

This construction is independent of the choice of the Cartan subalgebra $\mathfrak{h}$ due to Chevalley's Theorem.

Proposition 13.1.1. Let (V, Δ) be a root system. Then Δ is indecomposable if and only if there does not exist non-trivial decomposition $(V, \Delta) = (V_1, \Delta_1) \oplus (V_2, \Delta_2)$ where $V = V_1 \oplus V_2$, $V_1 \perp V_2$, $\Delta_i \subset V_i$, and $\Delta = \Delta_1 \cup \Delta_2$. Moreover, the decomposition of $\Delta = \sqcup \Delta_i$ into indecomposable sets corresponds to decomposition of the root system in the orthogonal direct sum of indecomposable root systems.

proof. $(\Leftarrow)$ Suppose we have a decomposition $\Delta = \Delta_1 \sqcup \Delta_2$, $\Delta_i \subset V$, $\alpha \in \Delta_1$, $\beta \in \Delta_2$. Therefore, $\alpha + \beta \notin \Delta \cup \{0\}$, hence $q = 0$. As well, clearly $-\beta \in \Delta_2$, so $p = 0$. Therefore, $\frac{2(\alpha, \beta)}{(\beta, \beta)} = 0$, hence $(\alpha, \beta) = 0$. Let $V_i = \text{span}(\Delta_i)$, then by above $V_1 \perp V_2$, and thus $V = V_1 \oplus V_2$.

$(\Rightarrow)$ Suppose $V = V_1 \oplus V_2$, $V_1 \perp V_2$. Choose $\Delta_i = \Delta \cap V_i$. We show $\Delta_1 \sqcup \Delta_2$ is a decomposition. In the contrary case, choose $\alpha \in \Delta_1, \beta \in \Delta_2$ and suppose $\alpha + \beta \in \Delta \cup \{0\}$. Then, $\alpha + \beta \neq 0$ since $(\alpha, -\alpha) \neq 0$, so $\alpha + \beta \in \Delta$. Without loss of generality, $\alpha + \beta \in \Delta_1 \subset V_1$, as $\beta \in \Delta_2 \subset V_2$ and $V_1 \perp V_2$, we have $0 = (\alpha + \beta, \beta) = (\alpha, \beta) + (\beta, \beta) = (\beta, \beta)$. This is a contradiction, hence $\alpha + \beta \notin \Delta$, so we have a decomposition.

By the above argument, it is clear that the decomposition into indecomposables corresponds to the orthogonal decomposition with respect to $(\cdot, \cdot)$. $\square$

Definition 13.1.2. (V, Δ) is called indecomposable if it cannot be decomposed into a non-trivial orthogonal direct sum. $r = \dim V$ is called the **rank of** (V, Δ) and elements of Δ are called **roots**.

RMK: We could find that $\dim_{\mathbb{F}} \mathfrak{h}^* = \dim_{\mathbb{Q}} \mathfrak{h}_{\mathbb{Q}}^* = \dim_{\mathbb{R}} V = r$, where $V = \mathbb{R} \otimes \mathfrak{h}_{\mathbb{Q}}^*$. Thus the definition of rank is consistent with which in definition 6.2.1.

Definition 13.1.3. An isomorphism of indecomposable root systems (V, Δ) and (V_1, Δ_1) is a vector space isomorphism $\varphi : V \rightarrow V_1$, such that $\varphi(\Delta) = \Delta_1$, and $(\varphi(\alpha), \varphi(\beta))_1 = c(\alpha, \beta)$ for all $\alpha, \beta \in \Delta$, where c is a positive constant, independent of α and β .

In particular, replacing $(\cdot, \cdot)$ by $c(\cdot, \cdot)$, where $c > 0$, we get, by definition, an isomorphic root system.

Example 13.1.2. There is only one root systems of rank 1 under isomorphism: $(\mathbb{R}, \Delta = \{\alpha, -\alpha\}), \alpha \neq 0, (\alpha, \beta) = \alpha\beta$. This root system is isomorphic to that of $\mathfrak{sl}(2, \mathbb{F}), \mathfrak{so}(3, \mathbb{F}),$ and $\mathfrak{sp}(2, \mathbb{F})$.

Proposition 13.1.2. Let (V, Δ) be an indecomposable root system with the bilinear form $(\cdot, \cdot)$. Then

(1) Any other bilinear form $(\cdot, \cdot)_1$ for which the string property holds (i.e. the α -string through β are same under two bilinear forms, for all $\alpha, \beta \in \Delta$.) is proportional to $(\cdot, \cdot)$, i.e. $(\alpha, \beta)_1 = c(\alpha, \beta)$ for some positive $c \in \mathbb{R}$, independent of α and β .

(2) If $(\alpha, \alpha) \in \mathbb{Q}$ for some $\alpha \in \Delta$, then $(\beta, \gamma) \in \mathbb{Q}$ for all $\beta, \gamma \in \Delta$.

proof. (1) Fix $\alpha \in \Delta$. Since (V, Δ) is indecomposable, for any $\beta \in \Delta$ there exists a sequence $\gamma_0, \gamma_1, \dots, \gamma_k$ such that $\alpha = \gamma_0, \beta = \gamma_k, (\gamma_i, \gamma_{i+1}) \neq 0$ for all $i = 0, \dots, k-1$.¹ Define c by $(\alpha, \alpha)_1 = c(\alpha, \alpha)$. By the string property $p - q = 2 \frac{(\alpha, \gamma_1)}{(\alpha, \alpha)} = 2 \frac{(\alpha, \gamma_1)_1}{(\alpha, \alpha)_1}$. Hence $(\alpha, \gamma_1)_1 = c(\alpha, \gamma_1)$. Likewise, by the string property, $\frac{2(\alpha, \gamma_1)}{(\gamma_1, \gamma_1)} = \frac{2(\alpha, \gamma_1)_1}{(\gamma_1, \gamma_1)_1}$. Hence $(\gamma_1, \gamma_1)_1 = c(\gamma_1, \gamma_1)$. Continuing this way, we show that $(\gamma_2, \gamma_2)_1 = c(\gamma_2, \gamma_2), \dots, (\beta, \beta)_1 = c(\beta, \beta)$. Since $\frac{2(\alpha, \beta)}{(\alpha, \alpha)} = \frac{2(\alpha, \beta)_1}{(\alpha, \alpha)_1}$, we conclude that $(\alpha, \beta)_1 = c(\alpha, \beta)$ for all $\beta \in \Delta$. And for $(\beta, \beta)_1 = c(\beta, \beta)$, we can do the same thing for β , thus $(\beta, \theta)_1 = c(\beta, \theta)$ for all $\theta \in \Delta$. Thus $(\alpha, \beta)_1 = c(\alpha, \beta)$ for all $\alpha, \beta \in \Delta$. Since Δ spans V , we conclude that (1) holds.

The same argument proves (2). $\square$

13.2 Root lattice

Definition 13.2.1. A lattice in an Euclidean space V is a discrete (A subset $\Delta \subseteq V$ is discrete if every bounded subset of V contains only finitely many points of Δ .) subgroup $(Q, +)$ of V , which spans V over $\mathbb{R}$, i.e. $\mathbb{R}Q = V$.

Example 13.2.1. For example, $\mathbb{Z}^n \subset \mathbb{R}^n$. $\{1, \sqrt{2}\} \subset \mathbb{R}$, then $\mathbb{Z}\{1, \sqrt{2}\}$ is not a discrete set. For we can consider the "subtraction with a remainder", the remainders can be represented as $a + b\sqrt{2}$, $a, b \in \mathbb{Z}$. Thus the remainder can be infinitely small, otherwise $\sqrt{2}$ is a rational number.

Proposition 13.2.1. If Δ is a finite set in an Euclidean space V , spanning V over $\mathbb{R}$, such that $(\alpha, \beta) \in \mathbb{Q}$ for all $\alpha, \beta \in \Delta$, then $\mathbb{Z}\Delta$ is a lattice in V .

proof. The only thing to prove is that $\mathbb{Z}\Delta$ is a discrete set. Choose a basis $\beta_1, \dots, \beta_r$ of V among the vectors of Δ . Then for any $\alpha \in \Delta$, we have $\alpha = \sum_{i=1}^r c_i \beta_i, c_i \in \mathbb{R}$. Hence, $(\alpha, \beta_j) = \sum_{i=1}^r c_i (\beta_i, \beta_j)$. We thus get a system of linear equations with j varying from 1 to r . But $((\beta_i, \beta_j))_{i,j=1}^r$ is a Gram matrix of a basis, hence it is non-singular. Hence the system of linear equations has a solution and c_i can be computed by Cramer's rule, so all $c_i \in \mathbb{Q}$. And since Δ is finite, we could find a $N \in \mathbb{Z}$, $Nc_i \in \mathbb{Z}$. Thus $\alpha \in \frac{1}{N}\mathbb{Z}\{\beta_1, \dots, \beta_r\}, \forall \alpha \in \Delta$ and we conclude that $\mathbb{Z}\Delta \subset \frac{1}{N}\mathbb{Z}\{\beta_1, \dots, \beta_r\}$, where N is a positive integer. But $\frac{1}{N}\mathbb{Z}\{\beta_1, \dots, \beta_r\}$ is discrete, hence $\mathbb{Z}\Delta$ is discrete. $\square$

Corollary 13.2.1. If (V, Δ) is a root system, then $Q := \mathbb{Z}\Delta$ is a lattice, called **the root lattice**.

proof. If (V, Δ) is a root system, then V is spanned by Δ over $\mathbb{R}$ naturally. The corollary holds if (V, Δ) is indecomposable for proposition 13.1.2., hence holds for any root system (V, Δ) . $\square$

We will list four series of root systems of rank r known to us. In all cases we assume $(\varepsilon_i, \varepsilon_j) = \delta_{ij}$, for definition 13.1.3., we know that bilinear forms differ by a constant multiple are regarded as identical up to isomorphism. We will give the results first. The space V is obvious for B, C, D. For A, we know $\dim V = n - 1$ and $\mathbb{R}\Delta \subseteq V$, thus V is what we need. We only need to prove Q , where Q of B is obvious.

Type	$\mathfrak{g}$	V	Δ	Q
A	$\mathfrak{sl}(r+1, \mathbb{F})$	$\{\sum_{i=1}^{r+1} a_i \varepsilon_i a_i \in \mathbb{R}, \sum_{i=1}^{r+1} a_i = 0\}$	$\{\varepsilon_i - \varepsilon_j 1 \leq i, j \leq r+1, i \neq j\}$	$\{\sum_{i=1}^{r+1} a_i \varepsilon_i a_i \in \mathbb{Z}, \sum_{i=1}^{r+1} a_i = 0\}$
B	$\mathfrak{so}(2r+1, \mathbb{F})$	$\{\sum_{i=1}^r a_i \varepsilon_i a_i \in \mathbb{R}\}$	$\{\pm \varepsilon_i \pm \varepsilon_j, \pm \varepsilon_i 1 \leq i, j \leq r, i \neq j\}$	$\{\sum_{i=1}^r a_i \varepsilon_i a_i \in \mathbb{Z}\}$
C	$\mathfrak{sp}(2r, \mathbb{F})$	$\{\sum_{i=1}^r a_i \varepsilon_i a_i \in \mathbb{R}\}$	$\{\pm \varepsilon_i \pm \varepsilon_j, \pm 2\varepsilon_i 1 \leq i, j \leq r, i \neq j\}$	$\{\sum_{i=1}^r a_i \varepsilon_i a_i \in \mathbb{Z}, \sum_{i=1}^r a_i \in 2\mathbb{Z}\}$
D	$\mathfrak{so}(2r, \mathbb{F}), r \geq 3$	$\{\sum_{i=1}^r a_i \varepsilon_i a_i \in \mathbb{R}\}$	$\{\pm \varepsilon_i \pm \varepsilon_j 1 \leq i, j \leq r, i \neq j\}$	$\{\sum_{i=1}^r a_i \varepsilon_i a_i \in \mathbb{Z}, \sum_{i=1}^r a_i \in 2\mathbb{Z}\}$

¹See corollary 11.2.1.

Proposition 13.2.2. For A , $\mathbb{Z}\Delta = Q$, where $Q = \{\sum_{i=1}^{r+1} a_i \varepsilon_i | a_i \in \mathbb{Z}, \sum_{i=1}^{r+1} a_i = 0\}$.

sketch of proof. Clearly, $\mathbb{Z}\Delta \subseteq Q$. We only need to prove the reverse inclusion. Notice that:

$$\begin{aligned} \sum_{i=1}^{r+1} a_i \varepsilon_i &= a_1(\varepsilon_1 - \varepsilon_2) + (a_1 + a_2)(\varepsilon_2 - \varepsilon_3) + \cdots \\ &\quad + (a_1 + a_2 + \cdots + a_r)(\varepsilon_r - \varepsilon_{r+1}) + (a_1 + \cdots + a_{r+1})(\varepsilon_{r+1}), \end{aligned}$$

□

Proposition 13.2.3. For C , $\mathbb{Z}\Delta = Q$, where $Q = \{\sum_{i=1}^r a_i \varepsilon_i | a_i \in \mathbb{Z}, \sum_{i=1}^r a_i \in 2\mathbb{Z}\}$.

proof. $\mathbb{Z}\Delta_{C_r} \subset Q_{C_r}$ as each member of Δ_{C_r} is of the form $a_1 \varepsilon_i + a_2 \varepsilon_j$ with $a_1 = \pm 1, a_2 = \pm 1$. So each element of $\mathbb{Z}\Delta_{C_r}$ is of the form $\sum_{i=1}^r a_i \varepsilon_i$ where $a_i \in \mathbb{Z}, \sum_{i=1}^r a_i \in 2\mathbb{Z}$.

$Q_{C_r} \subset \mathbb{Z}\Delta_{C_r}$ as any $\sum_{i=1}^r a_i \varepsilon_i$ with $a_i \in \mathbb{Z}, \sum_{i=1}^r a_i \in 2\mathbb{Z}$, can be written in the form

$$\begin{aligned} \sum_{i=1}^r a_i \varepsilon_i &= a_1(\varepsilon_1 - \varepsilon_2) + (a_2 + a_1)(\varepsilon_2 - \varepsilon_3) + \cdots \\ &\quad + (a_r + a_{r-1} + \cdots + a_1)(\varepsilon_r - \varepsilon_1) + (a_r + \cdots + a_1)\varepsilon_1, \end{aligned}$$

All these coefficients belong to $\mathbb{Z}$ and the final coefficient of ε_1 belongs to $2\mathbb{Z}$ so can be written as $\frac{a_r + \cdots + a_1}{2} \cdot 2\varepsilon_1$, hence $\sum_{i=1}^r a_i \varepsilon_i \in \mathbb{Z}\Delta_{C_r}$. □

Proposition 13.2.4. For D , $\mathbb{Z}\Delta = Q$, where $Q = \{\sum_{i=1}^r a_i \varepsilon_i | a_i \in \mathbb{Z}, \sum_{i=1}^r a_i \in 2\mathbb{Z}\}$.

sketch of proof. The proof for Q_{D_r} is identical except $2\varepsilon_i \in \Delta_{D_r}$, so write

$$\begin{aligned} \sum_{i=1}^r a_i \varepsilon_i &= a_1(\varepsilon_1 - \varepsilon_2) + (a_2 + a_1)(\varepsilon_2 - \varepsilon_3) + \cdots \\ &\quad + \frac{(a_r + a_{r-1} + \cdots + a_1)}{2}(\varepsilon_r - \varepsilon_1) + \frac{(a_r + \cdots + a_1)}{2}(\varepsilon_r - \varepsilon_2) \\ &\quad + \frac{(a_r + \cdots + a_1)}{2}(\varepsilon_1 + \varepsilon_2), \end{aligned}$$

as $r \geq 3$, and this belongs to $\mathbb{Z}\Delta_{D_r}$. □

Now, we have proved the table.

13.3 E_8, E_7 and E_6

Definition 13.3.1. A lattice Q is called **integral** (resp. **even**) if $(\alpha, \beta) \in \mathbb{Z}$ (resp. $(\alpha, \alpha) \in 2\mathbb{Z}$) for all $\alpha, \beta \in Q$. We have a direct corollary from the definition that a subgroup of an even lattice is an even lattice.

Note that an even lattice is always integral: if $\alpha, \beta \in Q$, Q even, then $(\alpha + \beta, \alpha + \beta) = (\alpha, \alpha) + (\beta, \beta) + 2(\alpha, \beta)$, $(\alpha, \alpha), (\beta, \beta) \in 2\mathbb{Z}$. Hence $2(\alpha, \beta) \in 2\mathbb{Z}$, so $(\alpha, \beta) \in \mathbb{Z}$.

Now, let's consider E_r , which is a lattice. r is a positive integer, $E_r \subset \mathbb{R}^r, (\varepsilon_i, \varepsilon_j) = \delta_{ij}$:

$$E_r = \left\{ \sum_{i=1}^r a_i \varepsilon_i \mid \text{either all } a_i \in \mathbb{Z} \text{ or all } a_i \in \mathbb{Z} + \frac{1}{2}, \text{ and } \sum_{i=1}^r a_i \in 2\mathbb{Z} \right\}$$

Proposition 13.3.1. E_r is an even lattice if and only if r is divisible by 8.

proof. We use that $a^2 \pm a \in 2\mathbb{Z}$ if $a \in \mathbb{Z}$, thus $a^2 \equiv a \pmod{2}$ and $\sum_{i=1}^r a_i^2 \equiv \sum_{i=1}^r a_i \pmod{2}$. Let $\alpha = \sum_{i=1}^r a_i \varepsilon_i \in E_r$.

Case 1: all $a_i \in \mathbb{Z}$, then $(\alpha, \alpha) = \sum_{i=1}^r a_i^2 \equiv \sum_{i=1}^r a_i \equiv 0 \pmod{2}$ by the condition of E_r , so $(\alpha, \alpha) \in 2\mathbb{Z}$.

Case 2: write $\alpha = \rho + \beta$ where $\rho = (\frac{1}{2}, \dots, \frac{1}{2})$ and β has integer coefficients b_i . Then $(\alpha, \alpha) = (\rho, \rho) + 2(\rho, \beta) + (\beta, \beta) = (\rho, \rho) + \sum_{i=1}^r (b_i^2 + b_i) = \frac{r}{4} + n, n \in 2\mathbb{Z}$. So (α, α) is even if and only if r is a multiple of 8. $\square$

We will use the following theorem to prove (Δ_{E_8}, V) is a root system. Having this theorem, we only need to verify the form of Δ and Δ spans V over $\mathbb{R}$.

Theorem 13.3.1. Let Q be an even lattice in an Euclidean space V , and assume the subset $\Delta = \{\alpha \in Q \mid (\alpha, \alpha) = 2\}$ spans V over $\mathbb{R}$. Then (V, Δ) is a root system.

proof. Axiom (i). of a root system is clear, as is axiom (iii): if $(\alpha, \alpha) = 2$, then $n\alpha \in \Delta$ iff $(n\alpha, n\alpha) = 2$ iff $n = \pm 1$. It remains to show the string property.

Reversing the sign of α if necessary, we may assume $(\alpha, \beta) \geq 0$. Note that for $\alpha, \beta \in \Delta$ we have:

$$0 \leq (\alpha - \beta, \alpha - \beta) = (\alpha, \alpha) - 2(\alpha, \beta) + (\beta, \beta) = 4 - 2(\alpha, \beta),$$

where $(\alpha, \beta) \in \mathbb{Z}, (\alpha, \beta) \geq 0$. So the only possibilities are $(\alpha, \beta) = 0, 1$ or 2 . In the last case, we have $(\alpha - \beta, \alpha - \beta) = 0$, i.e. $\alpha = \beta$, hence $q = 0, p = 2$, so $p - q = (\alpha, \beta) = 2$ and the string property is satisfied; for $(\alpha, \beta) = 1$, for $(\alpha - \beta, \alpha - \beta) = 2$, $(\alpha - \beta, \alpha - \beta) = 6$, we have $\alpha - \beta \in \Delta, \alpha + \beta \notin \Delta$, so $p = 1, q = 0$, and $p - q = (\alpha, \beta)$; for $(\alpha, \beta) = 0$, $\alpha + \beta \notin \Delta, \alpha - \beta \notin \Delta, p = 0, q = 0$, and $p - q = (\alpha, \beta)$. So the string property holds generally, and the theorem holds. $\square$

Proposition 13.3.2. (Δ_{E_8}, V) is a root system. And we claim it's indecomposable.

$$\begin{aligned} \Delta_{E_8} &:= \{\alpha \in E_8 \mid (\alpha, \alpha) = 2\} \\ &= \{\pm \varepsilon_i \pm \varepsilon_j \mid i \neq j\} \cup \left\{ \frac{1}{2}(\pm \varepsilon_1 \pm \dots \pm \varepsilon_8) \mid \text{even number of minus signs} \right\}, \end{aligned}$$

$$|\Delta_{E_8}| = 240, \mathbb{R}\Delta_{E_8} = V.$$

proof.

$$E_8 = \left\{ \sum_{i=1}^8 a_i \varepsilon_i \mid \text{all } a_i \in \mathbb{Z}, \sum_{i=1}^8 a_i \in 2\mathbb{Z} \right\} \cup \left\{ \sum_{i=1}^8 a_i \varepsilon_i \mid \text{all } a_i \in \mathbb{Z} + \frac{1}{2}, \sum_{i=1}^8 a_i \in 2\mathbb{Z} \right\}.$$

The elements from the first set satisfying $(\alpha, \alpha) = 2$ are clearly $\{\pm \varepsilon_i \pm \varepsilon_j \mid i \neq j\}$ as $(\varepsilon_i, \varepsilon_j) = \delta_{ij}$, and the second set must have all $a_i = \pm \frac{1}{2}$ else $(\alpha, \alpha) > 2$, and as $\sum_{i=1}^8 a_i \in 2\mathbb{Z}$, there must be an even number of minus signs.

$$\begin{aligned} |\Delta_{E_8}| &= |\{\varepsilon_i + \varepsilon_j \mid i \neq j\}| + |{-\varepsilon_i - \varepsilon_j \mid i \neq j\}| + |\{\varepsilon_i - \varepsilon_j \mid i \neq j\}| \\ &\quad + \left| \left\{ \frac{1}{2}(\pm \varepsilon_1 \pm \dots \pm \varepsilon_8) \mid \text{even number of minus signs} \right\} \right| \\ &= \frac{8 \cdot 7}{2} + \frac{8 \cdot 7}{2} + 8 \cdot 7 + 2^7 \\ &= 240. \end{aligned}$$

Clearly $\varepsilon_i = \frac{\varepsilon_i + \varepsilon_j}{2} + \frac{\varepsilon_i - \varepsilon_j}{2} \in \mathbb{R}\Delta_{E_8}$ and $\{\varepsilon_i\}$ form a basis of V , hence $\mathbb{R}\Delta = V$. $\square$

Proposition 13.3.3. (Δ_{E_7}, V) is a root system. And we claim it's indecomposable. Take $\rho = (\frac{1}{2}, \dots, \frac{1}{2})$ and let $\Delta_{E_7} = \{\alpha \in \Delta_{E_8} \mid (\alpha, \rho) = 0\}$, $Q_{E_7} = \{\alpha \in Q_{E_8} \mid (\alpha, \rho) = 0\}$, $V_{E_7} = \{v \in V_{E_8} \mid (v, \rho) = 0\}$. (V_{E_7}, Δ_{E_7}) is a root system of rank 7, and $|\Delta_{E_7}| = 126$.

proof. Clearly $\Delta_{E_7} = \{\alpha \in Q_{E_8} | (\alpha, \alpha) = 2 \text{ and } (\alpha, \rho) = 0\} = \{\alpha \in Q_{E_7} | (\alpha, \alpha) = 2\}$. Q_{E_7} is an even lattice in V_{E_7} as it is a subgroup of Q_{E_8} and clearly $\mathbb{R}Q_{E_7} = V_{E_7}$, so (V_{E_7}, Δ_{E_7}) is a root system of rank 7 as $V_{E_8} = V_{E_7} \oplus \mathbb{R}(\frac{1}{2}, \dots, \frac{1}{2})$, $V_{E_7} = (\frac{1}{2}, \dots, \frac{1}{2})^\perp$, and $\dim V_{E_8} = 8$.

$$\Delta_{E_7} = \{\varepsilon_i - \varepsilon_j | i \neq j\} \cup \left\{ \frac{\pm \varepsilon_1 \pm \dots \pm \varepsilon_8}{2} \mid 4 \text{ minus signs} \right\}.$$

Hence,

$$|\Delta_{E_7}| = 56 + \binom{8}{4} = 126.$$

□

Proposition 13.3.4. Let $\Delta_{E_6} = \{\alpha \in \Delta_{E_7} | (\alpha, \varepsilon_7 + \varepsilon_8) = 0\}$, $Q_{E_6} = \{\alpha \in Q_{E_7} | (\alpha, \varepsilon_7 + \varepsilon_8) = 0\}$, $V_{E_6} = \{v \in V_{E_7} | (v, \varepsilon_7 + \varepsilon_8) = 0\}$. (V_{E_6}, Δ_{E_6}) is a root system of rank 6, $|\Delta_{E_6}| = 72$ and is indecomposable.

proof. Clearly $\Delta_{E_6} = \{\alpha \in Q_{E_6} | (\alpha, \alpha) = 2\}$. Q_{E_6} is an even lattice in V_{E_6} as it is a subgroup of Q_{E_7} and clearly $\mathbb{R}Q_{E_6} = V_{E_6}$, so (V_{E_6}, Δ_{E_6}) is a root system of rank 6 as $V_{E_7} = V_{E_6} \oplus \mathbb{R}(\varepsilon_7 + \varepsilon_8)$, $V_{E_6} = (\varepsilon_7 + \varepsilon_8)^\perp$, and $\dim V_{E_7} = 7$.

$$\begin{aligned} \Delta_{E_6} = & \{\varepsilon_i - \varepsilon_j | 1 \leq i \neq j \leq 6, \text{ if } i = 7, j = 8, \text{ if } i = 8, j = 7\} \\ & \cup \left\{ \frac{\pm \varepsilon_1 \pm \dots \pm \varepsilon_8}{2} \mid 4 \text{ minus signs and } \varepsilon_7, \varepsilon_8 \text{ have opposite signs} \right\}. \end{aligned}$$

Hence,

$$|\Delta_{E_6}| = 6 \cdot 5 + 2 + 2 \binom{6}{3} = 72.$$

□

13.4 More indecomposable root systems: F_4 and G_2

Proposition 13.4.1. Define:

$$\begin{aligned} V &= \bigoplus_{i=1}^4 \mathbb{R}\varepsilon_i; (\varepsilon_i, \varepsilon_j) = \delta_{ij}; \\ Q_{F_4} &= \left\{ \sum_{i=1}^4 a_i \varepsilon_i \mid \text{all } a_i \in \mathbb{Z} \text{ or all } a_i \in \frac{1}{2} + \mathbb{Z} \right\}; \text{ and} \\ \Delta_{F_4} &= \{\alpha \in Q_{F_4} \mid (\alpha, \alpha) = 1 \text{ or } 2\}. \end{aligned}$$

(V, Δ_{F_4}) is an indecomposable root system of rank 4 with 48 roots.

proof. A simple verification shows $\Delta = \{\pm \varepsilon_i, \pm \varepsilon_i \pm \varepsilon_j : i \neq j\} \cup \{\frac{1}{2}(\pm \varepsilon_1 \pm \varepsilon_2 \pm \varepsilon_3 \pm \varepsilon_4)\}$, which has 48 elements.

All that is difficult is to check that the string property holds. For this, we tabulate enough α, β with the corresponding data $p, q, (\alpha, \beta), (\alpha, \alpha)$. I will use a shorthand for writing vectors—a string of four numbers $\eta_1 \eta_2 \eta_3 \eta_4$ represents the vector $\eta_1 \varepsilon_1 + \eta_2 \varepsilon_2 + \eta_3 \varepsilon_3 + \eta_4 \varepsilon_4$. Depending on how many numbers are presumed zero in such a string, we take the other numbers η to be ± 1 or $\pm \frac{1}{2}$ as appropriate.

In the last line here we have written δ_{k+} to mean 1 when there are exactly k positive η_i and zero otherwise. While considering each calculation of p and q , that the roots actually appear in strings without gaps is easily checked. Thus we have a root system.

To see that it is indecomposable, we need to note that all the roots are equivalent under the equivalence relation generated by $\alpha \sim \alpha'$ when $(\alpha, \alpha') \neq 0$. Note that $\varepsilon_i \sim \frac{1}{2}(\pm \varepsilon_1 \pm \varepsilon_2 \pm \varepsilon_3 \pm \varepsilon_4)$, so that $\{\varepsilon_i, \frac{1}{2}(\pm \varepsilon_1 \pm \varepsilon_2 \pm \varepsilon_3 \pm \varepsilon_4)\}$ is contained in an equivalence class. Next, $\pm \varepsilon_i \pm \varepsilon_j \sim \varepsilon_i$, showing that all roots are equivalent. Thus the root system is indecomposable.

α	β	p	q	$p - q$	(α, β)	(α, α)
1000	$\eta_1\eta_200$	$1 + \eta_1$	$1 - \eta_1$	$2\eta_1$	η_1	1
	$0\eta_2\eta_30$	0	0	0	0	1
	$\eta_1\eta_2\eta_3\eta_4$	$\frac{1}{2} + \eta_1$	$\frac{1}{2} - \eta_1$	$2\eta_1$	η_1	1
1100	η_1000	$\frac{1}{2}(1 + \eta_1)$	$\frac{1}{2}(1 - \eta_1)$	η_1	η_1	2
	$00\eta_30$	0	0	0	0	2
	$\eta_10\eta_30$	$\frac{1}{2}(1 + \eta_1)$	$\frac{1}{2}(1 - \eta_1)$	η_1	η_1	2
	$\eta_1\eta_2\eta_3\eta_4$	$\delta_{\eta_1=\eta_2=\frac{1}{2}}$	$\delta_{\eta_1=\eta_2=-\frac{1}{2}}$	$\eta_1 + \eta_2$	$\eta_1 + \eta_2$	2
$\frac{1}{2}(1111)$	η_1000	$\frac{1}{2}(1 + \eta_1)$	$\frac{1}{2}(1 - \eta_1)$	η_1	$\frac{1}{2}\eta_1$	1
	$\eta_1\eta_200$	$2\delta_{\eta_1=\eta_2=1}$	$2\delta_{\eta_1=\eta_2=-1}$	$\eta_1 + \eta_2$	$\frac{1}{2}(\eta_1 + \eta_2)$	1
	$\eta_1\eta_2\eta_3\eta_4$	$2\delta_{4+} + \delta_{3+} + \delta_{2+}$	$2\delta_{0+} + \delta_{1+} + \delta_{2+}$	$\sum \eta_i$	$\frac{1}{2}\sum \eta_i$	1

□

Proposition 13.4.2. *The following describes an indecomposable root system with 12 roots:*

$$\begin{aligned}
V_{G_2} &= V_{A_2}; \\
Q_{G_2} &= Q_{A_2}; \text{ and} \\
\Delta_{G_2} &= \{\alpha \in Q_{A_2} \mid (\alpha, \alpha) = 2 \text{ or } 6\}.
\end{aligned}$$

proof. $\Delta = \{(a_1, a_2, a_3) \mid a_1 + a_2 + a_3 = 0, a_i \in \mathbb{Z}, \sum a_i^2 \in \{2, 6\}\}$. Now in order to have square sum 2, exactly two of the a_i must equal ± 1 , so we have $\pm(1, -1, 0), \pm(1, 0, -1), \pm(0, 1, -1)$. In order to have square sum 6, exactly one must be ± 2 , and the other two must both be:

$$\mp 1 : \pm(2, -1, -1), \pm(-1, 2, -1), \pm(-1, -1, 2)$$

¹ We tabulate enough of the relevant quantities below to verify that G_2 is a root system, in the style of the previous exercise. (Here η stands for 1 or -1 .)

α	β	p	q	$p - q$	(α, β)	(α, α)
$(1, -1, 0)$	$(\eta, -\eta, 0)$	$1 + \eta$	$1 - \eta$	2η	2η	2
	$(\eta, 0, -\eta)$	$\frac{1}{2}(3 + \eta)$	$\frac{1}{2}(3 - \eta)$	η	η	2
	$(2\eta, -\eta, -\eta)$	$\frac{3}{2}(1 + \eta)$	$\frac{3}{2}(1 - \eta)$	3η	3η	2
	$(-\eta, -\eta, 2\eta)$	0	0	0	0	2
$(2, -1, -1)$	$(\eta, -\eta, 0)$	$\frac{1}{2}(1 + \eta)$	$\frac{1}{2}(1 - \eta)$	η	3η	6
	$(-\eta, 2\eta, -\eta)$	$\frac{1}{2}(1 - \eta)$	$\frac{1}{2}(1 + \eta)$	$-\eta$	-3η	6
	$(2\eta, -\eta, -\eta)$	$1 + \eta$	$1 - \eta$	2η	6η	6
	$(0, \eta, -\eta)$	0	0	0	0	6

Finally, to see that G_2 is irreducible, note that no two of the shorter roots are perpendicular, and each of the longer roots is not perpendicular to one of the shorter roots. □

¹This line feed is because of coding problem. Otherwise, this line will

14 Cartan Matrices and Dynkin Diagrams

14.1 Positive, negative, simple and highest root

Definition 14.1.1. Suppose (V, Δ) is a root system and $f : V \rightarrow \mathbb{R}^1$ is a linear map such that $f(\alpha) \neq 0$ for all $\alpha \in \Delta$. Then:

- (i) $\alpha \in \Delta$ is **positive** if $f(\alpha) > 0$ and **negative** if $f(\alpha) < 0$.
- (ii) A positive root is **simple** if it cannot be written as the sum of two positive roots.
- (iii) A **highest root** θ is a root where $f(\theta)$ is maximal; that is, $f(\theta) \geq f(\alpha)$ for all $\alpha \in \Delta$.

Definition 14.1.2. Here are some notations:

- Δ_+ is the set of positive roots, and Δ_- is the set of negative roots.
- $\Pi \subset \Delta_+$ is the set of simple roots.
- Π is **indecomposable** if it can't be written as a disjoint union of orthogonal sets $\Pi_1 \sqcup \Pi_2$ with $\Pi_1 \perp \Pi_2$.

Definition 14.1.3. Say Π is a basis if:

- Π is a basis of V
- every root β can be written as $\beta = \sum k_\alpha \alpha$, $\alpha \in \Pi$, and the coefficients k_α are all positive integers or all negative integers.

We will see that Π defined in definition 14.1.2 is a basis from the theorem below.

Theorem 14.1.1. (Dynkin). (a) If $\alpha, \beta \in \Pi$ and $\alpha \neq \beta$, then $\alpha - \beta \notin \Delta$ and $(\alpha, \beta) \leq 0$.

(b) Every positive root is a nonnegative integer linear combination of simple roots; i.e. $\Delta_+ \subseteq \mathbb{Z}_{\geq 0} \Pi$. Then we can define the **height** of a positive root, which is the sum of the coefficients of the combination.

(c) If $\alpha \in \Delta_+ \setminus \Pi$ then $\alpha - \gamma \in \Delta$ for some $\gamma \in \Pi$; moreover, then $\alpha - \gamma \in \Delta_+$.

(d) Π is a basis of V over $\mathbb{R}$ and of the lattice Q over $\mathbb{Z}$. Hence the integer linear combinations from part (b) are unique, which equals to the linear independence of Π .

(e) Δ is indecomposable if and only if Π is indecomposable.

proof. (a) This is a proof by contradiction. If $\alpha - \beta = \gamma \in \Delta$, then either

- $\gamma \in \Delta_+$, so $\alpha = \beta + \gamma$, which contradicts $\alpha \in \Pi$; or
- $\gamma \in \Delta_-$, so $\beta = \alpha + (-\gamma)$, which contradicts $\beta \in \Pi$.

Consider the α -string through β , we can find that $p = 0$ and $q \geq 0$, thus $(\alpha, \beta) \leq 0$.

(b) If α is simple then we're done. Otherwise, $\alpha = \beta + \gamma$ for some $\beta, \gamma \in \Delta_+$. Then $f(\alpha) = f(\beta) + f(\gamma)$, so both $f(\beta)$ and $f(\gamma)$ are strictly less than $f(\alpha)$. Repeat this process with β and γ until all summands are simple (which must happen in finitely many steps since Δ is finite), thus yielding α as a sum of simple roots.

(c) Suppose $\alpha \in \Delta_+ \setminus \Pi$. If $\alpha - \gamma \notin \Delta$ for all $\gamma \in \Pi$, then the string condition would imply $\frac{2(\gamma, \alpha)}{(\gamma, \gamma)} \leq 0$ for all $\gamma \in \Pi$ (specially as in (a)). Then by (b), we assume $\alpha = \sum_{\gamma \in \Pi} k_\gamma \gamma$:

$$(\alpha, \alpha) = \left(\alpha, \sum_{\gamma \in \Pi} k_\gamma \gamma \right) = \sum_{\gamma \in \Pi} (\alpha, k_\gamma \gamma) \leq 0,$$

which would imply $\alpha = 0$ and thus $\alpha \notin \Delta$. Hence $\alpha - \gamma \in \Delta$ for some $\gamma \in \Pi \subset \Delta$.

Now if $\alpha - \gamma = \beta \in \Delta_-$, then $\gamma = \alpha + (-\beta)$, which would contradict γ being simple.

(d) From (b) we have that Π spans Δ_+ over $\mathbb{Z}_{\geq 0}$. Then since $\Delta = \Delta_+ \cup \Delta_- = \Delta_+ \cup (-\Delta_+)$ and $Q = \mathbb{Z}\Delta$, Π spans Q over $\mathbb{Z}$ and thus spans V over $\mathbb{R}$.

To prove linear independence of Π , suppose the contrary—that there existed a nontrivial linear combination of simple roots $\sum_i k_i \alpha_i = 0$. Split this into positive parts $a_i, i \in I$ and negative parts $-b_j, j \in J$,

¹Here the function f is man-made.

where $b_j \geq 0$. Moving the negative parts to the other side and assuming $a_i = 0, i \in J$ and $b_j = 0, j \in I$, we then get two new series a_i and b_i . Thus we obtain:

$$\gamma := \sum_i a_i \alpha_i = \sum_i b_i \alpha_i,$$

where all a_i and b_i are non-negative and $a_i b_i = 0$ for all i . Since all α_i are positive, $f(\gamma) > 0$, so $\gamma \neq 0$ and $(\gamma, \gamma) > 0$. However, by (a) we also have

$$(\gamma, \gamma) = \left(\sum a_i \alpha_i, \sum b_j \alpha_j \right) \leq 0,$$

thus giving us a contradiction.

(e) If (V, Δ) is decomposable, assume $\Delta = \Delta_1 \cup \Delta_2$. We can find the basis Π_1 of Δ_1 and Π_2 of Δ_2 separately. Thus $\Pi_1 \cup \Pi_2$ is a basis of $\Delta_1 \cup \Delta_2 = \Delta$. For $(\Delta_1, \Delta_2) = 0$, $(\Pi_1, \Pi_2) = 0$ and thus the elements in $\Pi_1 \cup \Pi_2$ are linear independent. Π is a basis of Δ , thus can't be presented as a linear combination of elements in $\Pi_1 \cup \Pi_2$. However, $\Pi_1 \cup \Pi_2$ is a basis of $\Delta \supseteq \Pi$, thus $\Pi = \Pi_1 \cup \Pi_2$. Conversely, if Π decomposes as $\Pi_1 \sqcup \Pi_2$ with $\Pi_1 \perp \Pi_2$, then we will show $\Delta = (\mathbb{Z}\Pi_1 \cap \Delta) \cup (\mathbb{Z}\Pi_2 \cap \Delta)$.

Suppose the contrary—then $\alpha = \gamma_1 + \gamma_2$ for some $\alpha \in \Delta$ and $\gamma_i \in \mathbb{Z}_{\geq 0} \Pi_i$. By flipping the sign of α if necessary, we can assume $\alpha \in \Delta_+$. By (b), we can subtract simple roots until γ_1 is simple. Then $\Pi_1 \perp \Pi_2$ implies

$$p - q = \frac{2(\alpha, \gamma_1)}{(\gamma_1, \gamma_1)} = \frac{2(\gamma_1, \gamma_1)}{(\gamma_1, \gamma_1)} = 2$$

We must have $(p, q) = (2, 0)$, otherwise the γ_1 -string through α will have 5 roots, which contradicts to corollary 10.3.1. Thus we have $\beta := \alpha - 2\gamma_1 = \gamma_2 - \gamma_1$ is a root (it can't be zero since $\Pi_1 \perp \Pi_2$). Flipping the sign of β if necessary, we can assume $\beta \in \Delta_+$. However, the decomposition $\beta = \gamma_2 - \gamma_1$ (or $\gamma_1 - \gamma_2$ if we flipped the sign) can be made into an integer linear combination of simple roots with mixed signs (by expanding γ_2 in terms of simple roots). By (d), this linear combination is unique, so the non-negative linear combination guaranteed by (b) cannot exist. $\square$

Then we have a quick corollary using the string condition:

Corollary 14.1.1. Say θ is the highest root and α is a simple root, then for all α string through θ , $q=0$ and exists a α_0 , whose string through θ has $p \neq 0$. Thus we have the highest root subtract some simple root is a root.

Proposition 14.1.1. Prove that if (V, Δ) is an indecomposable root system and $f : V \rightarrow \mathbb{R}$ is a linear map such that $f(\alpha) \neq 0$ for all $\alpha \in \Delta$, then there exists a unique highest root $\theta \in \Delta$.

proof. Suppose first that $\theta = \sum \lambda_i \alpha_i$ is a highest root. We wish first to show that $\lambda_i > 0$ for all i (it is a basic property of root systems that $\lambda_i \geq 0$ for all i). Suppose on the contrary that $\lambda_j = 0$. Expanding $(\alpha_j, \theta) = \sum \lambda_i (\alpha_j, \alpha_i)$, as all the simple roots are at obtuse or right angles, we have $(\alpha_j, \theta) \leq 0$, with equality iff $\alpha_j \perp \alpha_i$ whenever $\lambda_i \neq 0$. Now as $\theta \pm \alpha_j$ cannot be a root¹, the string condition implies that $(\alpha_j, \theta) = 0$. Thus $\alpha_j \perp \alpha_i$ whenever $\lambda_i \neq 0$ and $\lambda_j = 0$. Thus, if $\lambda_j = 0$ for some j , the simple roots decompose into disjoint perpendicular subsets $\{\alpha_j | \lambda_j = 0\}$ and $\{\alpha_j | \lambda_j \neq 0\}$. This is impossible, as the root system is indecomposable.

Now suppose we have two distinct highest roots θ and $\tilde{\theta}$. Using the string condition, we have $(\theta, \tilde{\theta}) \leq 0$, as otherwise $\theta - \tilde{\theta}$ is a root with $f(\theta - \tilde{\theta}) = 0$, which is impossible. Thus $0 \geq (\theta, \tilde{\theta}) = \sum \lambda_i (\alpha_i, \tilde{\theta})$, showing that $(\alpha_i, \tilde{\theta}) = 0$ (For $\tilde{\theta}$ is the highest root, the string condition implies that $(\alpha_i, \tilde{\theta}) \geq 0$ for each i). Yet then $\tilde{\theta}$ is perpendicular to a basis of V , so $\tilde{\theta} = 0$, a contradiction. $\square$

¹ $\theta + \alpha$ is obviously not a root, for otherwise θ is not the highest root. $\theta - \alpha$ is not a root, because the coefficients of α must have the same signs.

14.2 Cartan matrix and Dynkin diagrams

Definition 14.2.1. Let $\Pi = \{\alpha_1, \dots, \alpha_r\}$ be the set of simple roots of Δ (corresponding to f). The matrix $A = \left(\frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \right)_{i,j=1}^r = \text{diag} \left(\frac{2}{(\alpha_i, \alpha_i)} \right)_{i=1}^r \cdot ((\alpha_i, \alpha_j))_{i,j=1}^r$ is called the **Cartan matrix**. Furthermore, we can know that the determinant of all principal minors are positive from the expression $\text{diag} \left(\frac{2}{(\alpha_i, \alpha_i)} \right)_{i=1}^r \cdot ((\alpha_i, \alpha_j))_{i,j=1}^r$.

We will show later that it is independent of choice of f .

Proposition 14.2.1. The Cartan matrix has all entries integers, and the following properties:

- (a) $A_{ii} = 2$ for all i .
- (b) If $i \neq j$ then $A_{ij} \leq 0$, and $A_{ij} = 0 \Leftrightarrow A_{ji} = 0$.
- (c) All principal minors of A are positive. In particular, $\det A > 0$.

proof. (a) is immediate, and (b) follows by Theorem 14.1.1 (a). For (c), we note that we may factorize A as $\text{diag} \left(\frac{2}{(\alpha_i, \alpha_i)} \right)_{i=1}^r \cdot ((\alpha_i, \alpha_j))_{i,j=1}^r$. The first term may be ignored, while the second is the Gram matrix for the inner product with respect to Π . The result follows by Sylvester's criterion. $\square$

Definition 14.2.2. If (V, Δ) is indecomposable, we have a unique highest root θ (by the above proposition). Let $\alpha_0 = -\theta$, and $\Pi_0 = \{\alpha_0, \alpha_1, \dots, \alpha_r\}$. The matrix $\tilde{A} = \left(\frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \right)_{i,j=0}^r$ is called the **extended Cartan matrix**.

Proposition 14.2.2. $\tilde{A}$ satisfies all of the properties of Proposition 14.2.1, except $\det \tilde{A} = 0$.

proof. The proof of properties (a) and (b) are exactly the same as for the standard Cartan matrix A . However, we need to see n th the principle minors are still all positive, except for the determinant (which is zero).

Again, $\tilde{A}$ factors as $\tilde{A} = \text{diag} \left(\frac{2}{(\alpha_i, \alpha_i)} \right)_{i=0}^r \cdot ((\alpha_i, \alpha_j))_{i,j=0}^r$, and the first matrix has all diagonal entries positive, so we only need to investigate the principle minors of $M = ((\alpha_i, \alpha_j))_{i,j=0}^r$.

Given any proper subset I of $\{0, \dots, r\}$, I is a subset of some subset I' of $\{0, \dots, n\}$ with n elements. Now the α_i with $i \in I'$ are a basis of V , as the highest root is $\sum \lambda_i \alpha_i$ with the λ_i nonzero. Thus the submatrix of M corresponding to I' is a Gram matrix for the inner product, and thus has all principle minors zero. In particular, the principle minor of M corresponding to I is nonzero (by Sylvester's criterion), and thus so is the corresponding principle minor of $\tilde{A}$. Of course, the determinant of M (and thus A) is zero as the $\alpha_i (i = 0, \dots, n)$ are not linearly independent. $\square$

Definition 14.2.3. A $r \times r$ matrix satisfying all of the properties of Proposition 14.2.1 is called an **abstract Cartan matrix**.

Now, let's classify the abstract Cartan matrices.

The only 1×1 such matrix is (2), the Cartan matrix of type A_1 .

There are more possibilities for 2×2 abstract Cartan matrices A . We know that $A = \begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix}$ for nonnegative integers a, b . Moreover, $4 - ab > 0$, so that $ab \leq 3$. There are four possibilities for A (up to taking the transpose):

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}.$$

Now, we should introduce the definition of Dynkin diagram.

Definition 14.2.4. The dynkin diagram of $D(A)$ of a Cartan matrix A :

- Using r points to represent simple roots $\alpha_1, \alpha_2, \dots, \alpha_r$
- Using $A_{ij} \cdot A_{ji} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \cdot \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)} \equiv k_{ij}$ lines connect the i -th point and the j -th point

- If $(\alpha_i, \alpha_i) < (\alpha_j, \alpha_j)$, draw an arrow from the j-th point to the i-th point.¹

Then we get the Dynkin diagram of A . We can find that $D(A)$ determines the Cartan matrix.

The Dynkin diagram $D(A)$ depicts the Cartan matrix A by a graph with r vertices in bijection with the simple roots. For any two distinct simple roots α_i and α_j , the corresponding 2×2 Cartan matrix will be one of the above four, or a transpose thereof. The vertices corresponding to a chosen pair of roots are joined as follows in each case

$$\begin{array}{cccccc} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} & \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} & \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix} & \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} & \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} & \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix} \\ \circ \quad \circ & \circ \text{---} \circ & \circ \Rightarrow \circ & \circ \Leftarrow \circ & \circ \Rrightarrow \circ & \circ \Lleftarrow \circ \end{array}$$

In each diagram, the left node corresponds to the first row and column of the matrix. Note that when the two roots in question have different lengths, the arrow points to the shorter root. The diagram formed in this way is the **Dynkin diagram** $D(A)$.

Remark. Any subdiagram of a Dynkin diagram is again a Dynkin diagram. Π is indecomposable if and only if the Dynkin diagram is connected.

14.3 Some problems

1、Let the simple root system of the complex semisimple Lie algebra $\mathfrak{g}$ be $\Pi = \{\alpha_1, \alpha_2\}$, and suppose the Cartan matrix is

$$A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}.$$

Try to find all the roots.

Does the general statement hold? That is, is the entire root system uniquely determined by the given simple root system $\{\alpha_1, \dots, \alpha_r\}$ and the Cartan matrix?²

proof. We just need to analyze the positive roots, for the negative roots are simply the inverse of the positive ones.

Consider the α_2 string through α_1 , we have known that $\alpha_1 - \alpha_2$ is not a root, thus $p=0$. We know that $p - q = \frac{2(\alpha_1, \alpha_2)}{(\alpha_2, \alpha_2)} = -2$, thus $q=2$ and $\alpha_1 + \alpha_2$ and $\alpha_1 + 2\alpha_2$ are roots. We can prove that $\alpha_1 + 2\alpha_2$ is the only root of height 3, for we can consider the α_1 string through $\alpha_1 + \alpha_2$ and find that $2\alpha_1 + \alpha_2$ is not a root. Then we can find that there are not root of height 4, for $\alpha_1 + 3\alpha_2$ is not a root using the string condition and $2\alpha_1 + 2\alpha_2 = 2(\alpha_1 + \alpha_2)$ is not a root. If $3\alpha_1 + \alpha_2$ is a root, then consider the α_2 string through $3\alpha_1 + \alpha_2$, we can find that $p=0$ and $q=10$, which is a contradiction, thus it's not a root. So all positive roots are $\{\alpha_1, \alpha_2, \alpha_1 + \alpha_2, \alpha_1 + 2\alpha_2\}$ and all roots are $\{\alpha_1, \alpha_2, \alpha_1 + \alpha_2, \alpha_1 + 2\alpha_2, -\alpha_1, -\alpha_2, -\alpha_1 - \alpha_2, -\alpha_1 - 2\alpha_2\}$.

For general cases, the root system can be find samely. $\square$

¹Using a division, $(\alpha_i, \alpha_i) < (\alpha_j, \alpha_j)$ is equal to $|A_{ij}| > |A_{ji}|$. Thus we can memorize like: The arrow is from j to i if $|A_{ij}| > |A_{ji}|$.

²The problem is not very hard, one just need to analyze the strings patiently.

15 Classification of Dynkin Diagrams

15.1 The Dynkin diagrams of simple Lie algs.

During this subsection, the different choice of Π may contributes to different Cartan matrices, but the Dynkin diagrams should not be different.

Proposition 15.1.1. *First, we put all the Dynkin diagrams here:*

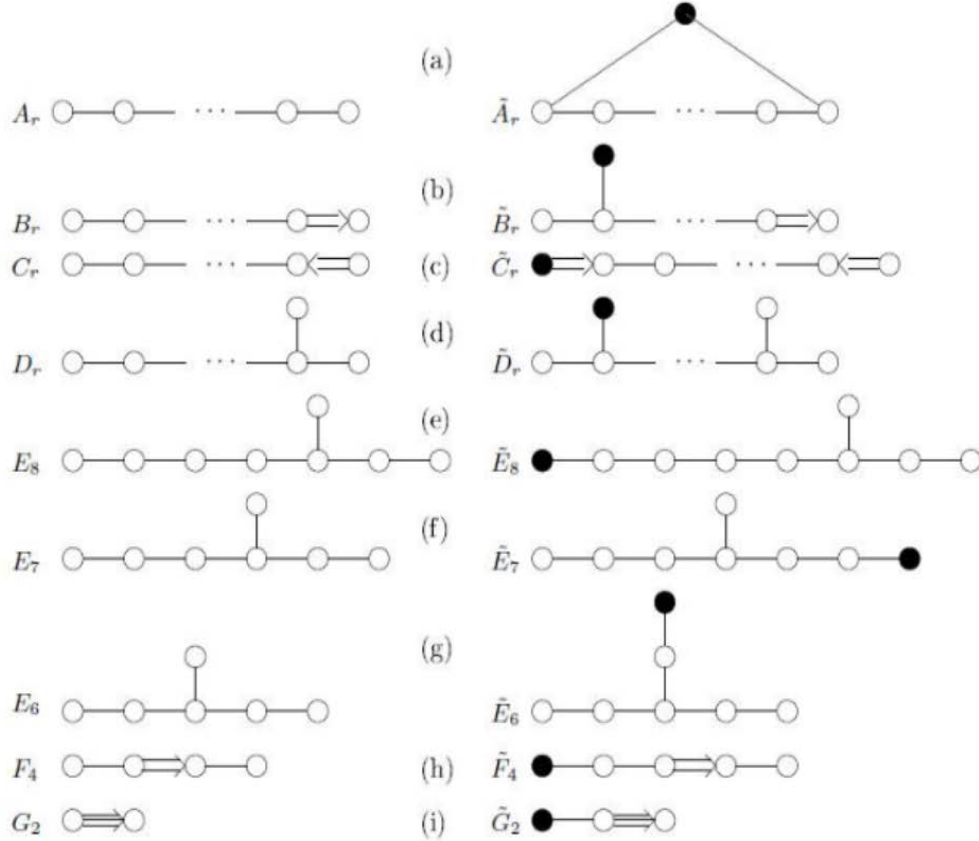


Figure 1: Dynkin diagrams and extended Dynkin diagrams of all the indecomposable root systems, from top to bottom, these correspond to: A_r , B_r , C_r , D_r , E_8 , E_7 , E_6 , F_4 , G_2 .

Example 15.1.1. $A_r (r \geq 2)$: This corresponds to $\mathfrak{sl}(r+1, \mathbb{F})$. We have $(\varepsilon_i, \varepsilon_j) = \delta_{ij}$ and $\Delta_{\mathfrak{sl}(r+1, \mathbb{F})} = \{\varepsilon_i - \varepsilon_j | i, j \in \{1, \dots, r+1\} \text{ and } i \neq j\} \subset V$, where V is the subspace of $\bigoplus_{i=1}^{r+1} \mathbb{R}\varepsilon_i$ on which the sum of coordinates (in the basis $\{\varepsilon_i | i \in \{1, \dots, r+1\}\}$) is zero.

Take $f \in V^*$ given by $f(\varepsilon_1) = r+1, f(\varepsilon_2) = r, \dots, f(\varepsilon_{r+1}) = 1$.

Then, $\Delta_+ = \{\varepsilon_i - \varepsilon_j | i < j\}$, $\Pi = \{\varepsilon_i - \varepsilon_{i+1} | i \in \{1, \dots, r\}\}$ and $\theta = \varepsilon_1 - \varepsilon_{r+1}$.

We know that its Cartan matrix and extended Cartan matrix are:

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 2 & -1 & 0 & 0 & \cdots & -1 \\ -1 & 2 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & & \vdots \\ 0 & 0 & -1 & 2 & \ddots & 0 \\ \vdots & \vdots & & \ddots & \ddots & -1 \\ -1 & 0 & \cdots & 0 & -1 & 2 \end{pmatrix}$$

Hence, we obtain the Dynkin diagrams in Figure 1 (a).

Example 15.1.2. $B_r(r \geq 3)$: This corresponds to $\mathfrak{so}(2r+1, \mathbb{F})$. We have $\Delta_{\mathfrak{so}(2r+1, \mathbb{F})} = \{\pm \varepsilon_i \pm \varepsilon_j, \pm \varepsilon_i | i, j \in \{1, \dots, r\} \text{ and } i \neq j\} \subset V = \oplus_{i=1}^r \mathbb{R} \varepsilon_i$.

Take f given by $f(\varepsilon_1) = r, \dots, f(\varepsilon_r) = 1$.

Then, $\Delta_+ = \{\varepsilon_i \pm \varepsilon_j, i < j; \varepsilon_i\}$, $\Pi = \{\varepsilon_i - \varepsilon_{i+1} | i \in \{1, \dots, r-1\}\} \sqcup \{\varepsilon_r\}$ and $\theta = \varepsilon_1 + \varepsilon_2$.

We know that its Cartan matrix and extended Cartan matrix are:

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & & \ddots & \ddots & -1 \\ 0 & \cdots & 0 & -2 & 2 \end{pmatrix} \text{ and } \left(\begin{array}{c|ccccc} 2 & 0 & -1 & 0 & 0 & 0 \\ \hline 0 & 2 & -1 & 0 & \cdots & 0 \\ -1 & -1 & 2 & -1 & & \vdots \\ 0 & 0 & -1 & 2 & \ddots & 0 \\ 0 & \vdots & & \ddots & \ddots & -1 \\ 0 & 0 & \cdots & 0 & -2 & 2 \end{array} \right)$$

Hence, we obtain the Dynkin diagrams in Figure 1 (b).

Example 15.1.3. $C_r(r \geq 1)$: This corresponds to $\mathfrak{sp}(2r, \mathbb{F})$. We have $\Delta_{\mathfrak{sp}(2r, \mathbb{F})} = \{\pm \varepsilon_i \pm \varepsilon_j | i, j \in \{1, \dots, r\}\} \subset V = \oplus_{i=1}^r \mathbb{R} \varepsilon_i$. We take the same f as in the previous case. Then, $\Pi = \{\varepsilon_i - \varepsilon_{i+1} | i \in \{1, \dots, r-1\}\} \sqcup \{2\varepsilon_r\}$ and $\theta = 2\varepsilon_1$. This gives the Cartan matrix and the extended Cartan matrix:

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & & \vdots \\ 0 & -1 & 2 & \ddots & 0 \\ \vdots & & \ddots & \ddots & -2 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix} \text{ and } \left(\begin{array}{c|ccccc} 2 & -1 & 0 & 0 & \cdots & 0 \\ \hline -2 & 2 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & & \vdots \\ 0 & 0 & -1 & 2 & \ddots & 0 \\ \vdots & \vdots & & \ddots & \ddots & -2 \\ 0 & 0 & \cdots & 0 & -1 & 2 \end{array} \right)$$

The corresponding Dynkin diagrams are shown in Figure 1 (c).

Example 15.1.4. $D_r(r \geq 4)$: This corresponds to $\mathfrak{so}(2r, \mathbb{F})$. We have $\Delta_{\mathfrak{so}(2r, \mathbb{F})} = \{\pm \varepsilon_i \pm \varepsilon_j | i, j \in \{1, \dots, r\} \text{ and } i \neq j\}$. Define $f \in V^*$ by $f(\varepsilon_1) = r-1, \dots, f(\varepsilon_r) = 0$.

Then $\Delta_+ = \{\varepsilon_i \pm \varepsilon_j | i < j\}$, $\Pi = \{\varepsilon_1 - \varepsilon_2, \dots, \varepsilon_{r-1} - \varepsilon_r, \varepsilon_{r-1} + \varepsilon_r\}$ and $\theta = \varepsilon_1 + \varepsilon_2$. This gives the Cartan matrix and the extended Cartan matrix:

$$\begin{pmatrix} 2 & -1 & 0 & \cdots & \cdots & 0 \\ -1 & 2 & -1 & & & \vdots \\ 0 & -1 & 2 & \ddots & & 0 \\ \vdots & & \ddots & 2 & -1 & -1 \\ \vdots & & 0 & -1 & 2 & 0 \\ 0 & \cdots & 0 & -1 & 0 & 2 \end{pmatrix} \text{ and } \left(\begin{array}{c|cccccc} 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ \hline 0 & 2 & -1 & 0 & \cdots & \cdots & 0 \\ -1 & -1 & 2 & -1 & & & \vdots \\ 0 & 0 & -1 & 2 & \ddots & & 0 \\ \vdots & \vdots & & \ddots & 2 & -1 & -1 \\ \vdots & \vdots & & 0 & -1 & 2 & 0 \\ 0 & 0 & \cdots & 0 & -1 & 0 & 2 \end{array} \right)$$

Hence, we have the Dynkin diagrams in Figure 1 (d).

Example 15.1.5. E_8 : We have $\Delta_{E_8} = \{\pm \varepsilon_i \pm \varepsilon_j | i, j \in \{1, \dots, 8\} \text{ and } i \neq j\} \sqcup \{\frac{1}{2}(\pm \varepsilon_1 \pm \varepsilon_2 \pm \cdots \pm \varepsilon_8) | \text{even number of } + \text{ signs}\}$ with $V = \oplus_{i=1}^8 \mathbb{R} \varepsilon_i$.

Let $f(\varepsilon_1) = 32, f(\varepsilon_2) = 6, f(\varepsilon_3) = 5, \dots, f(\varepsilon_7) = 1, f(\varepsilon_8) = 0$.

Then, $\Delta_+ = \{\varepsilon_i \pm \varepsilon_j | i < j\} \sqcup \{\frac{1}{2}(\varepsilon_1 \pm \varepsilon_2 \pm \cdots \pm \varepsilon_8) | \text{even number of } + \text{ signs}\}$. Also, $\Pi = \{\varepsilon_2 - \varepsilon_3, \varepsilon_3 - \varepsilon_4, \dots, \varepsilon_7 - \varepsilon_8, \varepsilon_7 + \varepsilon_8, \frac{1}{2}(\varepsilon_1 - \varepsilon_2 - \cdots - \varepsilon_7 + \varepsilon_8)\}$ and $\theta = \varepsilon_1 + \varepsilon_2$. We get the diagrams E_8 and $\tilde{E}_8$ in Figure 1 (e).

Example 15.1.6. E_6 : We have:

$$\Delta_{E_6} = \{\varepsilon_i - \varepsilon_j | i, j \in \{1, \dots, 6\} \text{ and } i \neq j\} \sqcup \{\pm(\varepsilon_7 - \varepsilon_8)\} \\ \sqcup \left\{ \frac{1}{2}(\pm\varepsilon_1 \pm \varepsilon_2 \cdots \pm \varepsilon_6) \pm \frac{1}{2}(\varepsilon_7 - \varepsilon_8) \mid \text{three} + \text{signs in the first 6 terms} \right\}.$$

Pick $f(\varepsilon_1) = 32, f(\varepsilon_2) = 7, f(\varepsilon_3) = 6, \dots, f(\varepsilon_7) = 2, f(\varepsilon_8) = 1$. We can then check that $\Pi = \{\varepsilon_2 - \varepsilon_3, \dots, \varepsilon_5 - \varepsilon_6, \varepsilon_7 - \varepsilon_8, \frac{1}{2}(\varepsilon_1 - \varepsilon_2 - \varepsilon_3 - \varepsilon_4 + \varepsilon_5 + \varepsilon_6 - \varepsilon_7 + \varepsilon_8)\}$, in particular, $\varepsilon_1 - \varepsilon_2 = \frac{1}{2}(\varepsilon_1 - \varepsilon_2 - \varepsilon_3 - \varepsilon_4 + \varepsilon_5 + \varepsilon_6 - \varepsilon_7 + \varepsilon_8) + \frac{1}{2}(\varepsilon_1 - \varepsilon_2 + \varepsilon_3 + \varepsilon_4 - \varepsilon_5 - \varepsilon_6 + \varepsilon_7 - \varepsilon_8)$. Also, $\theta = \varepsilon_1 - \varepsilon_6$ for obvious reasons. We obtain the Dynkin diagrams in Figure 1 (g).

Example 15.1.7. We have $\Delta_{E_7} = \{\alpha \in \Delta_{E_8} | \alpha_1 + \alpha_2 + \dots + \alpha_8 = 0\}$, so:

$$\Delta_{E_7} = \{\varepsilon_i - \varepsilon_j | i, j \in \{1, \dots, 8\} \text{ and } i \neq j\} \\ \sqcup \left\{ \frac{1}{2}(\pm\varepsilon_1 \pm \varepsilon_2 \cdots \pm \varepsilon_8) \mid 4 + \text{signs} \right\}.$$

Again, pick $f(\varepsilon_1) = 32, f(\varepsilon_2) = 7, f(\varepsilon_3) = 6, \dots, f(\varepsilon_7) = 2, f(\varepsilon_8) = 1$. Then, $\Pi = \{\varepsilon_2 - \varepsilon_3, \varepsilon_3 - \varepsilon_4, \dots, \varepsilon_7 - \varepsilon_8, \frac{1}{2}(\varepsilon_1 - \varepsilon_2 - \varepsilon_3 - \varepsilon_4 - \varepsilon_5 + \varepsilon_6 + \varepsilon_7 + \varepsilon_8)\}$ and $\theta = \varepsilon_1 - \varepsilon_8$ for obvious reasons. We obtain the Dynkin diagrams in Figure 1 (f).

Example 15.1.8. F_4 : We have $\Delta_{F_4} = \{\pm\varepsilon_i, \pm(\varepsilon_i \pm \varepsilon_j), \frac{1}{2}(\pm\varepsilon_1 \pm \varepsilon_2 \pm \varepsilon_3 \pm \varepsilon_4) | i, j \in \{1, \dots, 4\} \text{ and } i \neq j\}$.

Pick $f(\varepsilon_1) = 30, f(\varepsilon_2) = 3, f(\varepsilon_3) = 2, f(\varepsilon_4) = 1$. Then, $\Pi = \{\varepsilon_2 - \varepsilon_3, \varepsilon_3 - \varepsilon_4, \varepsilon_4, \frac{1}{2}(\varepsilon_1 - \varepsilon_2 - \varepsilon_3 - \varepsilon_4)\}$ and so $\theta = \varepsilon_1 + \varepsilon_2$. This produces the Dynkin diagrams of Figure 1 (h).

Example 15.1.9. G_2 : Finally, $\Delta_{G_2} = \{\pm(\varepsilon_1 - \varepsilon_2), \pm(\varepsilon_1 - \varepsilon_3), \pm(\varepsilon_2 - \varepsilon_3), \pm(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3), \pm(2\varepsilon_2 - \varepsilon_1 - \varepsilon_3), \pm(2\varepsilon_3 - \varepsilon_1 - \varepsilon_2)\}$.

Pick $f(\varepsilon_1) = 2, f(\varepsilon_2) = 1, f(\varepsilon_3) = 4$. Then, $\Pi = \{\varepsilon_1 - \varepsilon_2, -2\varepsilon_1 + \varepsilon_2 + \varepsilon_3\}$, and so $\theta = 2\varepsilon_3 - \varepsilon_1 - \varepsilon_2$. We obtain the Dynkin diagrams of Figure 1 (i).

15.2 Classification of Dynkin Diagrams.

Theorem 15.2.1. A complete non-redundant list of connected Dynkin diagrams is the following: $D(A_r)(r \geq 2), D(B_r)(r \geq 3), D(C_r)(r \geq 1), D(D_r)(r \geq 4), D(E_6), D(E_7), D(E_8), D(F_4)$ and $D(G_2)$. (See Figure 1 for these Dynkin diagrams and their extended versions.)

RMK: Note that $D(A_1) = D(B_1) = D(C_1), D(B_2) = D(C_2)$ and $D(D_3) = D(A_3)$.

We have now shown that any finite-dimensional simple Lie algebra yields one of a very restricted set of Dynkin diagrams (and hence Cartan matrices). The next step in the classification of semisimple Lie algebras will be to give a construction associating an abstract Cartan matrix to a Lie algebra, and hence to prove that these four classes plus five exceptional algebras are in fact the only finite-dimensional simple Lie algebras.